Ophthalmology

Workbook

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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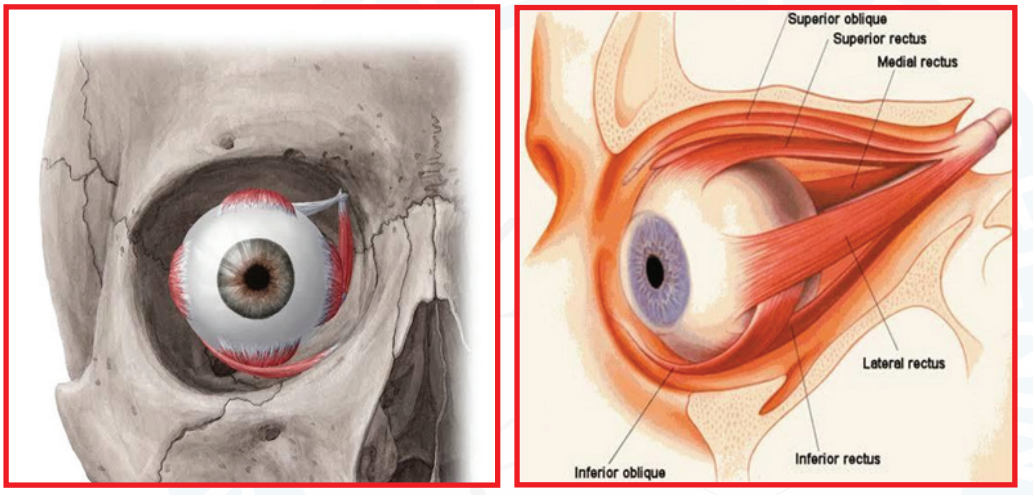
ANATOMY AND EMBRYOLOGY OF EYE

EYE ANATOMY



◀**FIG 2.1** Eye Anatomy

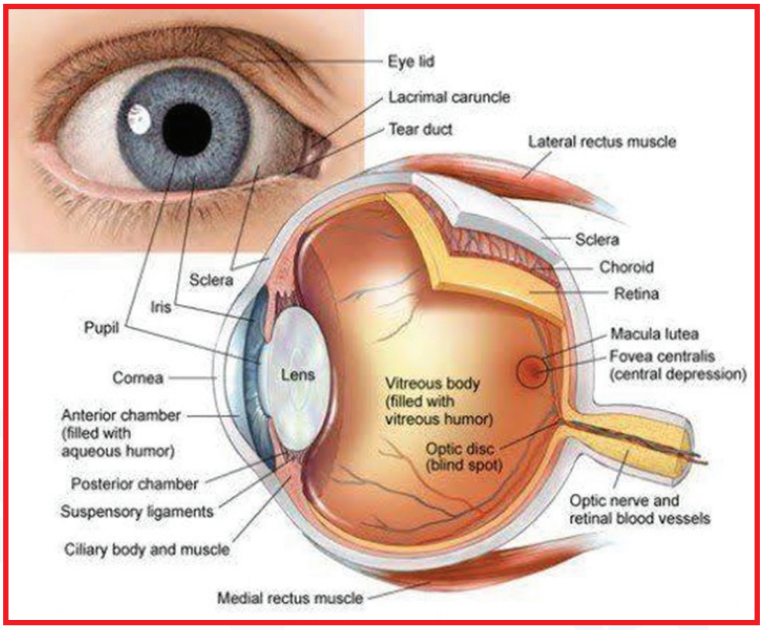
EYEBALL

- Axial length
  - Antero posterior - 24mm
- Volume
  - 6.5ml
- Cystic structure

3 COATS OF EYEBALL

- Fibrous coat
- Vascular coat
- Nervous coat

Active space



◀**FIG 2.2** Three layers of Eye

FIBROUS COAT

- Outermost coat
- Anterior 1/6th = Cornea (Main Refractive surface)
- Posterior 5/6th = Sclera (opaque)

VASCULAR COAT

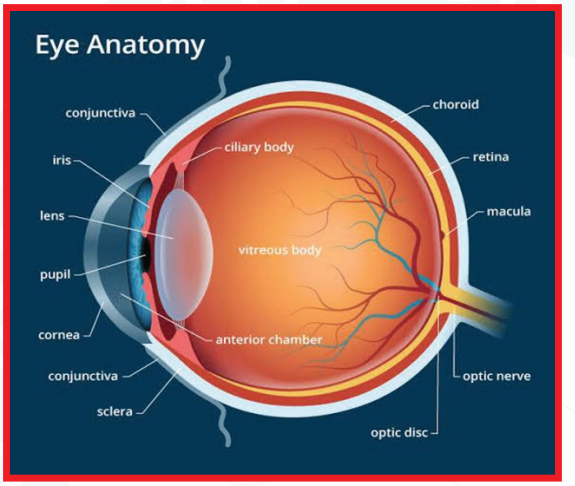
- Middle coat
- Uveal layer

1. Iris : central opening : Pupil
2. Ciliary Body
   - Pars Plicata → Ciliary Process .....
   - Pars Plana → .....
3. Choroid
   - Highly vascular
   - Supply nutrition to Outer layer of Retina

Active space

NERVOUS COAT

- Neural / Inner layer
- Retina
  - RPE (Retinal Pigment epithelium)
  - Neurosensory Retina
- Optic Nerve
  - Axons of the ganglion cells



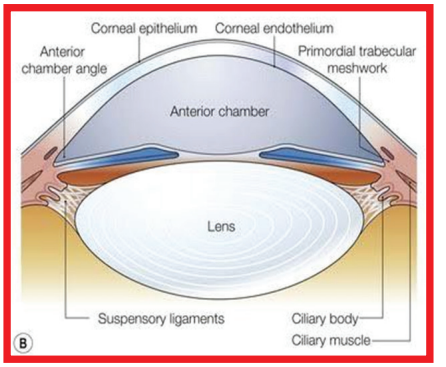
◀FIG 2.3 Eye Anatomy

SEGMENTS AND CHAMBERS OF EYEBALL

Posterior segment

⇨ Vitreous chamber

Anterior segment



◀FIG 2.4 Anterior Segment

Active space

1. Aqueous Humor
   - ✦ Ciliary Process of Pars Plicata
   - ✦ Provides nutrition to cornea & lens → Avascular
2. Vitreous Humor
   - ✦ Vitreous gel
   - ✦ 4ml
   - ✦ Provides nutrition
   - ✦ Maintains IOP

BLOOD SUPPLY

❑ **Ophthalmic artery**
- ✦ Central Retinal artery : Inner Retinal layers
- ✦ Ciliary Arteries

❑ **Cliary artery Branches**
1. Long Posterior ciliary artery → Suprachoroidal space
2. Short Posterior Ciliary Artery → Choroid
3. Anterior Ciliary Artery

EYE AT BIRTH

- ✦ Axial length – 16.5mm
- ✦ Hypermetropic
- ✦ Lens is spherical
- ✦ Macula is undifferentiated

Active space

EMBRYOLOGY OF THE EYE: IMPORTANT POINTS

1. Structures developing from Surface Ectoderm:
   Lens, Lacrimal Gland, Epithelium of Conjunctiva & Cornea

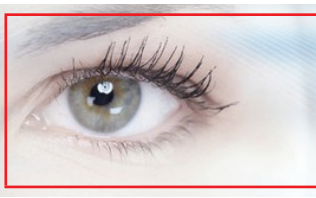
2. Structures developing from Neuroectoderm:
   Retina, Optic nerve, Secondary Vitreous, Epithelium of Iris & Ciliary body, Ciliary zonules, Sphincter pupillae & Dilator pupillae muscles

3. Structures developing from Neural Crest Mesenchyme:
   Sclera, Corneal layers except epithelium, Iris stroma, Choroid, Trabecular meshwork, Primary Vitreous

4. Structures developing from Mesodermal mesenchyme:
   Ciliary body, extraocular muscles, Iris stroma, Choroid

5. Uveal Melanocytes: Neural Crest Cells

Active space



PHYSIOLOGY OF VISION

HOW DO WE SEE

- **Visible Spectrum of eye = 380-750nm**
  - Phototransduction
    - By Photoreceptors of retina
    - Converts light signal → Neural signal
  - Transmission of signal
    - .....
    - .....
  - Visual perception

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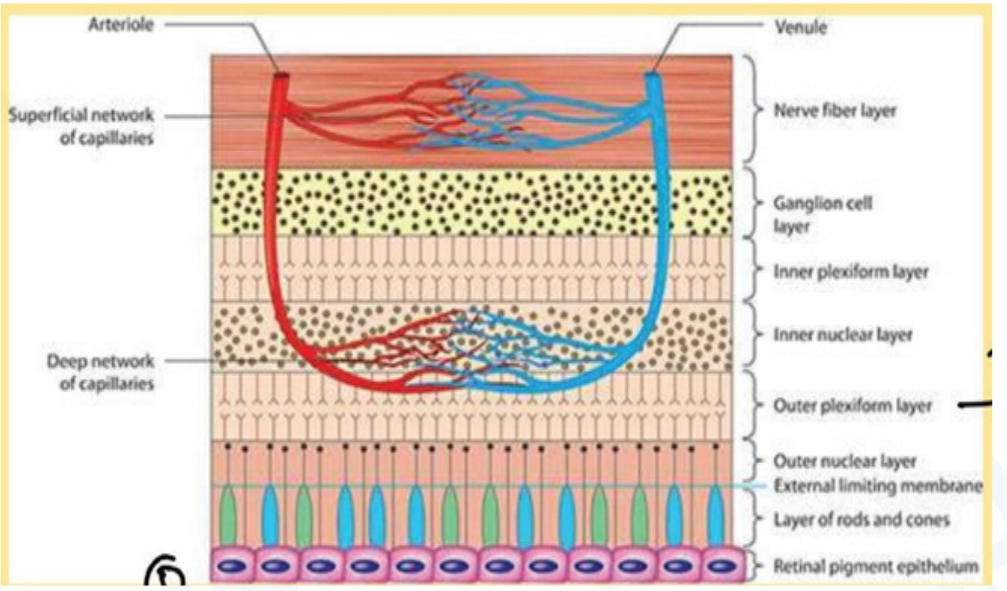
- Visual cortex



◀**FIG 3.2** Layers of Retina

RPE – 1 layer

Neurosensory Retina – 9 layers

Transduction – in Rods and cones

↓

Ganglion cell layer

↓

Axons of Ganglion cells form optic nerve

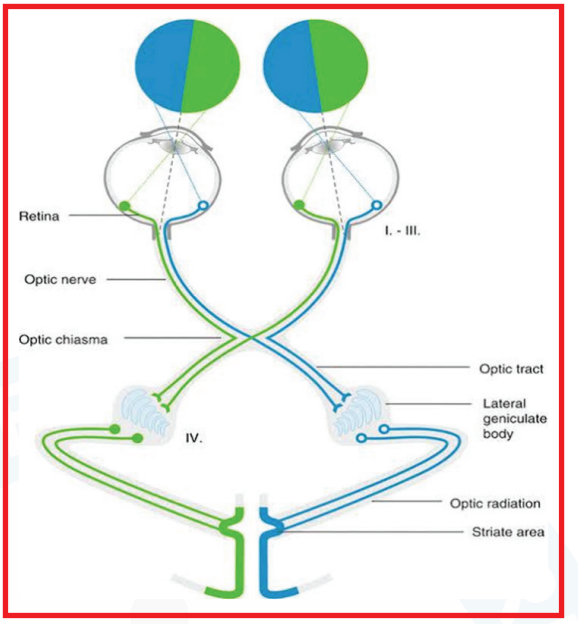
CHEMICAL BASIS OF VISUAL PROCESS

❑ Visual Pigments

Rods: Rhodopsin

Cones: Iodopsin, .

Active space



◀**FIG 3.3** Visual System

❑ **Primary Visual Cortex – Area no. 17**

.....

.....

TESTS FOR VISION

4 Components of Vision

- Light sense
- Form sense : Visual Acuity
- Color sense
- Contrast Sensitivity

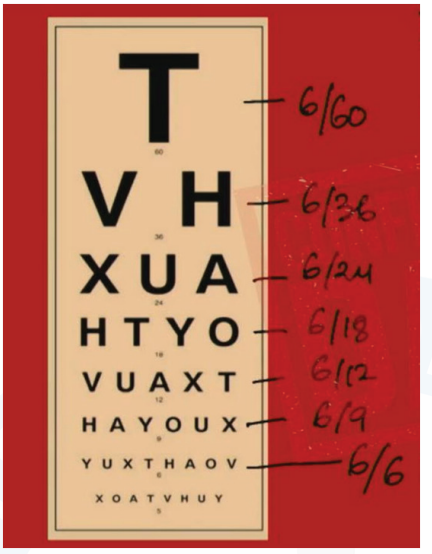
VISUAL ACUITY

Minimum angle of Resolution (MAR)

1 minute of Arc: Minimum angle which must be subtended by two points in space to be seen as separate point

Active space

DISTANCE VISION TESTING CHART



◀FIG 3.4 DistanceVision Testing Chart

SNELLEN'S CHART

Pt stands at 6m distance

6/6 → Normal value of Visual Acuity

Denominator – Distance at which normal person able to read

Numerator – Distance at which Pt reading

Pt not able to read the 1st line at 6m Distance

↓

5m distance – 5/60

4m Distance – 4/60

3m – 3/60

2m – 2/60

1m – 1/60

↓

Not visible

↓

Counting fingers

↓

Active space

MIST

Not visible

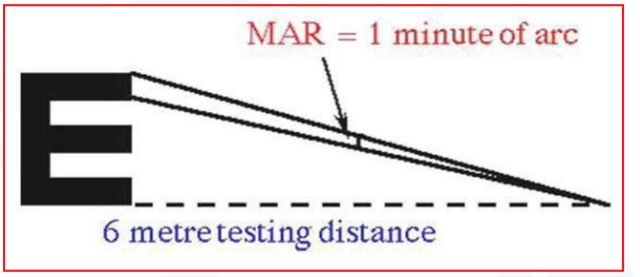
↓

Hand motions

↓

Perception of light

Projection of light Ray's



◀FIG 3.5

SNELLEN'S LETTER

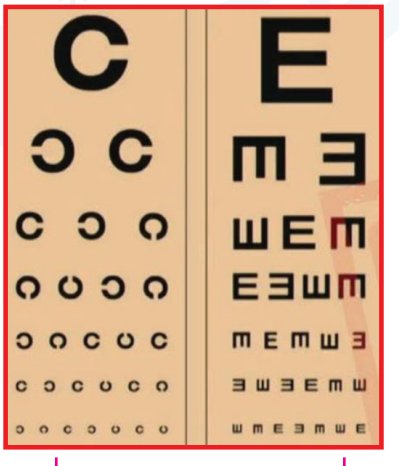
Each snellen's letter has 5 segments

Each segment subtends an angle of 1 min of Arc

Each segment – 1min of arc

Whole letter – .....

LANDOLT'S CHARTS

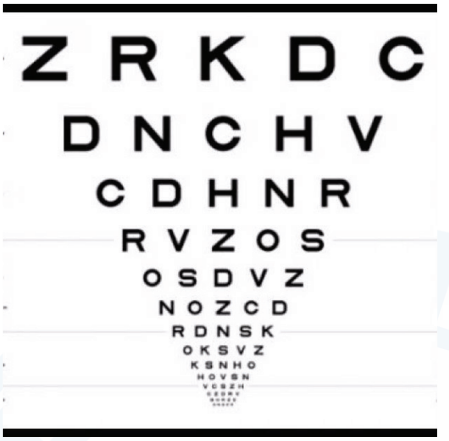


C- Chart E chart

◀FIG 3.6 Landolt's Charts

Active space

ETDRS CHART
EARLY TREATMENT OF DIABETIC RETINOPATHY STUDY



◀**FIG 3.7** ETDRS Chart

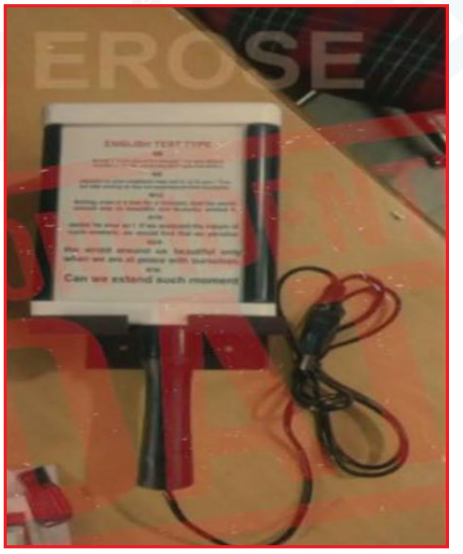
Each line has same number of letters

Testing Distance – 4m

.....

NEAR VISION CHARTS

- Snellen
- Jaeger
- Roman



◀**FIG 3.8** Near Vision Charts

Active space

VISUAL MILESTONES

- Fixation : Starts at 6 weeks , completes by 6 months
- Fusion : Develops by 6 months of age
- Steriopsis : Starts at 6 months
  Completes by 6 years
- Visual Acuity : 6/6 Develops by 6 years

VISUAL ACUITY IN CHILDREN

- Preverbal children : Objective
- Older children : Subjective

NEONATES (Objective Test)

- Teller preferential acuity test
- Visual evoked response
- Optokinetic nystagmus drum

PREFERENTIAL ACUITY TEST

◀**FIG 3.9** Perferential Acuity Test

Plain card

Pattern card – Visual Acuity is normal

Active space

OPTOKINETIC NYSTAGMUS

OPTOKINETIC DRUM (OKN DRUM)

◀FIG 3.10 OPTOKINETIC DRUM (OKN DRUM)

OPTOKINETIC NYSTAGMUS

↓

Physiological Nystagmus

1-2 YEARS

- Worth ivory ball test
- Sheridian ball test
- Boeck candy test
- Coin test

◀FIG 3.11 Ivory Clue Ball

1-4 years

- Cardiff acuity test (Monochromatic pictures)
- Allen picture cards

◀FIG 3.12 Allen Acuity Test

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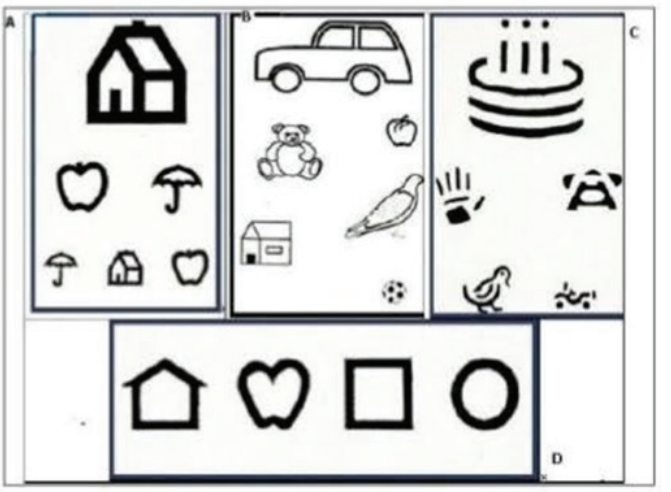
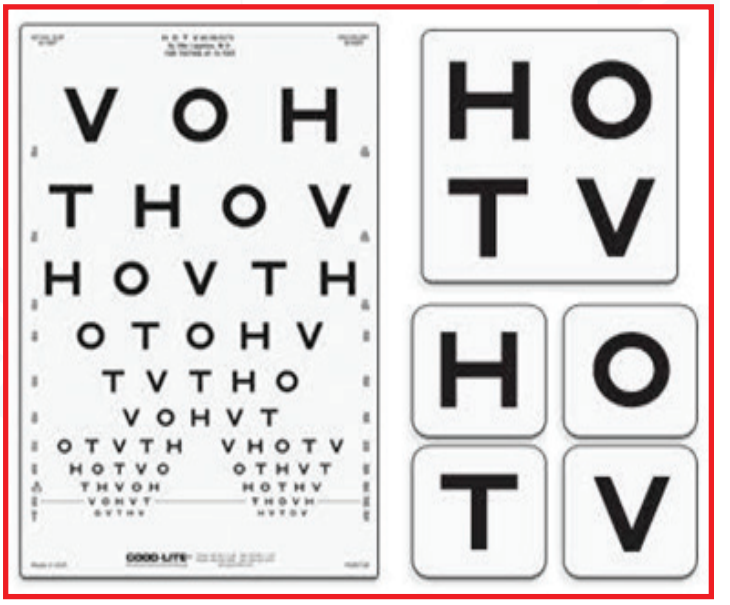
3-5 years

- Stycar (Screening test for young children and Retarded)



◀FIG 3.13 Stycar Test

SHERIDIAN HOTV



◀FIG 3.14 Sheridian Gardiner Chart

Active space

CONTRAST SENSITIVITY (Pelli Robson Chart)

◀FIG 3.15 Pelli Robson Chart

Charts for contrast Sensitivity

- Pelli – Robson Chart

COLOR VISION

- Normal color vision : Trichromatic
- Young – Helmholtz Trichromatic Theory of color vision
- 3 Primary colors : Red, Green, Blue
- 3 types of cones

COLOR VISION DECECTS

- **Anamalous: Partial color Blindness**
- **Anopia: Total Color Blindness**
  - Protanomalous (Red defective)
  - Dueteranomalous (Green defective)
  - Tritanomalous (Blue defective)
- **Dichromatic color vision: Protanopia / Deuteranopia/ Tritanopia**

Active space

MIST

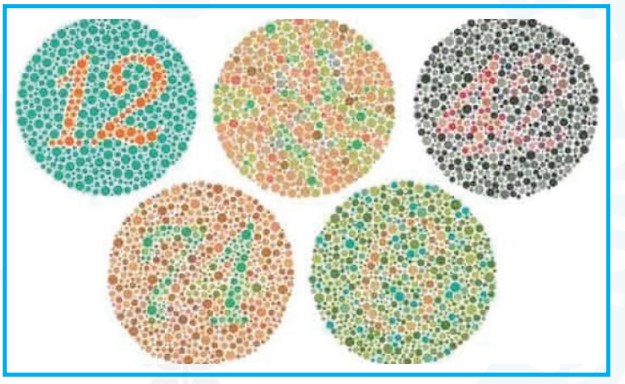
Congenital/ Acquired

- ❑ Blue – yellow defects: Retinal Diseases
- ❑ Red- green deficiency: Optic nerve Diseases

COLOR VISION TESTING

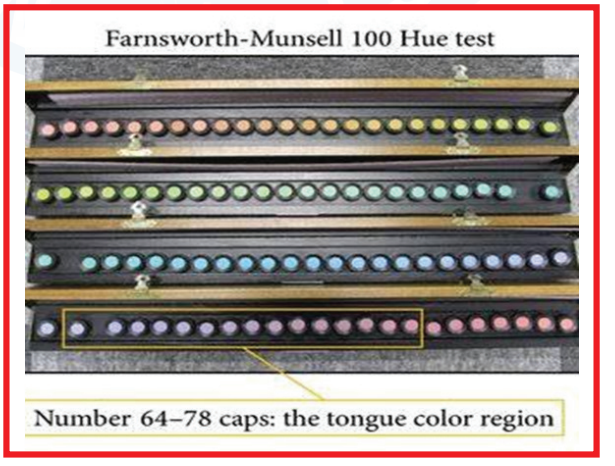
1. Pseudo isochromatic plates : Ishihara plates
2. Farnsworth Munsell (FM) 100- Hue test
3. Anomaloscopes
4. Lantren Test, wool test

Ishihara plates



◀FIG 3.16 Ishihara Plates

F- M 100 Hue Test



◀FIG 3.17 Farnsworth- Munsell 100 Hue test

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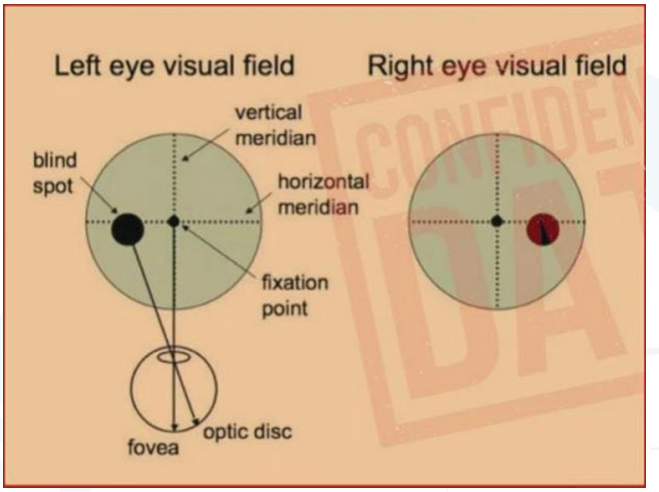
- Most specific color vision test
  Screening: Isochromatic charts (Ishihara test)
  Quantification: Fansworth Munsell 100 hue test
  Anamaloscope

VISUAL FIELDS



◀**FIG 3.18** Visual Fields

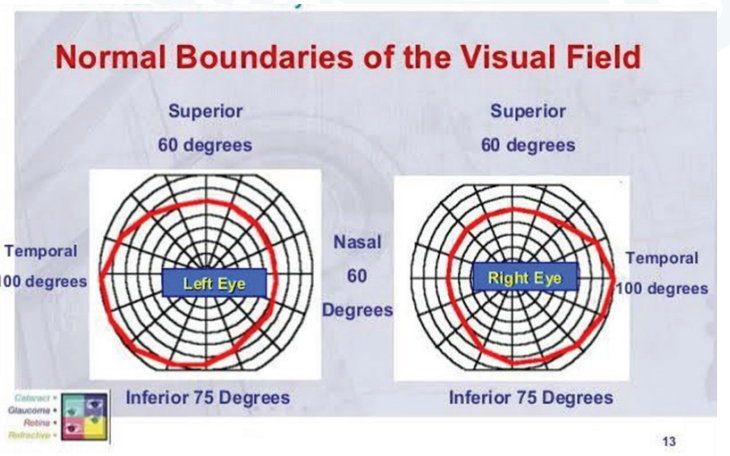
Fovea – Fixation point at centre

Blind spot – absolute Scotoma due to optic nerve

In Visual field – Blind Spot is temporal

Positive Scotoma: Floaters

Negative Scotoma: Absence of visibility



◀**FIG 3.19** Normal Boundaries of the Visual Field

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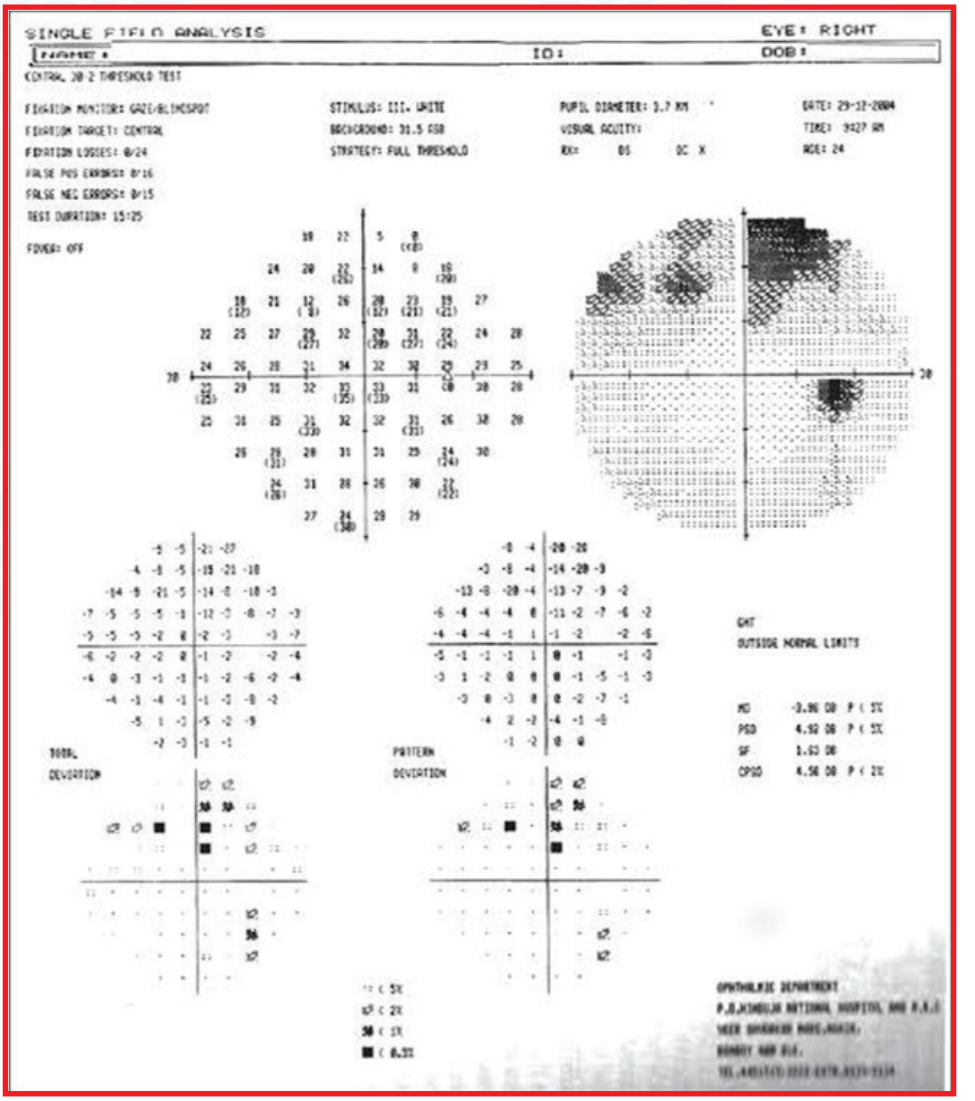
VISUAL FIELD TESTING: PERIMETRY

- Kinetic Perimetry : Moving Target
  Eg: Confrontation Method
  Goldmann perimeter
- Static Perimetry : Target stationary
  Eg: Humphrey's Perimeter
  Octopus
- Micro Perimetry : Macula area



◀**FIG 3.20** Humphrey's visual field

Active space

Workbook Questions

1. Q: The primary visual area in the cortex is:
   (a) Area no 17
   (b) Area no 18
   (c) Area no 19
   (d) None of the above MIST

2. Q: Avascular coat of the eye is
   (a) Uvea
   (b) Retina
   (c) Cornea
   (d) Choroid MIST

3. Q: Aqueous humour is secreted by
   (a) Pars plicata
   (b) Pars plana
   (c) Choroid
   (d) Iris MIST

4. Q: Refractive power of the eye is
   (a) 60D
   (b) 16D
   (c) 44D
   (d) 52D MIST

5. Q: Which of the following is not part of visual pathway
   (a) Optic nerve
   (b) Optic tract
   (c) Medial geniculate body
   (d) Optic radiations MIST

Answers: 1. (a), 2. (c), 3. (a) 4. (a), 5. (c).

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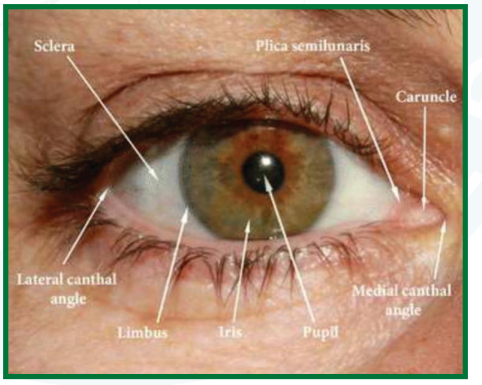
CLINICAL ANATOMY OF EYE

EXTERNAL SURFACE OF EYE



◀**FIG 4.2** External Surface of Eye

ANTERIOR SEGMENT OF EYE

◀**FIG 4.3** Anterior Segment of eye

Active space

- Instrument - Slit lamp examination
- Structures seen : Cornea, Iris, Aqueous humor, lens (Anterior & posterior surface) , Anterior vitreous

POSTERIOR SEGMENT OF EYE

◀**FIG 4.4** Posterior Segment of Eye

- Instrument used - Ophthalmoscopy
- Fundus examination → Fundoscopy
- Direct ophthalmoscopy – only central part of Retina seen
- Indirect ophthalmoscopy
- STRUCTURES : optic nerve, Optic veins & arterioles, Macula, Fovea, Mid – Peripheral part of Retina, Peripheral Retina (Ora serrata

EXAMINATION PROCEDURES

- Diffuse light examination
- Slit lamp examination
- Ophthalmoscopy

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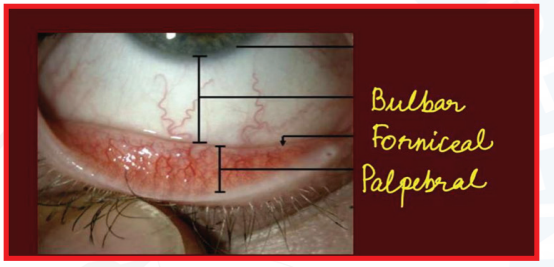
CONJUNCTIVA

ANATOMY AND CONJUNCTIVAL SIGNS

PARTS OF CONJUNCTIVA



◀**FIG 5.1** Bulbar Fornical Palpehral

SPECIALISED CELLS

✦ Goblet cells .....

- Max in number in Nasal Part of Bulbar conjunctiva

CONJUNCTIVITIS

❑ **Inflammation of conjunctiva**

✦ Allergic

✦ Infectious

☞ CLINICAL FEATURES

SYMPTOMS

1. Red eye/ Pink eye
2. Mild pain / Discomfort

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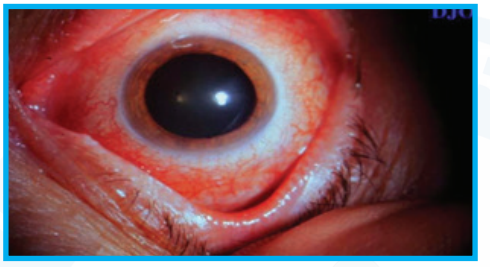
3. Grittiness
4. Photophobia
5. Watering & discharge

CLINICAL SIGNS

❑ 1. CONJUNCTIVAL CONGESTION



◀FIG 5.2 Conjunctival Congestion

- Superficial
- More in Fornices

❑ 2. DISCHARGE

- Watery : Viral
- Mucoid : Allergic
- Mucopurulent : Bacterial / Chlamydia
- Purulent : Severe Bacterial

❑ 3. PAPILLAE

◀FIG 5.3 Papillae

Giant Papillae → Cobble stone appearance

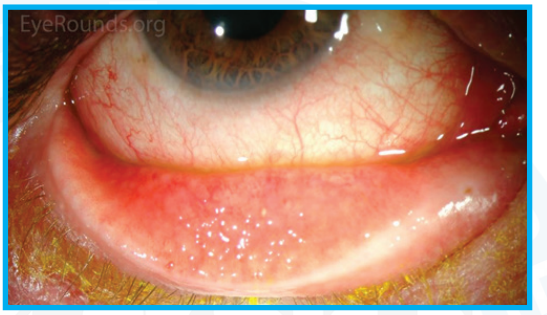
- Allergic conjunctivitis

Active space

- Bacteria
- Contact lens use

❑ 4. FOLLICLES

- Aggregation of lymphocytes
- Viral conjunctivitis
- Chlamydial conjunctivitis



◀**FIG 5.4** Follicles

❑ 5. MEMBRANE FORMATION

◀**FIG 5.5** True membrane

- True membrane → Corynebacteriumdiptheriae
- Pseudo membrane → Severe Conjunctivitis

◀**FIG 5.6** Pseudomembranous Conjunctivitis

Active space

INFECTIVE CONJUNCTIVITIS

VIRAL CONJUNCTIVITIS

◀FIG 5.7 Viral Conjunctivitis

- Self limiting

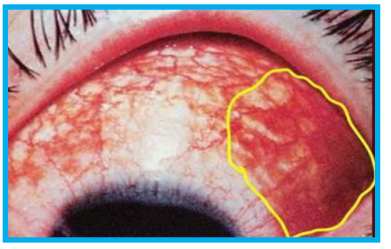
❑ **COMMON ORGANISMS**

- ✦ Adenoviral (Most common)
- ✦ Enterovirus
- ✦ Echovirus
- ✦ Coxsackie virus

ADENO VIRAL CONJUNCTIVITIS

1. Epidemic Keratoconjunctivitis (EKC)
   - ✦ Follicular, subconjunctival Hemorrhages, lymphadenopathy, Nummular lesions (80%)
2. Pharyngoconjunctival Fever
   - ✦ Pharyngitis
   - ✦ Fever
   - ✦ Conjunctivitis
   - ✦ Corneal invasion (30%)

ACUTE HEMORRHAGIC CONJUNCTIVITIS



◀FIG 5.8 Actue Hemorragic Conjuntivitis

Active space

- Enterovirus -70 (Most common)
- .....
- .....
- .....

BACTERIAL CONJUNCTIVITIS

❑ CAUSATIVE ORGANISIMS:

- Staph aureus (Most common)
- .....
- .....
- .....

◀FIG 5.9 Bacterial Conjunctivitis

❑ Membranous Conjunctivitis

- C.Diphtheria (anti diphtherial serum)
- Gonococcus

❑ Mucopurulent or Purulent

- Topical antibiotics

GONOCOCCAL CONJUNCTIVITIS

- Hyperacute Purulent discharge
- Lymphadenopathy
- Corneal ulceration
- Perforation

Gonococcus can penetrate intact cornea

Active space

☞ Treatment

- Topical antibiotics
- Cycloplegics
- Systemic

ADULT CHLAMYDIAL CONJUNCTIVITIS/ SWIMMING POOL CONJUNCTIVITIS

- ✦ Chlamydial trachomatis serotypes D-K
- ✦ Adenovirus

ANGULAR CONJUNCTIVITIS/DIPLOBACILLARY CONJUNCTIVITIS

- ✦ Inner and outer canthi
- ✦ Excoriation of skin
- ❑ **Maroxella axenfeld**

◀**FIG 5.10** Angular Conjunctivitis

☞ Treatment:

1. Oxytetracycline 1% ointment
2. Zinc oxide

OPHTHALMIA NEONATORUM

❑ **Conjunctivitis seen in neonates**

1. Within 28 days of life
2. Any discharge in a neonate is pathological

◀**FIG 5.11** Ophthalmia Neonatorum

CREDE'S METHOD

- ✦ Chemical : within hour → 1% silver Nitrate (obselete)
- ✦ Gonococcal : 1-4 days

MIST

Active space

- ✦ Chlamydial : 5-14 days
- ✦ Staph aureus : after 1 week
- ✦ Herpes simplex : after 1-2 weeks

☞ TREATMENT

❑ **PROPHYLAXIS**

- Povidone Iodine 2.5%
- Erythromycin ointment 0.5%
- Tetracycline ointment 1%

☞ TREATMENT

❑ **Most common: Chlamydial**

- Chemical : Eyewash , stop exposure
- Gonococcal : Most dangerous (cornea)
  - Bacitracin ointment / Atropine
  - Systemic antibiotic
- Chlamydial : Tetracycline ointment
- Staph aureus : Antibiotic ointment
- Herpes simplex : antiviral ointment

TRACHOMA

- ✦ Chronic Conjunctivitis of children (1-9years)
  - Major vector of Infection is flies
  - Chlamydial trachomatis serotypes A,B,Ba,C

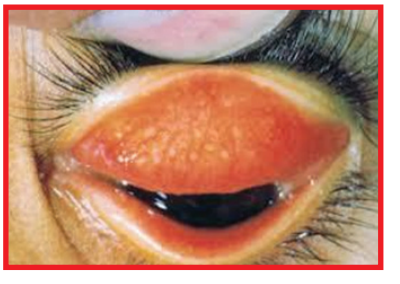
    D-k: adult inclusion Conjunctivitis

☞ CLINICAL FEATURES

- Intense itching
- FOLLICLES formation

  Tarsal conjunctiva

  Limbus



◀FIG 5.12 Trachoma

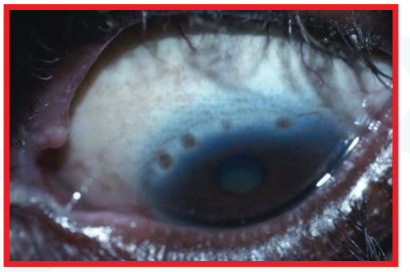
Active space

- Sago grain Follicles
- Limbus = HERBERT'S

FOLLICLES

↓ healing

HERBERT'S PITS

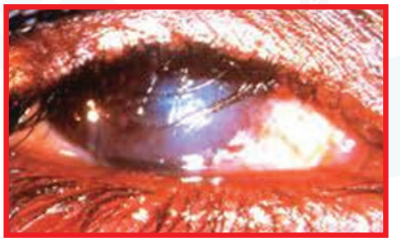


◀FIG 5.13 HERBERT'S PITS

❑ ARLT'S LINE

◀FIG 5.14 Art's Line

❑ TRICHIASIS → corneal ulcer → corneal opacity



◀FIG 5.15 Trichiasis

Vascularized corneal opacity → PANNUS FORMATION

❑ CORNEAL OPACITIES

❑ TRACHOMA: FISTO CLASSIFICATION (WHO)

☞ TREATMENT

1. Topical : 1% Tetracycline ointment

Active space

2. Systemic : DOC – Azithromycin

   Children = 20mg/kg body weight

   Adults = 1000mg

ALLERGIC CONJUNCTIVITIS

- Hallmark symptom – Itching
- Sign – Papillae formation

ACUTE ALLERGIC CONJUNCTIVITIS

- Seasonal
- Perennial

☞ Treatment

1. Topical anti Allergic mast cell stabilisers
2. Topical Antihistamines

VERNAL KERATOCONJUNCTIVITIS (VKC)/SPRING CATARRH

- Type 1 hypersensitivity to exogenous antigen
- No follicular, only Papillary reaction
- Common in ages 4-20yrs
- More common in males

Active space

☞ Clinical features

- Itching
- Redness
- Ropy discharge
- Foreign Body sensation

☞ Types

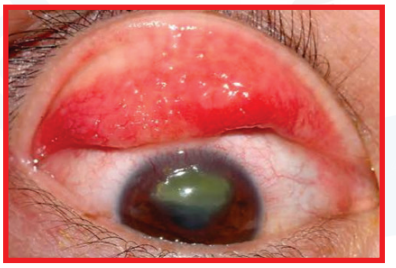
1. Bulbar
2. Palpebral
3. Mixed

◀FIG 5.16 PAPILLAE → Cobble stone appearance

❑ Horner Tranta Spots

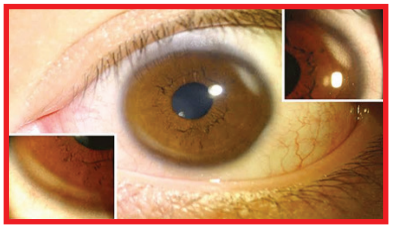
◀FIG 5.17 Horner Tranta Spots

❑ Shield Ulcer



◀FIG 5.18 Shield Ulcer

❑ Pseudo gerantaxon (Cupid's bow)



◀FIG 5.19 Pseudo gerantaxon

Active space

☞ Treatment

- Mild topical steroids and topical mast cell stabilisers

New drugs:

- Olopatadine, Epinastine (Mast cell stabiliser + Antihistamine)
- Beptostine

PHYLCTENULAR KERATOCONJUNCTIVITIS

- ✦ Mostly Unilateral
- ✦ Type 4 hypersensitivity to staphylococcal or TUBERCULAR (India) antigen.
- ✦ Affects children
- ❑ **Phylcten (yellowish nodule – limbus)**

◀**FIG 5.20** Phylcten (Yellowish Nodule)

☞ Complication:

- ❑ **Fascicular ulcer/ Sacrofulous ulcer**

CONJUNCTIVAL DEGENERATIONS

XEROPHTHALMIA

- ✦ Vitamin A Deficiency
- ✦ Keratinization of epithelium

Metaplasia of Conjunctival epithelium

(Non Keratinized → Keratinized)

Active space

☞ CLASSIFICATION

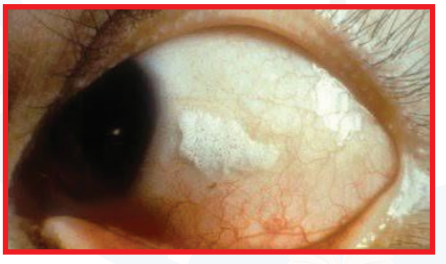
Vitamin A Deficiency

Xerophthalmia

- XN Night blindness
- X1A Conjunctival xerosis
- X1B Bitot spot
- X2 Corneal xerosis
- X3A Corneal ulceration, less than 1/3
- X3B Corneal ulceration, more than 1/3
- XS Corneal scar
- XF Xerophthalmic fundus

❑ BITOT SPOTS – Bulbar conjuctiva (Temporally)

White frothy / foamy



◀FIG 5.21 Bitot Spots seen on Bulbar Conjuctiva

❑ Keratomalacia

◀FIG 5.22 Keratomalacia

Active space

☞ Treatment

A. Topical Therapy: Lubricants

Keratomalacia: treatment of corneal ulcer

B. Vitamin A Therapy

Children <1yr of age = oral Dose of 1 Lac IU

Children >1yr = oral Dose of 2 lac IU

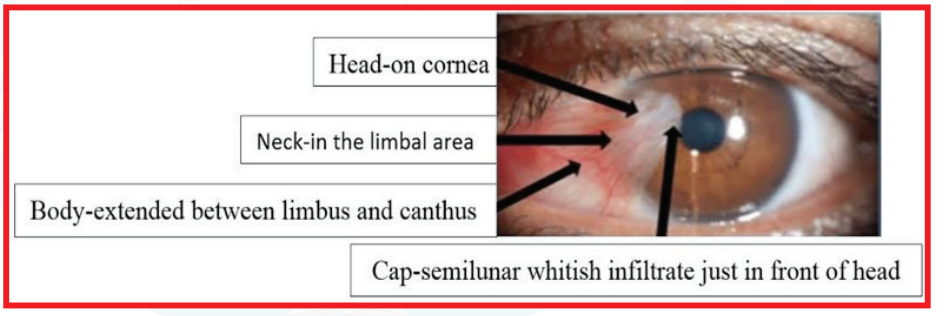
Injectable doses = $\frac{1}{2}$ of oral

CONJUNCTIVAL DEGENERATIONS

- Pterygium
- Pinguecula

PTERYGIUM

- Degenerative and hyperplastic condition (elastotic)
- Corneal Epithelium, Bowman's membrane and Stroma is destroyed
- ETIOLOGY : UV Ray's, dry heat, dusty environment



◀FIG 5.23 Pterygium

❑ **Stocker's Line**

☞ Clinical features

- Asymptomatic
- Cosmetic problem
- Astigmatism WTR
- Diplopia, Restriction of movement

Active space

- Decreased vision

☞ Treatment

- Asymptomatic is best left already alone.

☞ Surgery

- Simple excision or Bare Sclera technique

  (High recurrence rate → MMC)
- Conjunctival autograft
- Amniotic membrane graft

PINGUECULA

◀**FIG 5.24** Pinguecula

- Elastotic degeneration

Active space

Workbook Questions

1. Q: Drug of choice for trachoma is
   (a) Penicillin
   (b) Azithromycin
   (c) Tetracycline
   (d) Chloramphenicol MIST

2. Q: Horner Tranta's spots are seen in
   (a) Vernal Keratconjunctivitis
   (b) Active stage of trachoma
   (c) Healed stage of trachoma
   (d) Phylctenular conjunctivitis MIST

3. Q: Herbert's follicles are seen in
   (a) VKC
   (b) Active stage of trachoma
   (c) Healed stage of trachoma
   (d) Phylctenular conjunctivitis MIST

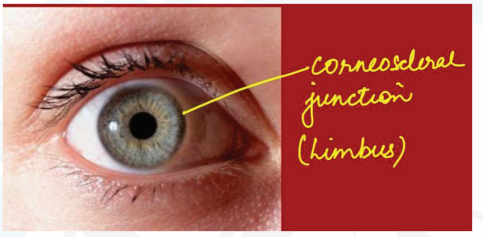
4. Q: Shield ulcer is seen in
   (a) Neurotrophic keratitis
   (b) Phylctenular conjunctivitis
   (c) Vernal Keratconjunctivitis
   (d) Herpetic keratitis MIST

5. Q: Which of the following causes acute haemorrhagic conjunctivitis?
   (a) Adenovirus
   (b) Staphylococcus
   (c) Herpes simplex
   (d) Hemophilus MIST

Answers: 1. (b), 2. (a), 3. (b) 4. (c), 5. (a).

Active space

ANATOMY AND PHYSIOLOGY



◀**FIG 6.1** Corneoscleral Junction

DIMENSIONS

- Horizontal diameter – 11.7mm
- Vertical diameter – 11mm
- Anterior surface is elliptical
- Posterior surface is circular

  Horizontal = Vertical = 11.5mm
- Diameter < 10mm → .....
- Diameter ≥13mm → .....
- Most important Refractive surface = Air – Cornea interface
- Refractive index – 1.376
- Refractive power : +43D (Total Power : + 60D)
- Average corneal thickness = 540 μ,

LAYERS OF CORNEA

1. Epithelium

Active space

2. Bowman membrane
3. Stroma
4. Dua's layer
5. Descemet membrane
6. Endothelium

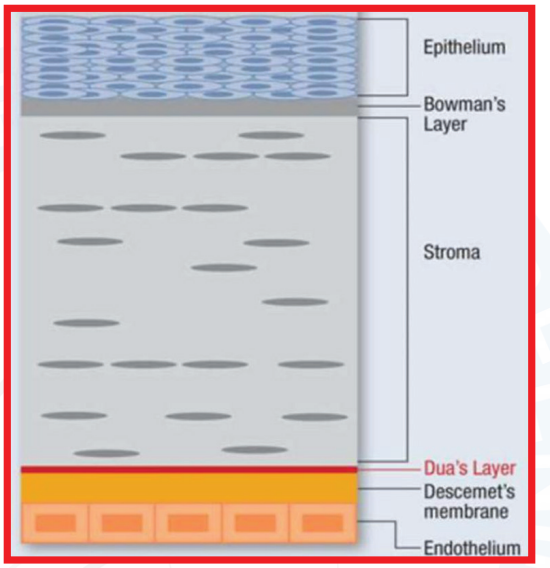


FIG 6.2 Cornea Layers

EPITHELIUM

- Stratified squamous
- Non Keratinized
- Barrier function
- Can Regenerate
- Limbal stem cells

BOWMAN MEMBRANE

- Not true membrane
- Condensed anterior Stroma
- Once injured, cannot Regenerate

↓

Leaves Scarring

Active space

STROMA

- >90% corneal thickness
- Collagen fibrils

  ↓

  Lamellae
- Contains keratocytes

DUA'S LAYER

- Pre Descemet layer
- 15microns , Toughest layer

DESCEMET MEMBRANE

- True membrane
- Secreted by endothelial cells
- Elastic membrane
- Descematocele
- Can Regenerate

ENDOTHELIUM

- Single layer of Hexagonal cells
- Uniform cells
- Na+ - K+ ATP ase pump

  ↓

  Helps maintaining dehydrated star
- Metabolically active (Most active)

❑ **Endothelial cells cannot regenerate**

- Adult = 2500- 3000cells/ mm2
- No.< 500cells/mm2 = Endothelial Decompensation → Corneal Edema

Active space

❑ Factors Responsible for corneal Transparency

- ✦ Lamellar arrangement of collagen fibres in Stroma
- ✦ Intrastromal pressure
- ✦ Pump function of Endothelium
- ✦ Barrier function of Epithelium & Endothelium
- ✦ Avascularity of cornea

NORMAL PHYSIOLOGY

- ✦ Metabolism : Aerobic (Major)
  Anaerobic conditions → Lactic acid
- ✦ Nutrition : Aqueous Humor
- ✦ Oxygen : From air in precorneal tear film
- ✦ Nerve supply : Trigeminal Nerve

CORNEAL INVESTIGATIONS

KERATOMETRY

- ✦ Measures corneal curvature of anterior surface
- ✦ Bausch & Lomb KERATOMETER

CORNEAL TOPOGRAPHY

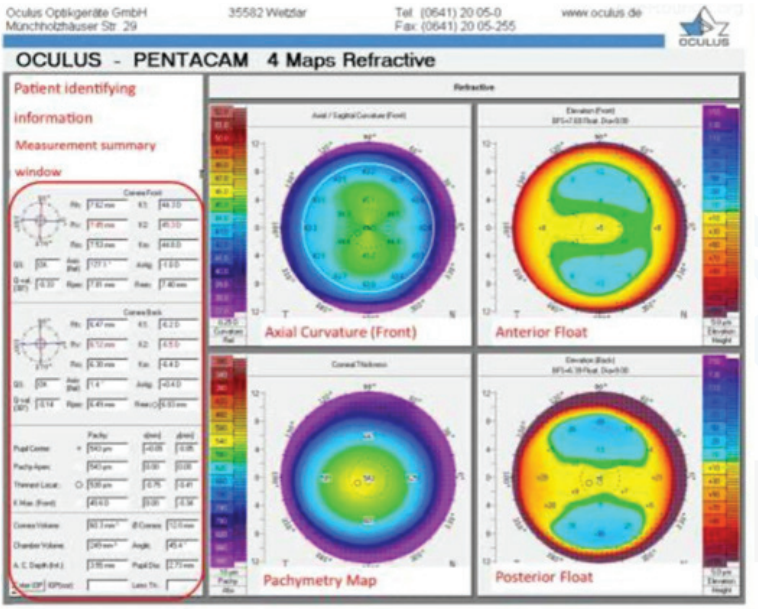
- ✦ Mapping of anterior corneal surface
- ✦ Based on Placido Disc

◀FIG 6.3 Corneal Topography

Active space

CORNEAL TOMOGRAPHY

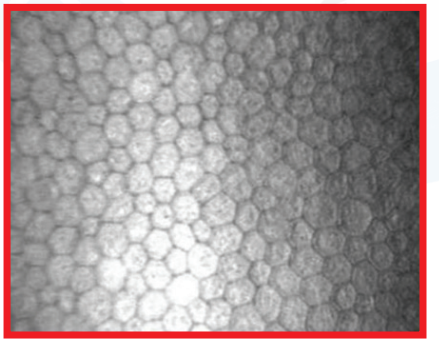
- Anterior surface And posterior surface of cornea
- Diagnosis of Keratoconus



◀FIG 6.4 Corneal Tomography

SPECULAR MICROSCOPY

- Number of Endothelial cell (count)
- Morphology of Endothelial cells



◀FIG 6.5 Specular Microscopy

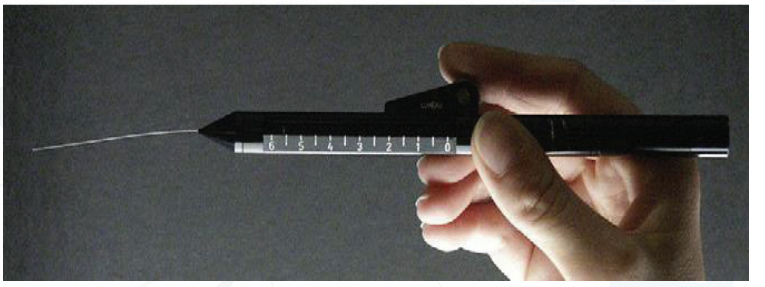
PACHYMETRY

- Measurement of corneal thickness
- Ultrasonic Pachymeter = Gold standard

Active space

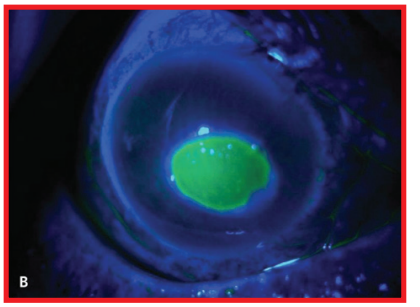
1. Corneal Reflex – Cotton Wisp Test

2. Corneal Aesthesiometer – Cochet Bonnet aesthesiometer



Vital Dyes for Corneal Staining

1. Sodium Fluroscein dye : Orange colored
   ↓ Cobalt blue filter
2. Rose Bengal Dye
   Red colored Dye
   ↓ White light



◀**FIG 6.6** Fluorescein Corneal Staining

Active space

- Fluorescein stain: Stains areas of Epithelial abrasion / Debridement
- Rose Bengal Stain: Stains Dead cells, Mucus, Dry eye

◀FIG 6.7 Rose Bengal staining

INFECTIVE KERATITIS

- Inflammation of Cornea secondary to infective Process.

CORNEAL ULCER

Epithelial defect associated with stromal infiltrate and necrosis

- Bacterial
- Fungal
- Viral
- Protozoal

Causes and Risk factors

Predisposing factors

- Corneal abrasion
- Dry eye
- Steroid abuse, anaesthetic abuse
- Chronic blepharitis
- Bullous keratopathy
- Immunosuppression

Developed countries – Contact lens use (Most common Risk Factor)

Developing countries – Trauma

Active space

Symptoms of Corneal Ulcers

- Photophobia
- Redness
- Pain
- Decreased vision
- Discharge

BACTERIAL KERATITIS

- Most common : Pneumococcus (India)
- World wide cause : Staph aureus

☞ Signs & Symptoms

- Bacterial Corneal ulcer

Organisms which can penetrate the intact Cornea

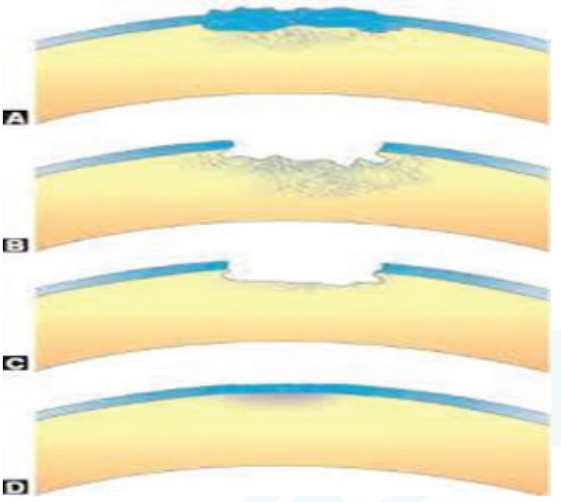
A - Acanthamoeba
S - Shigella
L - Listeria
U
N - Neisseria → Gonococcus, Meningococcus
C - Corynebacterium Diptheriae
H - Hemophilus

❑ **Pseudomonas cannot penetrate intact Cornea**

Active space

Pathology of corneal ulcer

Polymorph infiltrates
Ulceration (Superficial → Deep)



Vascularization → Healing process → Fibrosis → Opacity

Stromal Necrosis → Impending Perforation → Corneal perforation

Features of Bacterial Keratitis

- Wet Infiltrate
- Ulcer margins defined
- Hypopyon is sterile, Mobile
- Early Vascularization (+)
- Perforation (+)

FIG 6.8 Bacterial Keratitis

Active space

Management

Corneal Scrapings

Smear → Stain with Gram stain

Culture → Blood agar

- ✦ Topical Antibiotics : Fortified Antibiotics

  Fortified Cefaziline + Tobramycin

  Or

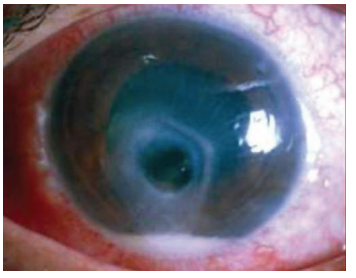
  Fortified Vancomycin
- ✦ Cycloplegics, Vitamin C oral
- ✦ Lubricants, No PATCHING

Complicated Corneal Ulcer

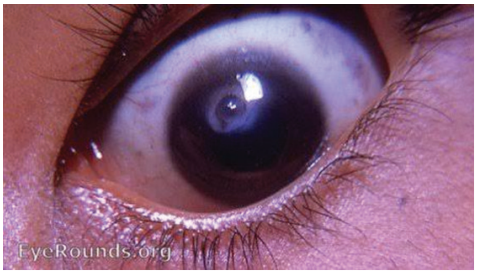
1. Non Healing Ulcer
   - Bandage contact lens
   - Epithelial Debridement
   - Chemical Cauterization (TCA)
   - Retained Foreign Body
2. Stromal Necrosis → Impending Perforation
3. Corneal Perforation

   ↓

   Iris Prolapse
4. Pseudo cornea formation

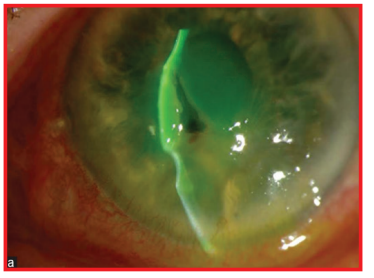


◀FIG 6.9 Stromal Thinning



◀FIG 6.10 Descematocele

Active space



◀FIG 6.11 Corneal Perforation

Treatment of Impending Perforation

1. Bandage contact lens
2. Not to strain (No coughing & No Sneezing)
3. Antiglaucoma Treatment
4. Cyano acrylate Glue (Tissue adhesive) →
   Impending Perforation & Small Sized
5. Large Perforation → Therapeutic Keratoplasty

◀FIG 6.12 Cyanoacrylate glue for small perforation

Pneumococcal Ulcer/ Hypopyon Ulcer

PSEUDOMONAS KERATITIS

- Rapidly Progressive
  ↓
  Within 48hours
  ↓
  Corneal perforation
- Most common cause of Keratitis inContact lens Users

Active space

Sequelae of Corneal Ulcer

1. Corneal opacity
2. Adherent Leucoma
3. Anterior Staphyloma

Corneal Opacities

1. Nebular Opacity → Most superficial, most visuallly disturbing
2. Macular Opacity → Iris details not visualised
3. Leucomatous Opacity → Iris not visible
4. Adherent leucoma

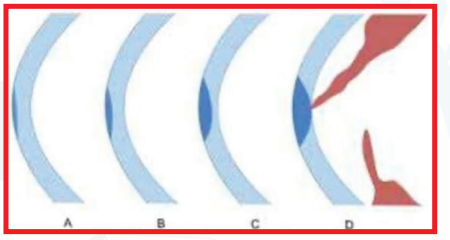


FIG 6.13 Corneal Opacities

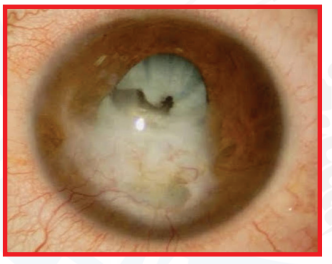


FIG 6.14 Adherent Leucoma

FIG 6.15 Anterior Staphyloma

FUNGAL KERATITIS

CAUSES & RISK FACTORS

Most common Cause – Filamentous Fungi

(Fusarium & Aspergillus)

Candida: In Immunocompromised

Active space

Risk Factors

- Trauma with organic & vegatative matter

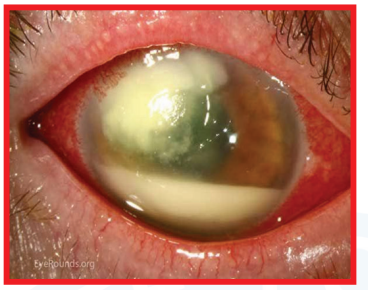


FIG 6.16 Fungal Keratitis

Features of Fungal Ulcer

- Dry, Raised infiltrate
- Feathery, ill defined margins
- Satellite lesions
- Hypopyon is Large, Dome shaped
  - Non Sterile
  - Fixed
- Vascularization & Perforation Uncommon

Management

- Wet KOH mount / Giemsa stain
- Saburaud's dextrose agar

DOC: Filamentous Fungi: Topical Natamycin

Yeast: Nystatin ointment, Amphotericin

Deep fungal Keratitis: Oral Flucanazole

VIRAL KERATITIS

- Reduced Corneal sensation
- Active infection
- Immune Reaction to viral Antigen

Active space

HERPES SIMPLEX KERATITIS

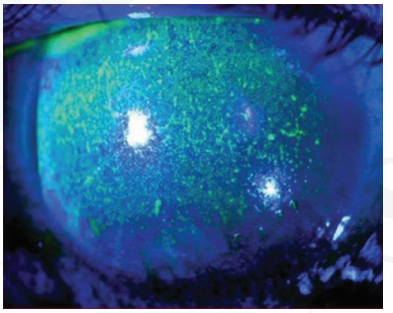
Primary lesions: Children, Conjunctivitis

Epithelial involvement

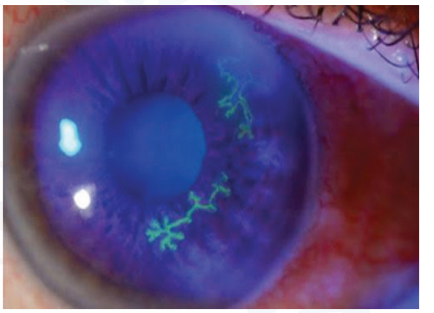
Recurrent Disease: Severe

1. Epithelium
2. Stroma
3. Endothelium

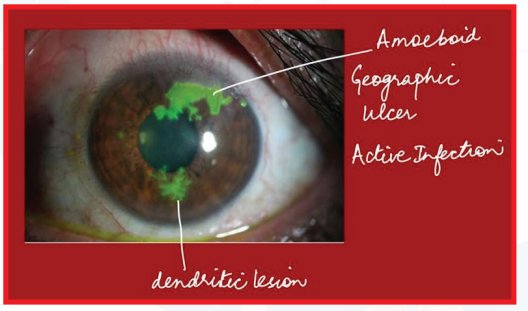
Epithelial Punctate lesions (Punctate Keratopathy)



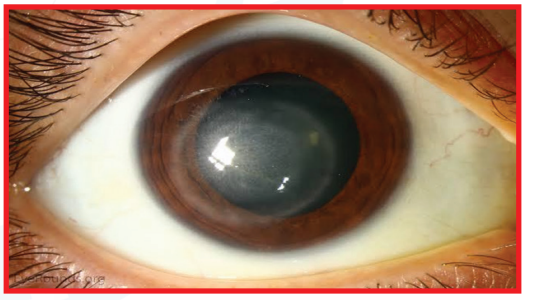
◀**FIG 6.17** Punctate Keratitis



◀**FIG 6.18** Dendritic Ulcer (Active infection)



◀**FIG 6.19** Geographic Ulcer (Active Infection)



◀ Fig 6.20 Disciform Keratitis

TREATMENT OF HERPES SIMPLEX KERATITIS

- Epithelial Keratitis (Active infection)
  - Acyclovir 3% ointment
  - Steroids are Contra indicated
- Disciform Keratitis = Immune mediated

Active space

- Topical steroids with Acyclovir 3%

✦ Metaherpetic Keratitis

- Non healing Neurotrophic ulceration

No antiviral/ No Steroids

Lubricants, BCL

HERPES ZOSTER OPHTHALMICUS

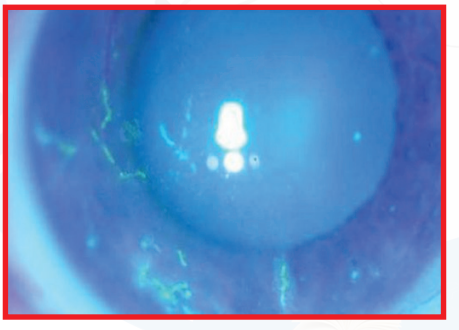
✦ Elderly

✦ Immunocompromised

✦ Gasserian ganglion Vth nerve

1. Skin eruptions
2. Fever
3. Neuralgia
4. Ocular involvement

◀FIG 6.21 Hutchinson's sign



◀FIG 6.22 Pseud odendrites

Epithelial: Punctate → Pseudo – Dendrites

❑ **STROMA**

1. Nummular lesions

   ↓

   Subepithelial infiltrate
2. Disciform keratitis
3. Kerato uveitis

MIST

Active space

TREATMENT OF HERPES ZOSTER KERATITIS

1. Systemic Acyclovir
2. Topical Acyclovir ointment
3. Cycloplegics
4. Disciform keratitis

↓

Topical steroids

MISCELLANEOUS KERATITIS

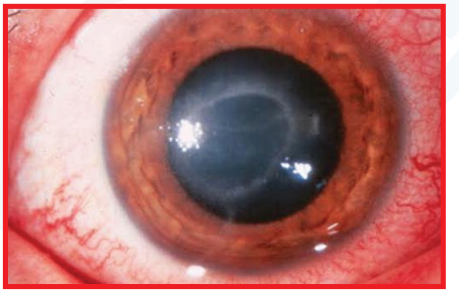
Acanthamoeba Keratitis

- Protozoa

Risk Factors

- Contact lens users
- Trauma with vegetative matter
- Use of muddy water

❑ **Severely Painful ulcer = Kerato neuritis**



◀**FIG 6.23** Acanthamoeba keratitis with ring ulcer

- Ring ulcer
- Kerato neuritis

Treatment:

DOC: Polyhexamethylene Biguanide (PHMB)

+

Propamidine

Active space

INTERSTITIAL KERATITIS

Stromal involvement without epithelial involvement

Immune mediated

☞ Treatment: Topical steroids

COGAN SYNDROME

- B/L interstitial keratitis
- Vestibulo – auditory symptoms

☞ Treatment: Systemic + Topical steroids

MOORAN'S ULCER/ RODENT ULCER

- Idiopathic
- Elderly – mild variant
- Young – Aggressive
- Immune mediated

◀FIG 6.24 Rodent Ulcer

☞ Treatment

1. Steroids
2. Immuno suppressants
3. Keratoplasty

NEUROPARALYTIC ULCER

◀FIG 6.25 Neuroparalytic Ulcer

Active space

- 7th nerve palsy
- Lagophthalmos
- Painful ulcer

Causes of Exposure Keratitis

- 7th nerve palsy
- Proptosis
- Coma

☞ Treatment: Lubricants
Patching
Tarsorrhaphy

NEUROTROPHIC ULCER

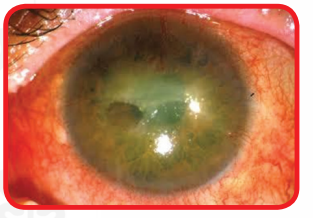


FIG 6.26 Neurotrophic Ulcer

- Painless ulcer
- Reduced Corneal sensation

Causes of Neurotrophic Ulcer

- 5th Nerve palsy
- Diabetes
- Leprosy
- Herpes
- CVA
- Brain tumors

Active space

☞ Treatment: Lubricants
Patch
Tarsorrhaphy

ECTATIC CORNEAL DISEASES

- Keratoconus
- KeratoGlobus
- Pellucid Marginal degeneration

KERATOCONUS

- Non inflammatory
- Bilateral
- More common in males
- Starts at Puberty
- Progressive Stromal Thinning

ASSOCIATIONS OF KERATOCONUS

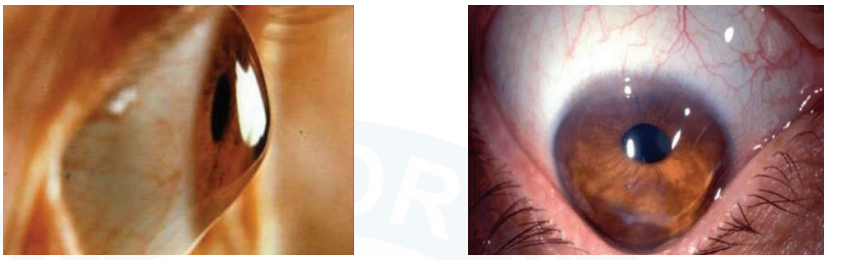
- **Ocular:**
  - Retinitis Pigmentosa
- **Systemic:**
  - Marfan syndrome
  - Down's syndrome
  - Ehler Danlos syndrome

CLINICAL SIGNS

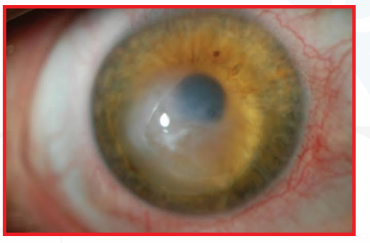
- Irregular Astigmatism – Scissoring Reflex on Retinoscopy –RIZZUTI SIGN
- Oil Droplet reflex – DDO
- Irregular Cornea on Placido Disc

Active space

- Munson's sign
- Stromal Thinning on Slit lamp
- Vogt's Stria – Vertical in Descemet's membrane
- Fleisher Ring – Iron Ring in Epithelium



◀FIG 6.27 Munsen's sign



◀FIG 6.28 Corneal Hydrops

❑ CORNEAL HYDROPS

- Sudden Rupture of Descemet's membrane
- Corneal Scarring

❑ IOC: Corneal Tomography

☞ TREATMENT

1. RGP contact lenses in Mild cases

◀FIG 6.29 Contact lenses

Active space

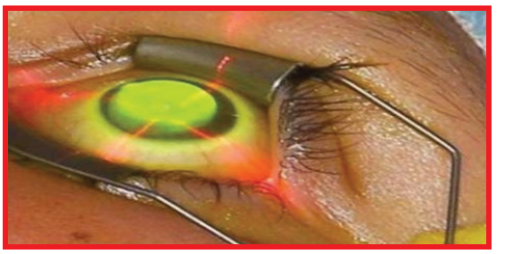
2. Collagen Cross linking C3R

Riboflavin Dye

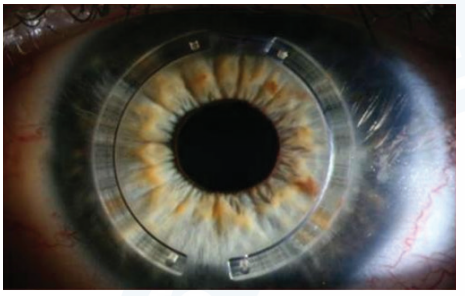
UV-A Rays

↓

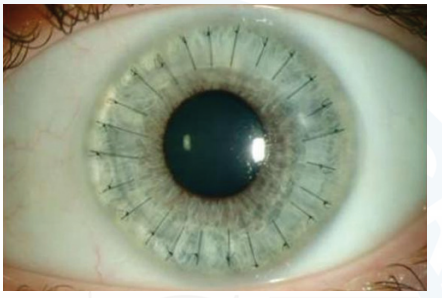
Reduce Progression



◀FIG 6.30 Collagen Cross linking C3R



(A) INTACS



(B) KERATOPLASTY

◀FIG 6.31

3. Intra Stromal corneal Ring segments
4. Penetrating Keratoplasty (PK)
5. Deep Lamellar anterior Keratoplasty

KERATOGLOBUS

- Bilateral Congenital

◀FIG 6.32 Keratoglobus

Active space

CORNEAL DYSTROPHY

- Hereditary
- Progressive
- Bilateral
- Non – Inflammatory

ANTERIOR DYSTROPHY

STROMAL DYSTROPHY

ENDOTHELIAL DYSTROPHY

COGAN MICROCYSTIC DYSTROPHY/ Map Dot- Finger print

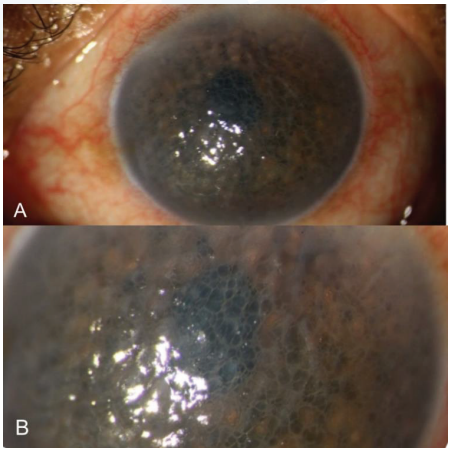
- History of Recurrent Erosions
- Most common anterior dystrophy

STROMAL CORNEAL DYSTROPHIES

- LATTICE DYSTROPHY – Most common stromal
- GRANULAR DYSTROPHY
- MACULAR DYSTROPHY

FUCH'S ENDOTHELIAL DYSTROPHY

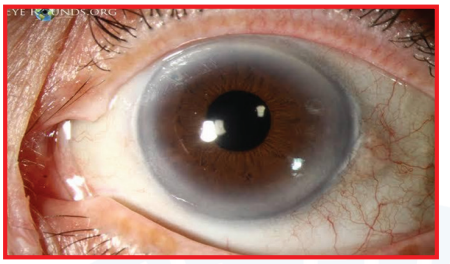
- Elderly Patient
- Loss of Endothelial cells



Active space

CORNEAL DEGENERATIONS

ARCUS SENILIS



◀FIG 6.33 Arcus Senilis

- Lipid deposition
- Starts in superior and Inferior part
- ARCUS JUVENILIS

BAND SHAPED KERATOPATHY

- Calcium deposition

◀FIG 6.34 Band Shaped Keratopathy

❑ Causes

- Uveitis in JRA
- Chronic inflammatory
- Phthisis Bulbi
- Silicone oil induced Keratopathy
- Hypercalcemia

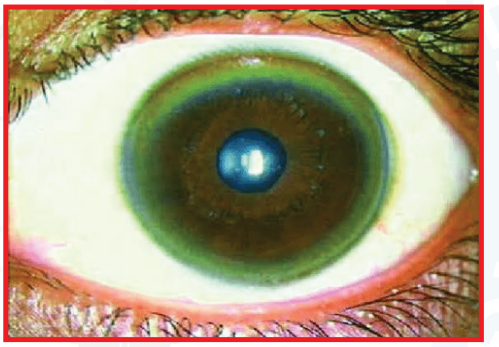
Active space

☞ Treatment

- Chelation with EDTA
- Phototherapeutic Keratectomy (PTK)
- Lamellar Keratoplasty

DEPOSITS IN CORNEA

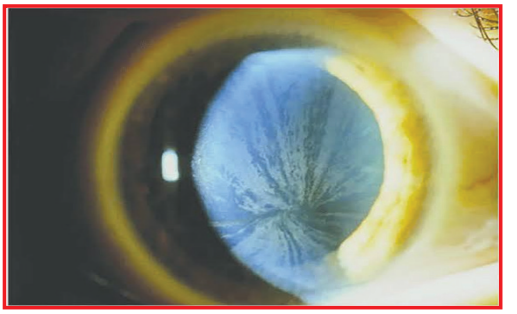
- ✦ Fleisher Ring → Keratoconus
- ✦ Stocker's line → Pterygium
- ✦ Keyser Fleisher (KF) ring
  - Copper deposition = Wilson's Disease
  - Chalcosis = copper deposition



◀FIG 6.35 Gonioscopy – Best method for early KF ring

CORNEA VORTICELLATA

- ✦ Vortex Keratopathy
- ✦ Whorl like pattern



◀FIG 6.36 Cornea Vorticellata

Active space

☞ Causes:

- Chloroquine
- Amiodarone
- Indomethacin
- Fabry's disease

KERATOPLASTY

✦ Corneal Transplantation

✦ Allograft : Replacing diseased cornea by Healthy Graft

☞ TYPES

☞ OPTICAL: Restore Vision

- Keratoconus, Scarring, Corneal dystrophy

☞ TECTONIC

- Descemetocele, Corneal Perforation

☞ THERAPEUTIC

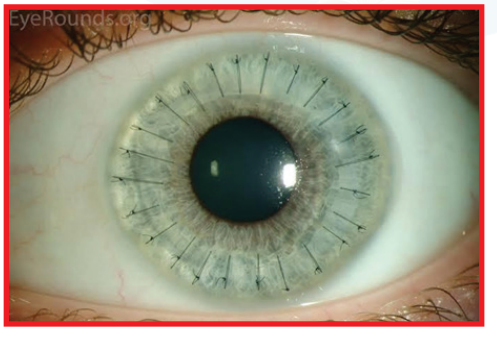
- Corneal ulcer

❑ **ALLOGRAFT**

Donor cornea should be harvested within 6hours (Upto 12hours)

✦ Penetrating Keratoplasty – Full thickness graft

✦ Lamellar – Partial Thickness

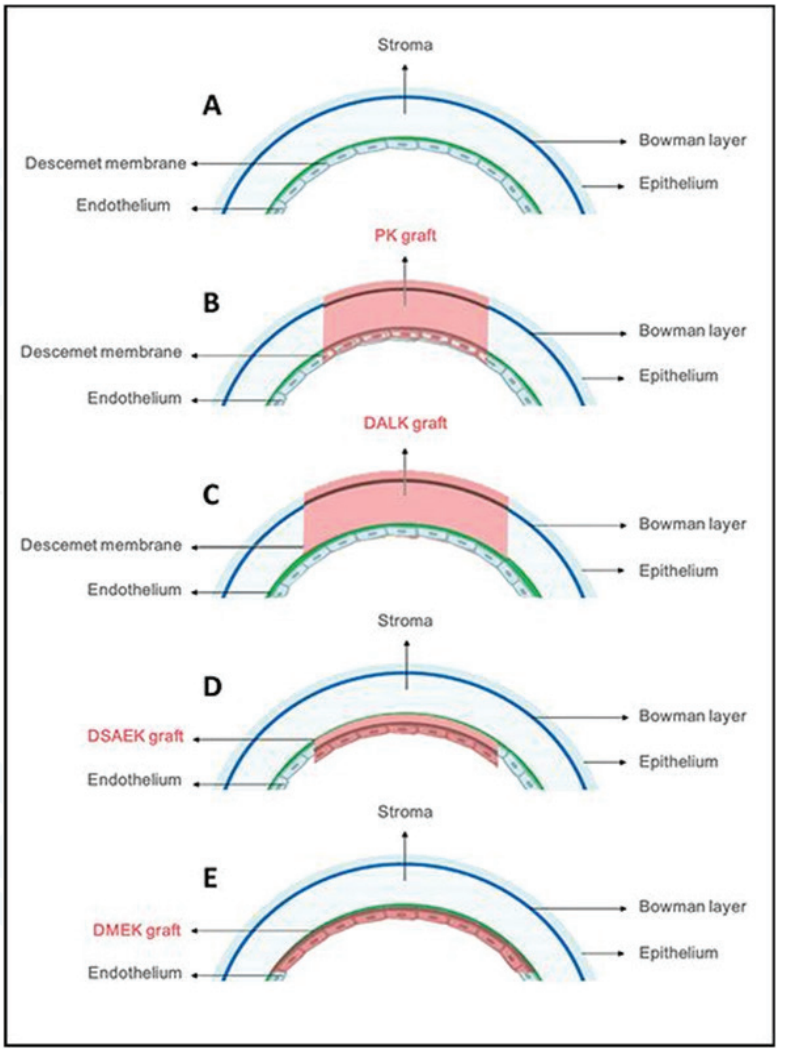


◀**FIG 6.37** Penetrating Keratoplasty

Active space

LAMELLAR KERATOPLASTY

- Anterior Lamellar – Host Descemet's membrane, Normal Endothelium
- Posterior Lamellar – Endothelial diseases



◀**FIG 6.38** Evolution of Posterior Lamellar Keratoplasty

Active space

Methods of Corneal Preservation

1. Short Term Storage:
   a. Moist Chamber method: whole globe is stored at 4*C upto 24 hours.
   b. McCarey Kaufman (M-K) medium: Donor corneoscleral button is stored upto 96 hours.

Most commonly used medium.

Consists of Tissue culture-199, 5% Dextran-40, HEPES buffer and gentamycin.

2. Intermediate Term Storage:

Storage upto 2 weeks.

Contains chondroitin sulphate.

Most common : Optisol GS, Cornisol.

3. Long-term preservation:

- Glycerine preservation upto 3-4 weeks.
- Organ culture method upto 35 days.
- Cryopreservation upto 1 year

Active space

Workbook Questions

1. Q: The most metabolically active layer of cornea is
   (a) Endothelium
   (b) Epithelium
   (c) Stroma
   (d) Descemet's membrane MIST

2. Q: Most common cause of corneal perforation in a rapidly progressive ulcer
   (a) Pneumococcus
   (b) Pseudomonas
   (c) Staph aureus
   (d) Diphtheria MIST

3. Q: Most common infection after soft contact lens use is
   (a) Pseudomonas
   (b) Pneumococcus
   (c) Acanthamoeba
   (d) Aspergillosis MIST

4. Q: All of the following are clinical signs of Keratoconus except
   (a) Vogt striae
   (b) Kayser-Fleischer ring
   (c) Rizzuti sign
   (d) Munsen's sign MIST

Active space

5. Q: A 30 year old farmer comes with pain, watering and decreased vision in the right eye for 2 days following injury with a twig. On examination, there is a corneal ulcer with elevated margins, satellite lesions and hypopyon. What is the most likely causative organism in this case?

   (a) Pseudomonas

   (b) Aspergillus

   (c) Acanthamoeba

   (d) Herpes simplex MIST

Answers: 1. (a), 2. (b), 3. (a) 4. (b), 5. (b).

Active space

DISEASES OF SCLERA

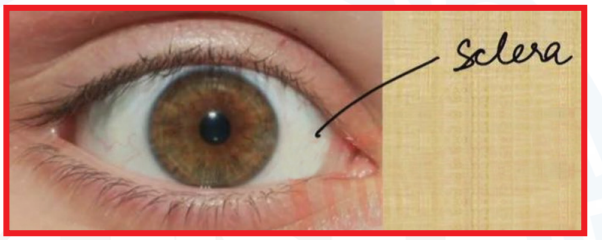


FIG 7.1 Sclera

Sclera is covered by

- Tenon's capsule
- Anterior part also covered by Bulbar conjunctiva

Thinnest – posterior to insertion of Rectus muscle

Prone to Rupture due to blunt trauma

Thickest - Peripapillary area

Inflammations of Sclera

- Episcleritis
- Scleritis

☞ Causes

- Collagen Vascular diseases
- Idiopathic
- Infectious
- Sarcoid

Active space

EPISCLERITIS

◀FIG 7.2 Diffuse Episcleritis

◀FIG 7.3 Nodular Episcleritis

☞ Clinical Features

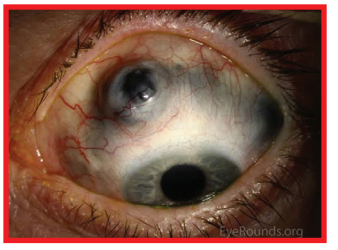
- ✦ Mild pain
- ✦ Superficial congestion

☞ Treatment

- ✦ NSAID'S Topical
- ✦ Self limiting

SCLERITIS

- ✦ Severe inflammation
- ✦ Severe pain

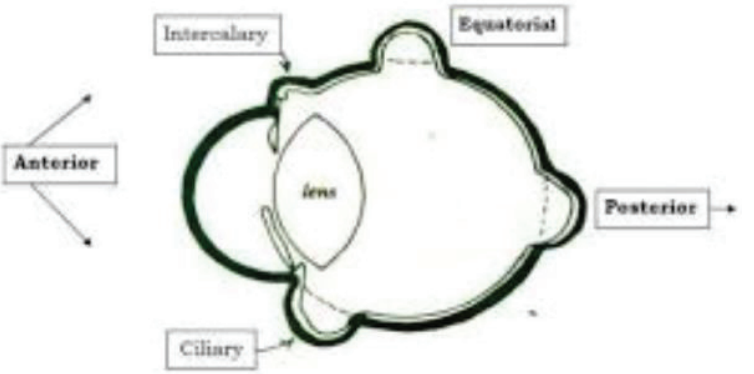


◀FIG 7.4 SCLEROMALACIA PERFORANS

Active space

STAPHYLOMA

- Ectasia of outer coats of the eyeball with incarceration of underlying uveal tissue.
  1. Anterior – Perforated corneal ulcer
  2. Intercalary – In peripheral ulcerative keratitis
  3. Ciliary – Healed scleritis
  4. Equatorial – Healed scleritis, high myopia
  5. Posterior – Pathological myopia



Active space

OPTICS AND REFRACTIVE ERRORS

NORMAL OPTICS OF THE EYE

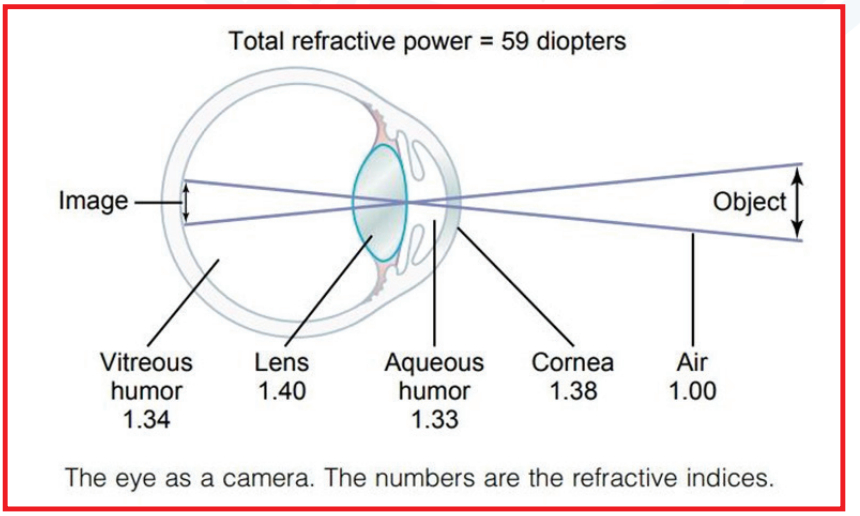
Total Refractive power of eye = + 60D

Cornea = 44-45D

Lens = 15-16D

❑ REFRACTIVE INDICES:

Maximum Refraction – Air cornea interface



The eye as a camera. The numbers are the refractive indices.

Active space

AMETROPIA

Ray's of light are not focusing at a single point on the Fovea

1. Myopia
2. Hypermetropia / Hyperopia
3. Astigmatism

MYOPIA

Short sightedness

Near sightedness

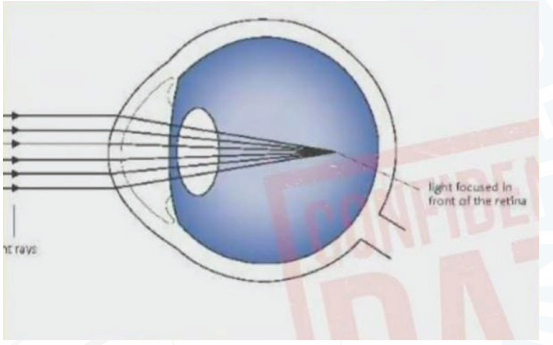


FIG 8.2 Myopia

TYPES OF MYOPIA

1. Axial Myopia – axial length of eye more (Most common)
2. Curvature Myopia – Curvature of cornea ↑
   - Lenticonus
3. Index Myopia – Refractive index of lens ↑
   Nuclear sclerosis
4. Positional Myopia – Displacement of lens

CLINICAL FEATURES

- Axial length increased
- Steep cornea
- AC depth increased

Active space

- Large optic disc
- Myopic retinal changes

SYMPTOMS

- Blurred vision for Distance
- Exophoria
- Floaters

◀FIG 8.3 Myopic Disc

Increase in Cup disc ratio = physiologically large cup

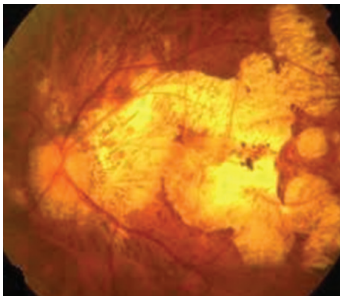
PATHOLOGICAL MYOPIA

High Myopia > -6D

Pathological Myopia = High Myopia + Chorioretinal degeneration

❑ **Fundus exam**

1. Tesselated/ Tigroid Fundus
2. Temporal Crescent
3. Chorioretinal scarring,
   Posterior Staphyloma



◀FIG 8.4 Peripapillary atrophy

◀FIG 8.5 Temporal crescent

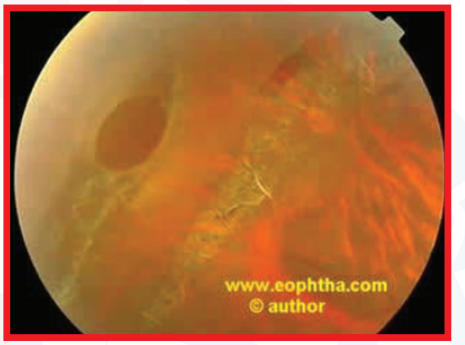
Active space

4. Lacquer cracks – Rupture of Bruch's membrane

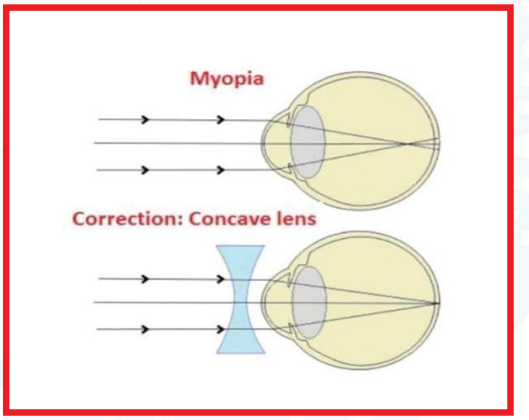
↓

5. Chorioidal Neovascular membrane (CNVM)
6. Foster Fuch spots
7. **Lattice degeneration in Peripheral Retina**

✦ Retinal Hole, Retinal Detachment



◀**FIG 8.6** Lattice Degeneration in Peripheral Retina



◀**FIG 8.7** MYOPIA CORRECTION

Concave lens: Minus lens

Diverging lens

Minification

1D = 2% Minification

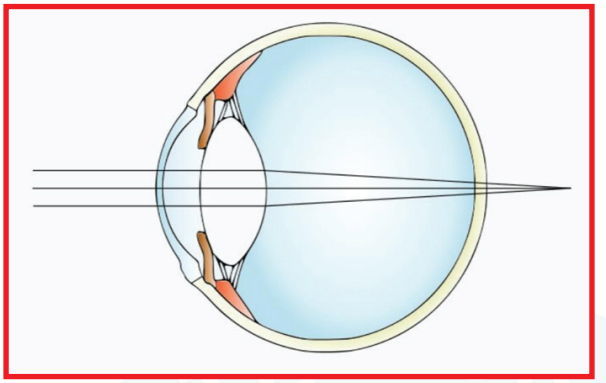
HYPERMETROPIA

Farsightedness

Long-sightedness

Active space

A/k Hyperopia



◀FIG 8.8 Hypermetropia

- Asthenopia
- More stye / Chalazion
- Esophoria

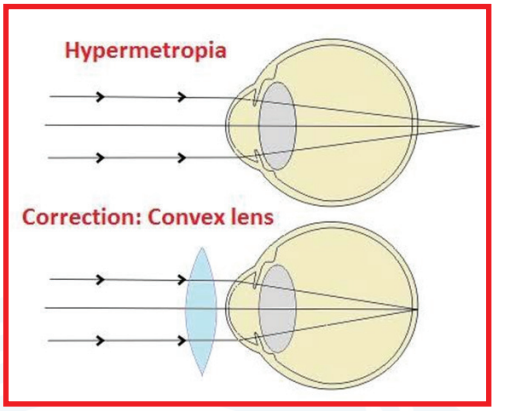
CLINICAL FEATURES

- Short axial length
- Flat cornea
- AC depth Decreased
- Small optic disc – Pseudo papillitis
- Silk shot appearance of fundus

◀FIG 8.9 Pseudo Papillitis

Active space

❑ CORRECTION

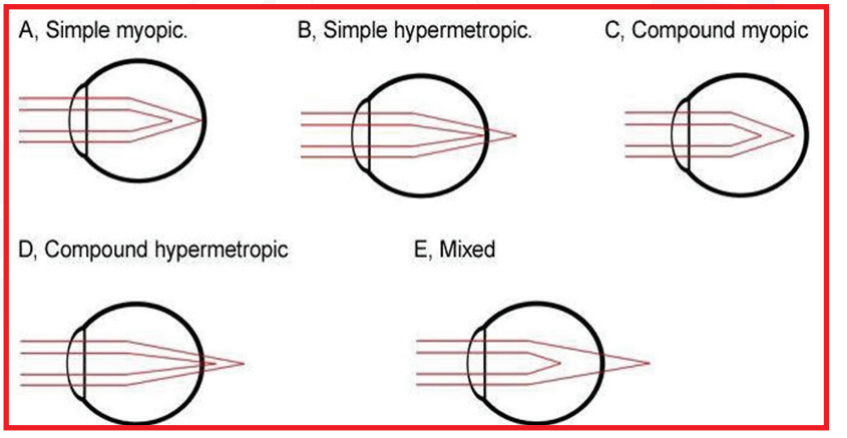


◀FIG 8.10 Hypermetropia

❑ Convex Lens

- ✦ Plus lens
- ✦ Converging
- ✦ 1D = 2% magnification of image

ASTIGMATISM



◀ FIG 8.11 Astigmatism

↓

At 360° = Same Refractive power at all points at all axis
No astigmatism

❑ 2 Parts

- ✦ Regular astigmatism
- ✦ Irregular Astigmatism

Active space

REGULAR ASTIGMATISM

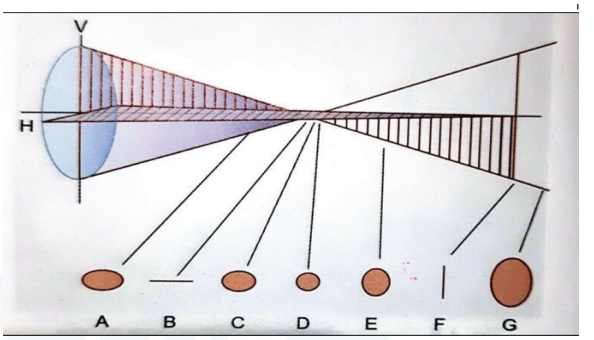
2 Principal meridia

- 90° or vertical meridian
- 180° or Horizontal meridian

CORRECTION

- Toric / Cylindrical glasses

STRUM'S CONOID



ACCOMODATION

- Helps us to focus at near objects
- Part of near reflex
  - Convergence
  - Pupillary constriction (Miosis)
  - Accomodation
    - Contraction of Ciliary muscle
    - Relaxation of Zonules
    - Increase in curvature of Anterior surface of lens

PRESBYOPIA

Physiological Decrease in capacity of accomodation with age

- Decreased tone of Ciliary muscle
- Rigidity of Zonules
- Decreased flexibility of lens

CORRECTION OF PRESBYOPIA

Convex lens (Plus lens)

40yrs = +0.5D

45yrs = +1.0D

50yrs = +1.5D

55yrs = +2.0D

>60yrs = +2.5 to +3.0D

Active space

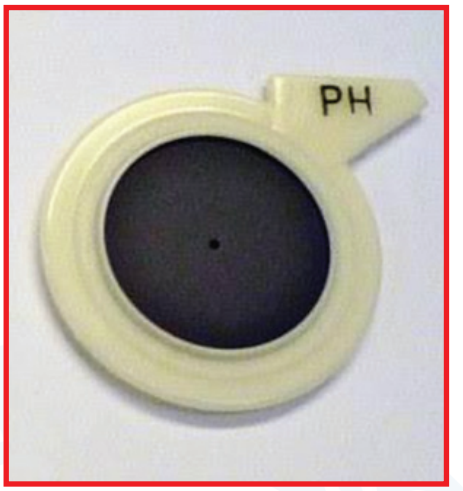


FIG 8.12 PINHOLE

1mm opening – only central light Ray's passing

PINHOLE

↓

1. If vision improves

↓

Refractive Error

2. If Vision Decreases

↓

Macular disease

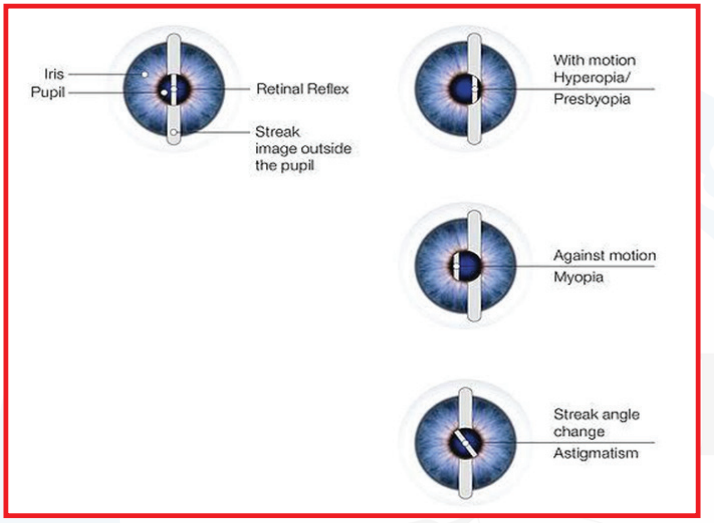
Central Cataract

Central Corneal opacity

RETINOSCOPY

- Objective
- To measure Refractive error
- Dark room procedures

Active space



◀**FIG 8.13** Reflex movement on retinoscopy

❑ RETINOSCOPY:

Same movement

Hypermetropia

Emmetropia

Myopia < -1D

Against motion = Myopia > -1D

Active space

❑ USE OF CYCLOPLEGIA

- ✦ Children
- ✦ Young adults
- ✦ Mydriasis – Elderly patient
- ✦ After 20yrs of age
- ✦ Pseudophakic, Aphakic

❑ CYCLOPLEGIC DRUGS

1. Atropine 1%
   - Strongest cycloplegic
   - Children
   - Drops & ointment
   - Ointment → children
   - Action lasts 2weeks
   - Preferred Upto 5yrs
2. 5-8 yrs → Homatropine

   ↓

   Action: 3days
3. Cyclopentolate 1%

   ↓

   8-20yrs

   Effect – 24hrs
4. Tropicamide

   Fastest Acting cycloplegic

   Shortest duration
5. Phenylephrine

   Not cycloplegic

   Only mydriatic

❑ GLASS PRESCRIPTION

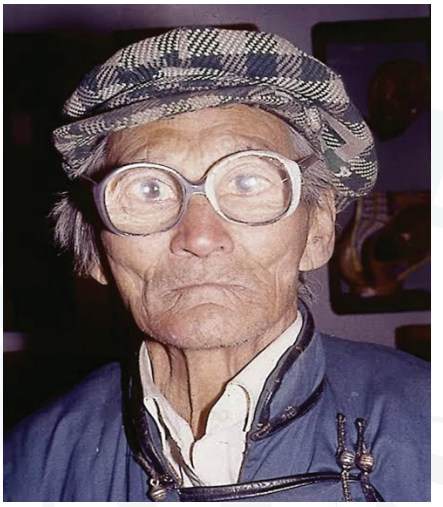
Convert Retinoscopy → Glass prescription

Active space

❑ APHAKIA

Absent lens

- ✦ Hypermetropic +15 to 16D
- ✦ High plus glasses (+8 to +10D)



◀**FIG 8.14** Aphakia Symptoms

☞ Disadvantage:

1. Magnification of image
2. Field of Vision
3. Ring scotoma
4. Pin cushion affect
5. Unilateral aphakia

↓

Diplopia

Active space

FIG 8.15

- **ROVING RING SCOTOMA**
- **CORRECTION OF APHAKIA**
  - Contact lens
  - Secondary IOL Implantation
    - AC IOL
    - Scleral fixated IOL

CORRECTION OF REFRACTIVE ERROR

- Spectacles
- Contact lens
- Refractive surgery

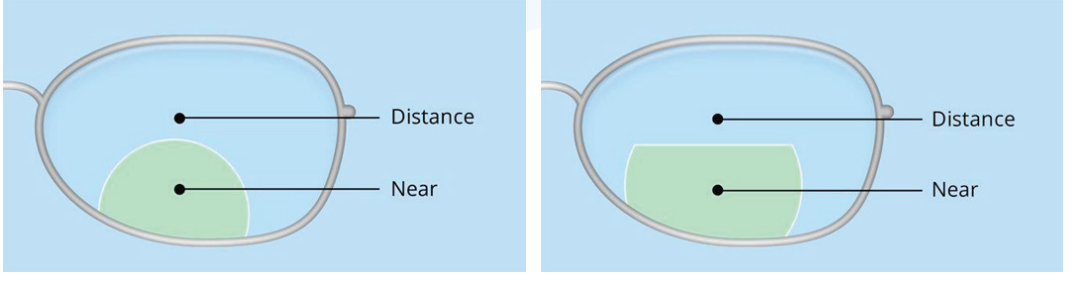


FIG 8.16 Bifocal glasses

- **Trifocal glasses**
  - Progressive glasses

Active space

CONTACT LENSES

- Cosmetic indication
- Better for Anisometropia (>2.5D)
- Anisokenia (different size of image)
- Aphakia
- Irregular Astigmatism
- Infection – Pseudomonas (MC)
- Giant Papillary conjunctivitis

REFRACTIVE SURGERY

REFRACTIVE SURGERY FOR MYOPIA

- Radial keratotomy RK – Radial cuts in cornea
- Photorefractive Keratectomy PRK
- Laser assisted in situ keratomileusis LASIK
- Small Incision Lenticule extraction SMILE
- Phakic intra ocular lens

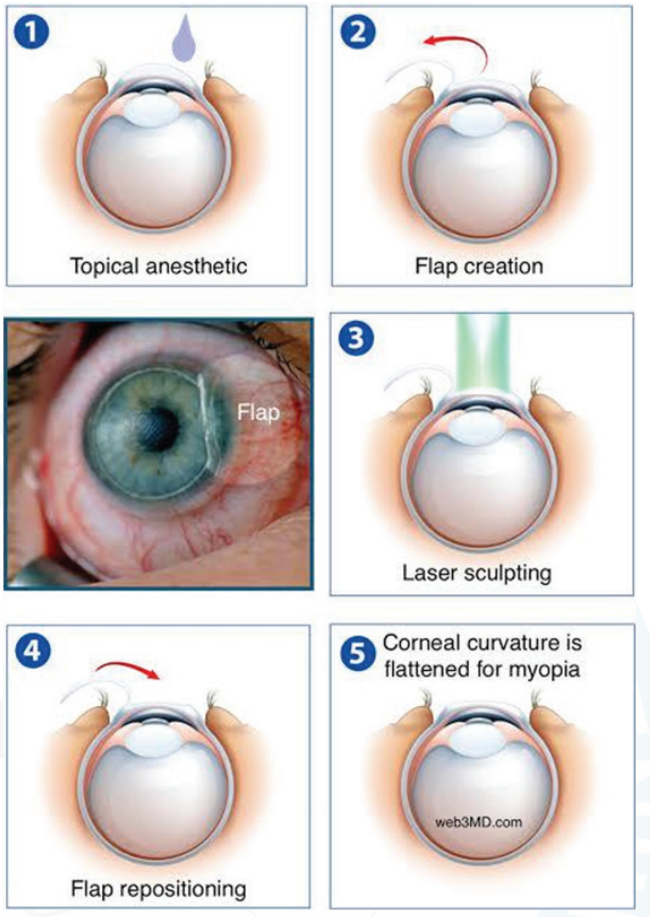
RANGE OF CORRECTION

- PRK -4 to -6D
- LASIK Upto -12D

  SMILE Upto -12D
- Phakic IOL >-12D

Photoablative laser – Excimer = 193nm

Active space



◀FIG 8.17 Lasik Eye Surgery

❑ LASIK

Criteria:

Pt age >21yrs

Stable Refraction

☞ C/I:

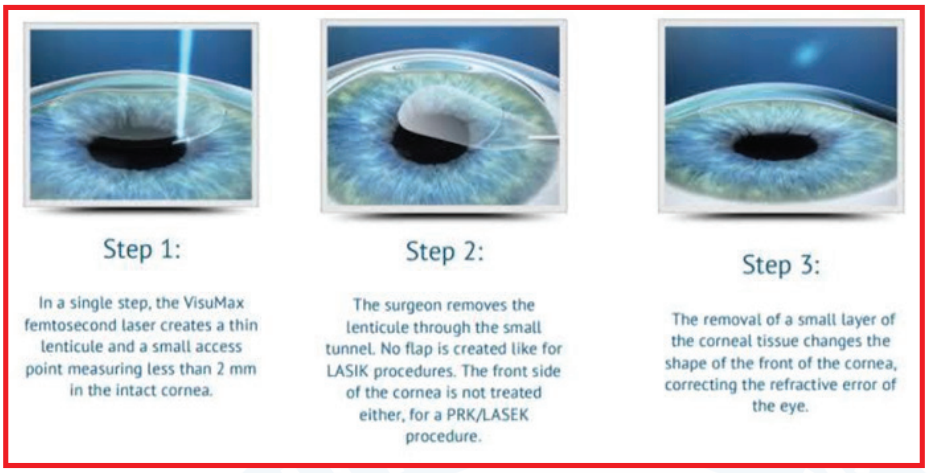
Keratoconus

Unstable Refraction

Thin cornea

Residual Corneal thickness – 270u

Active space



◀**FIG 8.18** SMILE procedure

- **SMILE:**

  There is no flap required

- **PHAKIC IOL**

  ICL: Implantable collamer lens

- **REFRACTIVE PROCEDURES FOR HYPERMETROPIA**
  - Laser assisted in situ keratomileusis LASIK

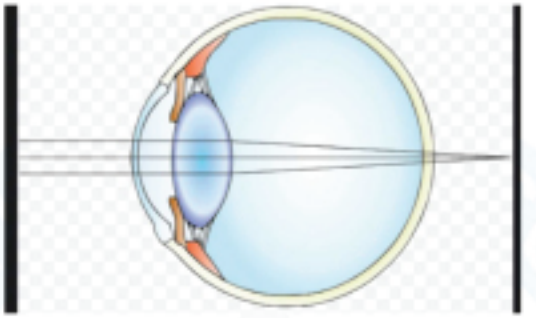
    Upto +4D
  - Conductive keratoplasty

    Non ablative procedure

Active space

Workbook Questions

1. Q: Identify the refractive error shown



(a) Myopia

(b) Astigmatism

(c) Aphakia

(d) Hypermetropia MIST

2. Q: The maximum refraction in eye occurs at

(a) Air – corneal interface

(b) Posterior surface of cornea

(c) Anterior surface of lens

(d) Centre of lens MIST

3. Q: A 6 month old child is brought to you by the parents who suspect the child having defective vision. Which of the following methods is not suitable for assessing the visual acuity in this child?

(a) Visual evoked potential

(b) Teller's preferential acuity cards

(c) Optokinetic nystagmus

(d) Snellen's E chart MIST

Active space

4. Q: All of the following are cycloplegics except

(a) phenylephrine

(b) atropine

(c) homatropine

(d) tropicamide MIST

5. Q: When refractions of two eyes are considerably different, the condition is termed as:

(a) Emmetropia

(b) Anisometropia

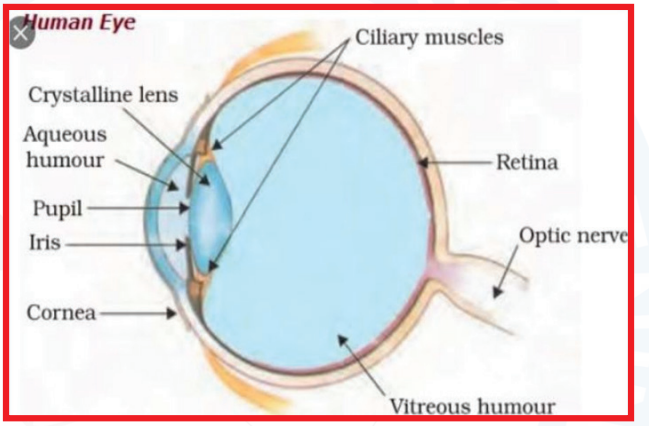
(c) Optic neuritis

(d) Anisokenia MIST

Answers: 1. (d), 2. (a), 3. (d) 4. (a), 5. (b).

Active space

ANATOMY AND PHYSIOLOGY



◀FIG 9.1 Human Anatomy Eye

CRYSTALLINE LENS

- BICONVEX - Anterior surface is less curved than posterior surface.
  Radius of curvature - Anterior Surface - 10mm
  Posterior Surface - 6mm
- Refractive Index
  - Nucleus = 1.40
  - Cortex = 1.386
- Refractive Power (14-16D)

❑ PARTS OF LENS

- Lens capsule
- Cortex

Active space

- Nucleus

  Lens proteins – Crystallines

  (Alpha, Beta, Gamma)

  ↓

  (Max)

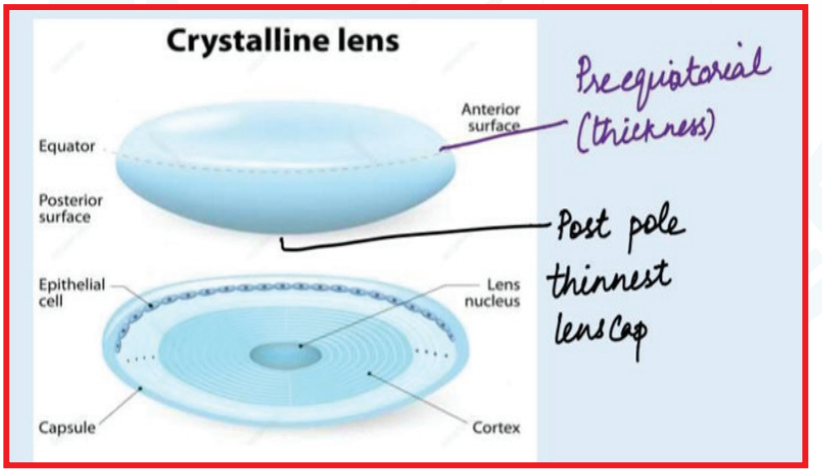
❑ LENS CAPSULE

- Anterior = Lens epithelium

- Posterior = No epithelium

  Thickness = Thickest – Pre Equatorial

  Thinnest – Posterior Pole



◀FIG 9.2 Crystalline Lens

- Lens grows throughout life

Active space

PHYSIOLOGY OF LENS

- Avascular
- Nutrition – Aqueous Humor & Vitreous Humor
- Anaerobic Metabolism

ANTIOXIDANT PRESENT IN LENS

- Glutathione
- Superoxide dismutase
- Catalase
- Vitamin C
- Vitamin E

CONGENITAL CATARCT

Any Opacity of lens

CAUSES OF CONGENITAL CATARCT

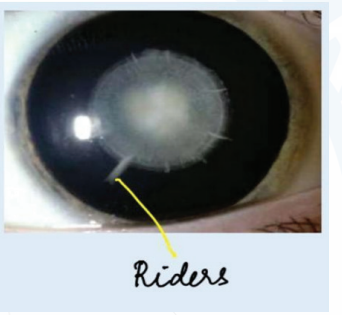
1. Idiopathic
2. Hereditary – Autosomal Dominant
   CRY Gene
3. Maternal Factors
   - TORCH infections
   - Malnutrition
   - Drugs → Steroids, Thalidomide
4. Malnutrition – Vit D Deficiency
   Hypocalcemia
5. Birth Trauma
6. Chromosomal abnormality – Down syndrome
7. Syndromic association- Lowe's Syndrome
   Marfan syndrome

Active space

MORPHOLOGICAL TYPES OF CONGENITAL CATARCT

- Punctate Cataract : Blue dot cataract (Most common)
- Zonular/ Lamellar (Commonest visually significant)
- Nuclear = congenital Rubella syndrome
- Coronary = Cortical club Opacities
- Sutural
- Polar

◀**FIG 9.3** Blue Dot Contract



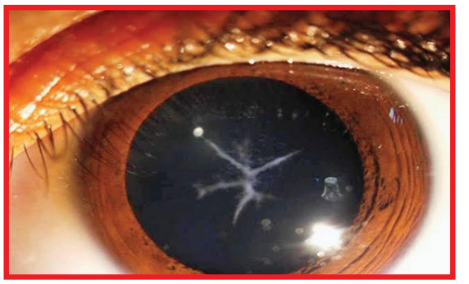
◀**FIG 9.4** Zonular / Lamellar Cataract

❑ PEARLY NUCLEAR CATARACT

- Congenital Rubella

◀**FIG 9.5** Pearly Nuclear Cataract

Active space



◀FIG 9.6 SUTURAL CATARACT

❑ ACQUIRED CATARACT

SENILE CATARACT

Age related

❑ CHANGES WITH AGEING

1. Conversion of Soluble Crystalline proteins into Insoluble Crystalloids

↓

Increased Refractive Index

↓

Index Myopia (Nuclear Type)

2. Decrease in Proteins and increase in Sodium with hydration of lens

↓

Loss of Transparency

↓

Cortical Cataract

❑ TYPES OF SENILE CATARACT

1. Nuclear Senile Catarct

   Nuclear Sclerosis

2. Cortical Catarct

Active space

3. Subcapsular Catarct

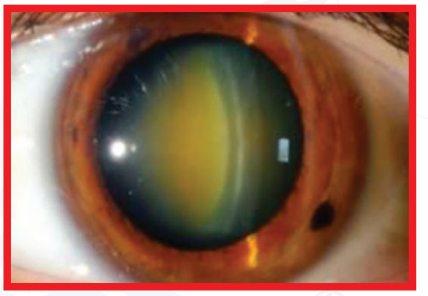
↓ ↓

Anterior Posterior

NUCLEAR SCLEROSIS

↓

Second sight of Old age



◀FIG 9.7 Nuclear Sclerosis Symptoms

✦ Yellowish

↓

Brown cataract

Cataract Brunescence

↓

Cataracta Nigra

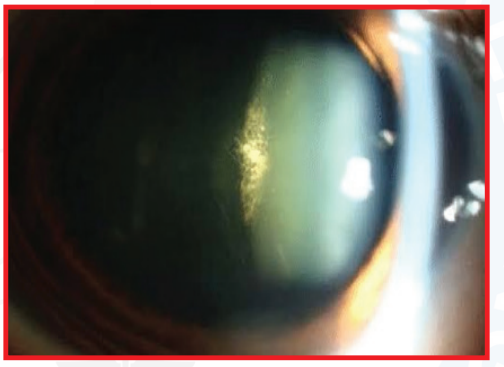
❑ **CORTICAL CATARACT**

CUNEIFORM TYPE

Active space

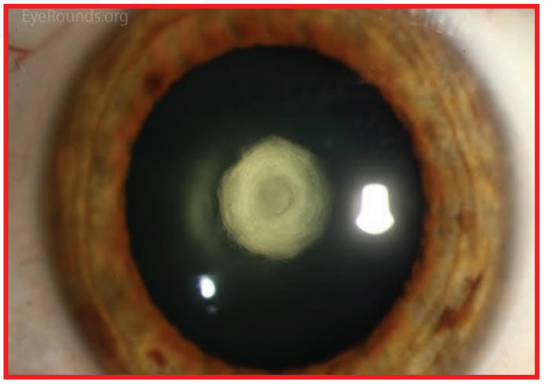
◀**FIG 9.8** Cortical Cataracts

- Colored Halos (Disappear with Stenopaeic slit) (Fincham Test)
- Mono ocular Diplopia
- Glare/ Diminution of vision in dim light



◀**FIG 9.9** Posterior Subcapsular Cataract (PSC)

- Visually significant in Early stages
- Glare in Bright Light



◀**FIG 9.10** Posterior Polar Cataract

Active space

✦ Onion Peel appearance

Cataract is adherent to posterior capsule

↓

Increased Risk of Posterior capsule Rupture during Surgery

❑ STAGES OF MATURITY

1. Stage of Lamellar separation (Reversible)

2. Incipient / Early Senile Cataract

↓

Colored Halos/ Monocular Diplopia

3. Immature Senile Cataract

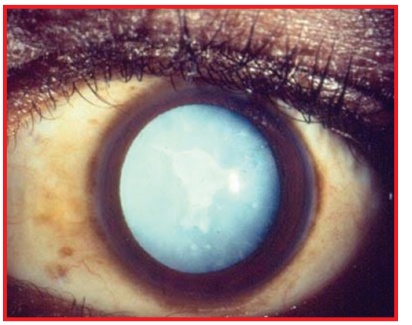
4. Stage of Mature Cataract (Total Cataract)
(White Cataract)

5. Stage of Hypermaturity

↓ ↓

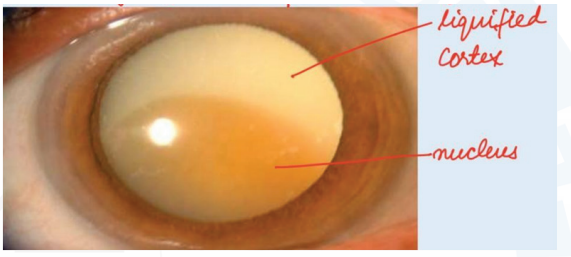
MORGAGNIAN CATARACT	SCLEROTICHYPERMATURE CATARACT
➢ Cortex liquified ➢ Lens sinks down ➢ Cortex leaks through lens capsule	➢ Shrinkage of lens ➢ Wrinkling lens capsule ➢ Weakness of Zonules

Active space



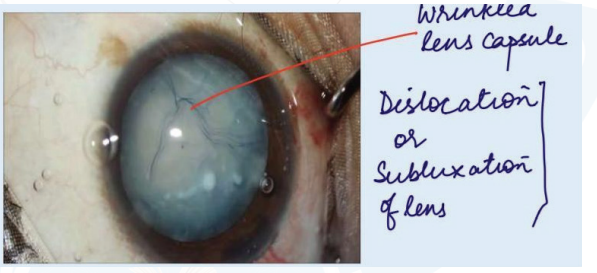
◀FIG 9.11 Mature Cataract

MORGAGNIAN CATARACT

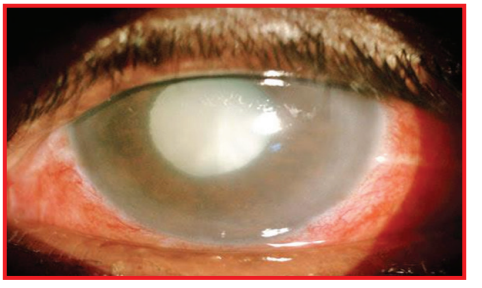


◀FIG 9.12 Morgagnian Cataract

❑ SCLEROTIC HYPERMATURE CATARACT



◀FIG 9.13 SCLEROTIC HYPERMATURE CATARACT



◀FIG 9.14 INTUMESCENT CATARACT

Active space

❑ **Phacomorphic Glaucoma**

Mature Cataract

↓

Hydration

↓

Pushes lens – Iris Diaphragm

↓

Intra ocular pressure increases

OTHER ACQUIRED CATARACTS

- ✦ Traumatic Cataract
- ✦ Metabolic Cataract
- ✦ Complicated Cataract
- ✦ Toxic/ Drug induced Cataract
- ✦ Pre Senile Cataract

TRAUMATIC CATARACT

1. Radiation Related Cataract

   Most Radiosensitive part of eye = LENS

- ✦ Posterior Subcapsular Cataract - (PSC)

2. Thermal Injury

- ✦ Glass Blower's Cataract = PSC
- ✦ True Exfoliation of lens capsule

3. Electric Injury

- ✦ Anterior Subcapsular Cataract

Active space

ROSETTE CATARACT

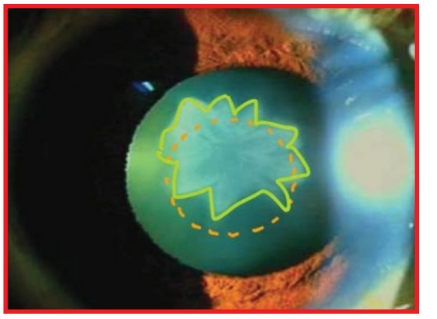
- Blunt Trauma
- Vossius Ring

Penetrating Trauma

↓

Anterior Capsular or

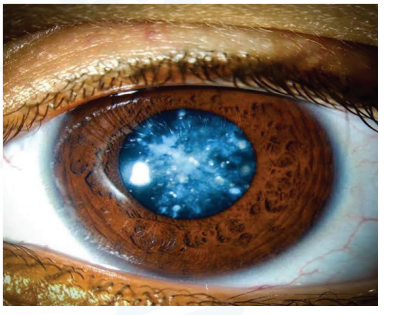
Subcapsular Cataract



◀FIG 9.15 Rosette Cataract

METABOLIC CATARACT

- Diabetes Mellitus
- Galactosemia
- Wilson's Disease



◀FIG 9.16 DIABETIC CATARACT

❑ **Snow flake Cataract**

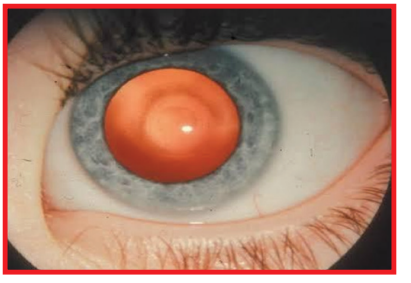
↓

Glucose →→→→→ Sorbitol

Aldose reductase

Fluctuating Refractive Error

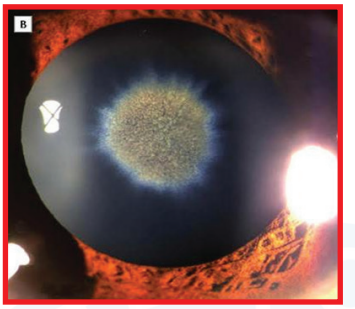
- Age related Cataract in diabetes – Nuclear Sclerosis



◀FIG 9.17 GALACTOSEMIA

Active space

- Deficiency of Galactose 1Phosphate Uridyl Transferase
- Oil Droplet Cataract → Reversible Cataract



◀FIG 9.18 WILSON'S DISEASE

- Sunflower Cataract

↓

Deposition of copper

- Keyser Fleisher (KF) Ring

- .....
- .....
- .....
- .....

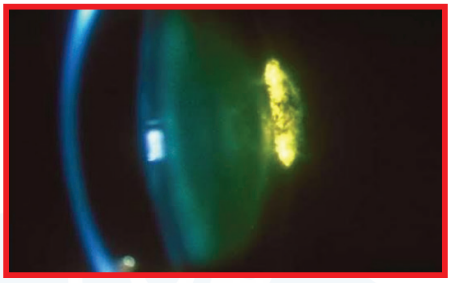
❑ COMPLICATED CATARACT

(Secondary Cataract)

☞ Causes

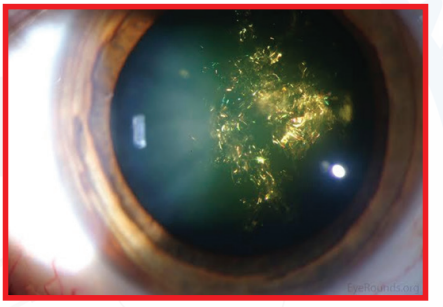
1. Uveitis	Posterior
2. High Myopia	Subcapsular
3. Retinitis Pigmentosa	Cataract
4. Acute Angle Closure Glaucoma	

Active space



◀FIG 9.19 COMPLICATED CATARACTS

- Bread Crumb appearance
- Polychromatic Lustre



◀FIG 9.20 MYOTONIC DYSTROPHY

- Christmas Tree Cataract

DRUG INDUCED CATARACT

POSTERIOR SUBCAPSULAR CATARACT	ANTERIOR SUBCAPSULAR CATARACT
➢ Systemic CorticoSteroids ➢ Busulphan ➢ Chloroquine	➢ Amiodarone ➢ Allopurinol ➢ Chlorpromazine (Stellate shaped) ➢ Gold ➢ Anticholinergics ➢ (Ecothiopate)

Active space

CLINICAL FEATURES OF CATARACT

- Difficulty in Vision
- Glare (PSC)
- Colored Halos (Incipient Cataract)
- Monocular Diplopia (Early)
- Decreased Visual Acquity

MANAGEMENT OF CATARACT

Surgical Management

BIOMETRY

- Process of calculation of power of Intra- ocular lens
  1. Axial length of the eyeball
  2. Keratometry

     ↓

     Keratometer (2 Principal meridian)

     K- Reading (average)

❑ **IOL POWER CALCULATION FORMULAE**

1. SRK II Formula

   P = A - 2.5 L - 0.9 K

   P - power of IOL

   A - A constant

   L - Axial length

   K - Keratometric Reading

   AL = 22-25mm
2. SRK -T Formula

   AL > 25mm (High Myopia)
3. Hoffer Formula

   AL< 22mm (Hypermetropic)

Active space

4. Haigis Formula } Post LASIK
   Holladay II Formula } (Post Refractive)

☞ TYPES OF CATARACT SURGERY

↓ ↓

Intracapsular Cataract Extraction (ICCE) → Whole lens with Capsular bag is removed

Extracapsular Cataract Extraction (ECCE) → Opening in Capsular bag (CAPSULOTOMY)
- Lens Removed
- Capsular bag (+)

❑ **INTRACAPSULAR CATARACT EXTRACTION (ICCE)**

↓

Cryoextraction of Lens

Only indication : Subluxated / dislocated lens

☞ Dis advantage

✦ Large Incision = 12mm
✦ Aphakia → Secondary IOL
✦ Disturbance of Vitreous
↓Retinal Detachment
✦ Increased Risk of glaucoma

❑ **EXTRACAPSULAR CATARACT EXTRACTION (ECCE)**

☞ Types

1. Conventional ECCE
2. Manual Small Incision Cataract Surgery (SICS)
3. Phacoemulsification
4. Femtosecond laser assisted Cataract Surgery

Active space

☞ Conventional ECCE

- 8-9mm limbal incision (Sutures applied)

☞ SICS

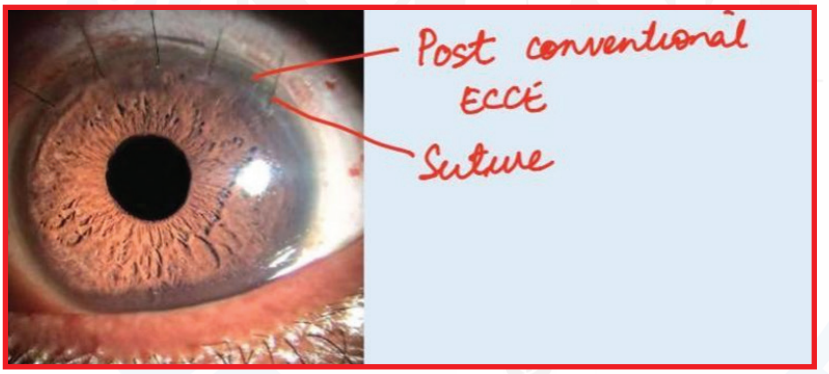
- 6-7mm CorneoScleral tunnel → Self sealing (Sutureless)

☞ Phacoemulsification

- ✦ 2.8-3.2 mm
- ✦ Triplanar Incision

  ↓

  Self sealing, Sutureless
- ✦ Micro Incision Surgery (MICS) = 2.5mm



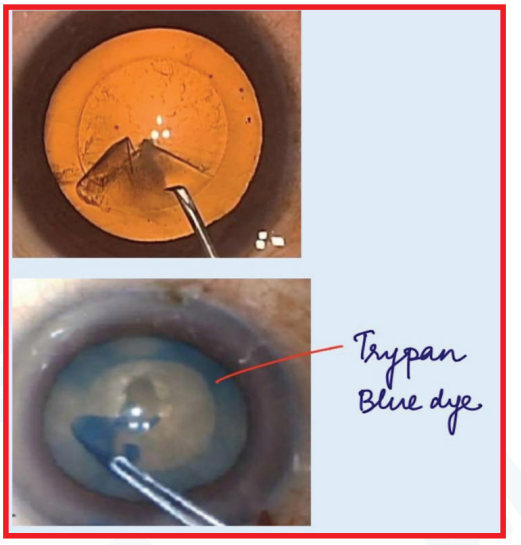
◀FIG 9.21 Phacoemulsification

❑ STEPS OF CATARACT SURGERY (Phacoemulsification)

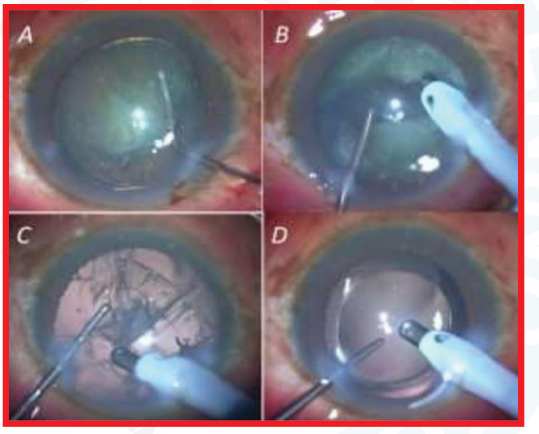
1. Entry wound (valvular incision)
2. Injection of Viscoelastic Substance
3. Capsulotomy → Continuous Curvilinear Capsulorrhexis (CCC)
4. Hydrodissection → Separate lens from Capsule
5. Fragmentation of nucleus and it's removal – Phaco machine
6. Aspiration of Cortical matter
7. Implantation of IOL in the Capsular Bag → Ideal site
8. Wash out Viscoelastic

Active space

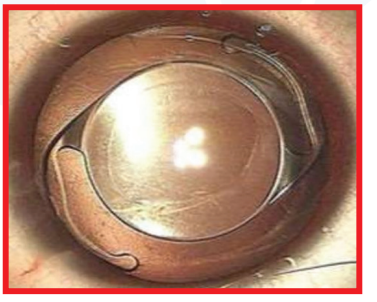
9. Hydration of entry wound (Self sealing)



◀FIG 9.22 Cataract Surgery



◀FIG 9.23 Cataract Surgery



◀FIG 9.24

FEMTOSECOND LASER ASSISTED CATARACT SURGERY (FLACS)

✦ Wavelength = 1053nm

Active space

☞ STEPS

1. Making of Corneal incisions
2. Capsulotomy (CCC)
3. Fragmentation of Nucleus

✦ MANAGEMENT OF CONGENITAL CATARCT

✦ Timing of Cataract Surgery →

✦ UNILATERAL Vs Bilateral Cataract

↓ ↓

More amblyopia Within 4 weeks of age

At the earliest

✦ TYPE OF THE SURGERY:

Lens aspiration/Phaco-aspiration with primary posterior capsulotomy with anterior vitrectomy

IOL power selection: (Consider Myopic Shift)

INTRAOCULAR LENS (IOL)

❑ **NON - FOLDABLE IOL (Rigid)**

PMMA

❑ **FOLDABLE IOL**

Acrylic = Best, Ideal

Silicone

Hydrogel

1. Monofocal IOL = Correct distance Vision
   Glasses for near vision
2. Multifocal IOL = Corrects both distance and
   Near vision
3. Toric IOL = Astigmatism

Active space

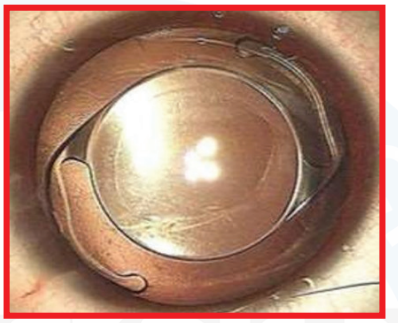
❑ SITE OF IMPLANTATION

1. Posterior Chamber IOL – PCIOL
2. Anterior Chamber IOL – ACIOL

↓

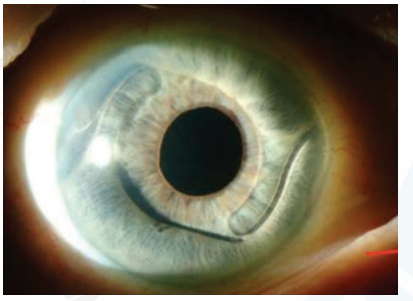
Intra operative Rupture of Posterior Capsule

ICCE



◀FIG 9.25 MONOFOCAL PCIOL

Ideal Site of Implantation – Within Capsular Bag



◀FIG 9.26 KELMAN 4-LOOP ACIOL

COMPLICATIONS OF CATARACT SURGERY

- ✦ Intraoperative – During Surgery
- ✦ Early Postoperative – Within 7 days
- ✦ Late Post operative

INTRA OPERATIVE

Rupture of Posterior Capsule (PC)

↓

Loss of Vitreous

Active space

Drop of nucleus into Vitreous Cavity

EARLY POSTOPERATIVE

Early Postoperative Endopthalmitis

Most dreaded complication

Most common cause – Staph epidermidis

POST OP ENDOPHTHALMITIS

↓

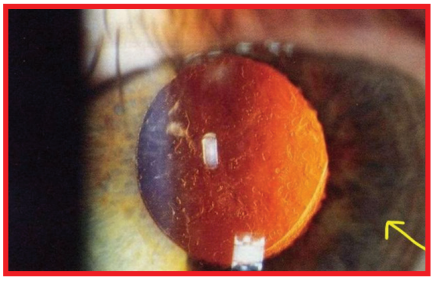
Intra Vitreal antibiotics

◀FIG 9.27 POST OP ENDOPHTHALMITIS

Preventive Measures → Prophylaxis 5% Povidone – Iodine Topical

❑ LATE POSTOPERATIVE

1. Late Post-op ENDOPHTHALMITIS
   Most common Cause – Propionibacterium acne, Fungi
2. Most common complication = Posterior Capsular opacification (After Cataract)
3. Cystoid Macular edema
4. Retinal Detachment



◀FIG 9.28 AFTER CATARACT

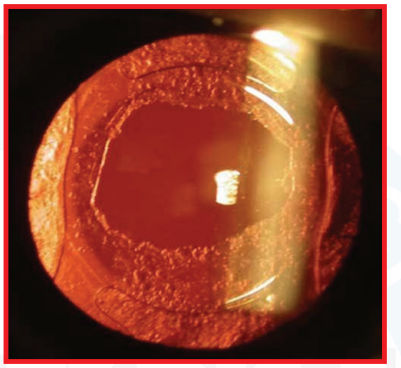
Active space

❑ **Posterior Capsular Opacification (PCO)**

↓

1. Elschnig Pearls
2. Sommering Ring

Most common delayed Complication of Cataract Surgery



◀**FIG 9.29** Nd-Yag Laser Capsulotomy

✦ Photodisruptive

✦ 1064nm

❑ **CONGENITAL ANOMALIES OF LENS**

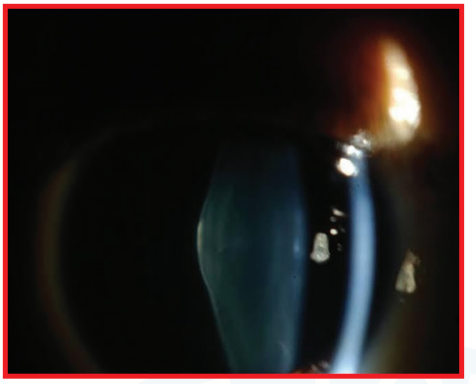
Lenticonus

↓ ↓

Anterior Posterior

◀**FIG 9.30** ANTERIOR LENTICONUS

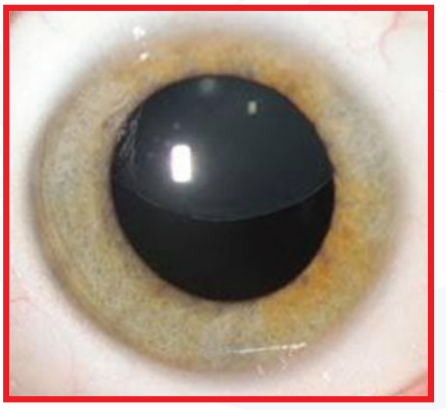
Active space



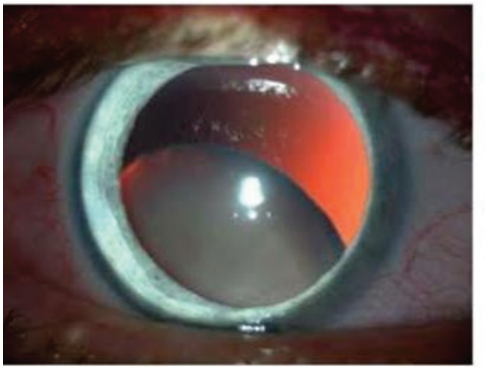
◀FIG 9.31 POSTERIOR LENTICONUS

ECTOPIA LENTIS

- **Subluxated / dislocated (Congenital)**
  1. Marfan's syndrome: Upwards & temporal
  2. Homocystinuria : Down & nasally
  3. Weil Marchesani syndrome :
     Microsphreophakia – Anterior Subluxation

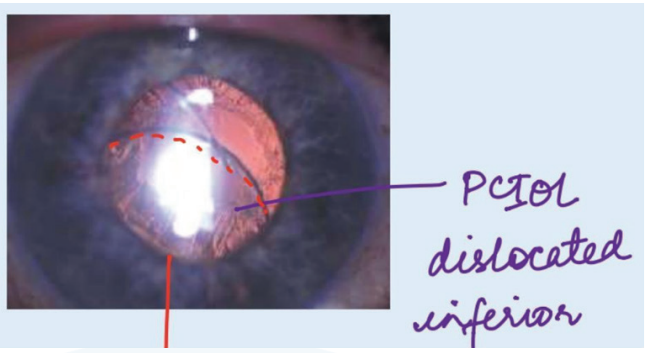


◀FIG 9.32 MARFAN'S SYNDROME



◀FIG 9.33 HOMOCYSTINURIA

Active space



◀**FIG 9.34** SUNSET SYNDROME

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

Active space

Workbook Questions

1. Q: Onion peel appearance is seen in which type of cataract?
   (a) Posterior subcapsular cataract
   (b) Posterior polar cataract
   (c) Anterior polar cataract
   (d) Nuclear cataract MIST

2. Q: Polychromatic luster is a feature of
   (a) Sunflower cataract
   (b) Rosette cataract
   (c) Complicated cataract
   (d) Nuclear cataract MIST

3. Q: Which laser is used in laser assisted cataract surgery?
   (a) Excimer laser
   (b) Nd Yag laser
   (c) Femtosecond laser
   (d) Diode laser MIST

4. Q: Intraocular lens (IOL) used to correct astigmatism
   (a) Multifocal IOL
   (b) Monofocal IOL
   (c) Toric IOL
   (d) ACIOL MIST

5. Q: Most common type of congenital cataract which is visually significant is
   (a) Capsular cataract
   (b) Blue dot cataract
   (c) Zonular cataract
   (d) Cuneiform cataract MIST

Answers: 1. (b), 2. (c), 3. (c) 4. (c), 5. (c).

Active space

GLAUCOMA

DEFINITION AND BASICS

Glaucoma

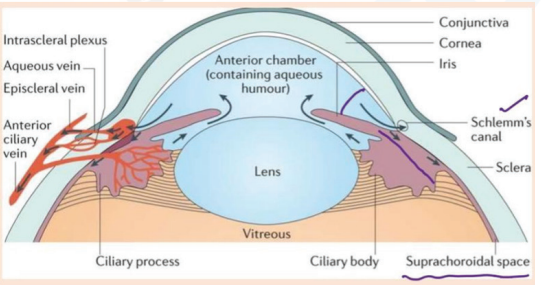
Optic Neuropathy Characterized by

1. Optic Disc changes,
2. Visual field defects
3. With increased Intra ocular Pressure as major Risk Factor
   (10% with Normal IOP)

NORMAL PHYSIOLOGY OF AQUEOUS HUMOR PART OF CILIARY BODY

From ciliary Process of Pars Plicata

Rate: 2-2.5uL/min



◀**FIG 10.1** FORMATION AND OUTFLOW OF AQUEOUS HUMOR

❑ OUTFLOW PATHWAYS

1. Conventional / Trabecular
   - Major 90%
2. Uveoscleral
   - 10%

Active space

❑ INTRA OCULAR PRESSURE IOP

Normal IOP: 10-21mmHg

Diurnal Variation: Upto 5mmHg

❑ Factors affecting IOP

- ✦ Age (elderly)
- ✦ Supine
- ✦ Exercise
- ✦ Valsalva manuver
- ✦ Large amount of Drinking water

INVESTIGATIONS IN GLAUCOMA

1. IOP measurement
2. Angle of Anterior Chamber
3. Optic Disc Evaluation
4. Visual fields

INVESTIGATIONS IN GLAUCOMA PART 1 - TONOMETRY

TONOMETRY = clinical
(Indirect measurement)

❑ TONOMETERS

1. Indentation Tonometer

- Schiotz

2. Applanation Tonometer : Goldmann Applanation Tonometer

- Imbert – Fick law
- Pressure = Force/area

◀FIG 10.2 SCHIOTZ INDENTATION TONOMETER

Active space

☞ Disadvantage

- ✦ Depend on Scleral Rigidity
- ✦ User dependent

GOLDMANN APPLANATION TONOMETER

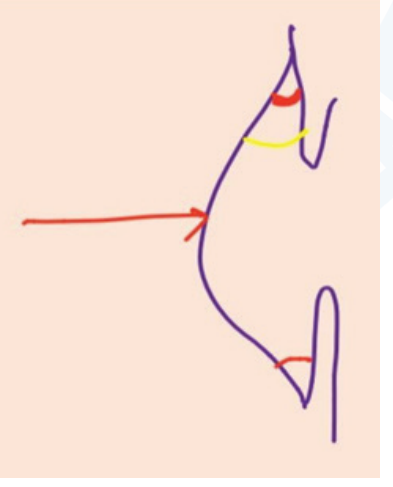
- ✦ Gold standard for measuring IOP
- ✦ Reading dependent upon Corneal thickness

  Thick cornea = False high IOP

  Thin Cornea = False low IOP

◀FIG 10.3 Measuring Intraocular Pressure

GONIOSCOPY (GONIOSCOPES)



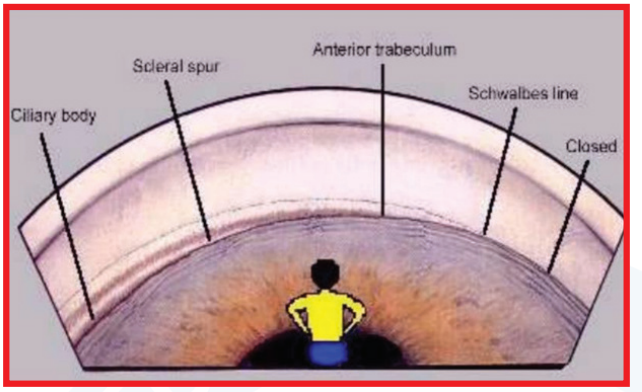
◀FIG 10.4 ANGLE OF ANTERIOR CHAMBER

❑ **Anterior → Posterior**

1. Schwalbe's line
2. Trabecular Meshwork

Active space

3. Scleral spur
4. Ciliary body and root of Iris



◀**FIG 10.5** SHAFFER'S GRADING OF ANGLE

INVESTIGATIONS IN GLAUCOMA PART - 2

EVALUATION OF OPTIC DISC OR OPTIC NERVE HEAD

Death of Ganglion cells → Mechanical Theory

↓ → Perfusion/ Ischemic

↓

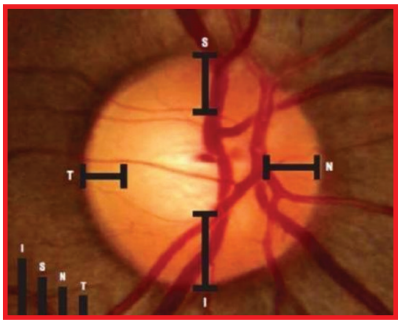
Loss of Nerve fibre layer

↓

Loss of Neuroretinal Rim

↓

Glaucomatous optic disc changes



◀**FIG 10.6** Normal Optic Disc or Optic Nerve Head (ONH)

Active space

I S N T

I - Inferior

S - Superior

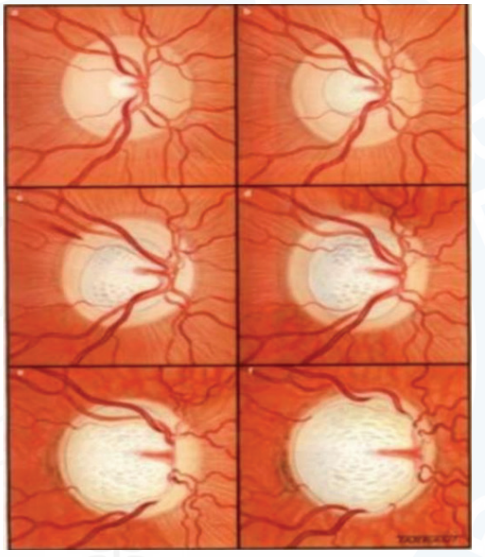
N - Nasal

T - Temporal

Normal eye → I> S> N>T

Glaucoma → Violates I S N T (×)

C:D = 0.3 (Cup Disc Ratio)

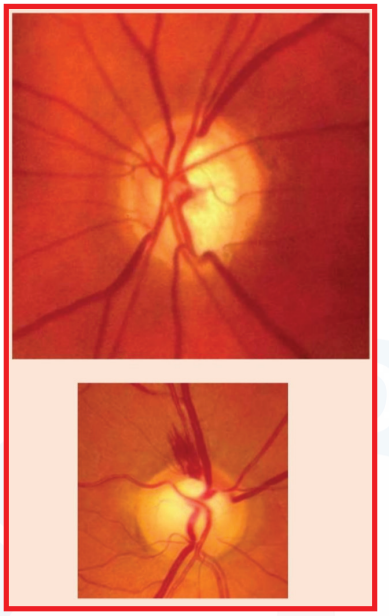


◀**FIG 10.7** Glaucomatous Optic Nerve Changes

GLAUCOMATOUS OPTIC NERVE CHANGES

1. Vertical Cupping (Inferior Rim is lost)
2. Increased Cupping (C:D> 0.7)
3. Asymmetrical (C:D >0.2 between 2 eyes)
4. NRR notching
5. Baring of Circumlinear vision
6. Nasalisation of vessels
7. Bayonetting of vessels
8. Laminar dot sign
9. Splinter Hemorrhage on Disc

Active space



◀FIG 10.8

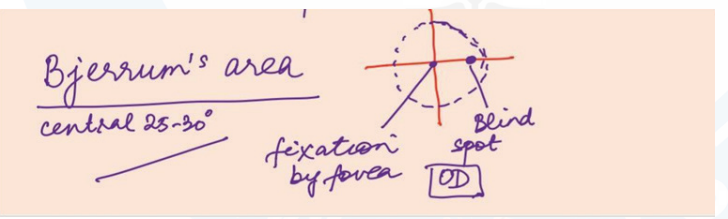
FIELD DEFECTS IN GLAUCOMA

1st Change – Generalised Constriction

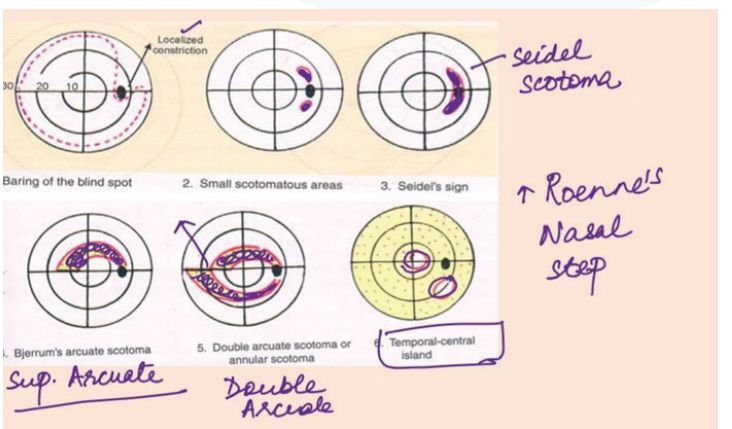
Baring of Blind Spot

1st Field Defect – Paracentral Scotoma

(Superonasal)



◀FIG 10.9 FIELD DEFECTS IN GLAUCOMA



◀FIG 10.10 Field Defects in Glaucoma

Active space

CLASSIFICATION OF GLAUCOMA

- Congenital Glaucoma
- Open angle Glaucoma
- Closed angle Glaucoma

CONGENITAL GLAUCOMA

- Autosomal Recessive
- Developmental anomaly of Angle of AC
- Trabeculo dysgenesis

True Congenital Glaucoma – At Birth

Infantile Glaucoma – till 3yrs age

Juvenile Glaucoma – After 3yrs – Adolescence

◀FIG 10.11 CONGENITAL GLAUCOMA

- Blepharospasm
- Photophobia
- Lacrimation

Buphthalmos – Large sized eye balls

❑ **Examination Under Anesthesia**

- Megalocornea >/ = to 13mm
- Haab's stria – Breaks in Descemet's Membrane (Horizontal)
- Cornea Hazy or Clear (Ground glass appearance)
- AC deep
- Thin Sclera, Bluish

Active space

- ✦ Lens is flat, May Subluxated
- ✦ IOP is increased
- ✦ Increased C:D ratio → May be reversible

❑ TREATMENT OF CONGENITAL GLAUCOMA

Surgery

1. Goniotomy – .....

   Possible in cases of Clear cornea

2. Trabeculotomy – .....

   Can be done in Hazy cornea

PRIMARY GLAUCOMA

PRIMARY OPEN ANGLE GLAUCOMA (POAG)

❑ Risk Factors

- ✦ Hereditary : Optineurin, Myocilin
- ✦ Age – Common in Elderly
- ✦ Myopes
- ✦ Diabetes
- ✦ Hypertension
- ✦ Cigarette smoking
- ✦ Thyrotoxicosis

ROLE OF FAMILY HISTORY

1st degree Relatives – increased Risk

Siblings have 10% Risk (4 times more prone)

Offsprings have 4% Risk (2 times more prone)

Active space

☞ Clinical Features

- Asymptomatic
- Frequent change of Presbyopic glasses

O/E : IOP is increased or May be normal range

Diurnal Variation of IOP

Upto 5mmHg – Normal

5-8mmHg – Borderline

>8mmHg – POAG

- AC deep, Angles are open
- OD = Changes (+), Visual Field (+)

CLINICAL SPECTRUM

1. Increased IOP , OD (Normal), Visual Field (Normal) → Ocular Hypertension
2. Increased IOP, OD changes (+), VF defects (+) → POAG
3. Normal IOP, OD Changes (+), VF defects (+) → Normal Tension Glaucoma (NTG)

☞ TREATMENT MODALITIES

1. First line Modality : Anti Glaucoma Medication

   DOC:Prostaglandin Analogues
2. Laser Trabeculoplasty
3. Surgery – Trabeculectomy

ANGLE CLOSURE GLAUCOMA

PRIMARY ANGLE CLOSURE GLAUCOMA

☞ Risk Factors

- ✦ Age
- ✦ Female Sex
- ✦ Hypermetropia (Shallow AC)
- ✦ Small eyes

Active space

- Decreased Corneal diameter
- Flat Iris (Plateau Iris)

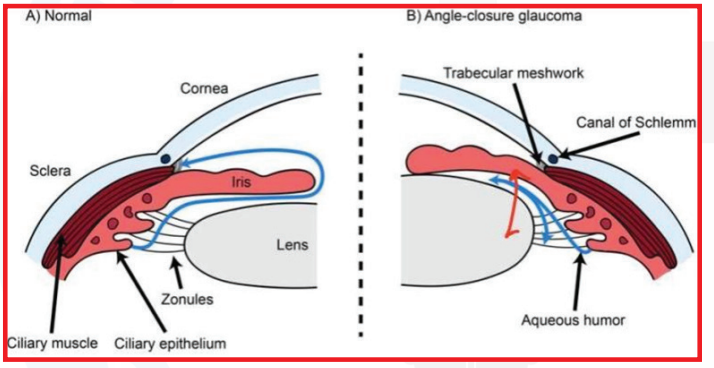
☞ Clinical Features

Asymptomatic

Episodes of Headache, Decreased Vision and coloured Halos

O/E :

- IOP may be normal or increased
- AC shallow
- Angle narrow or Closed
- Optic Nerve → May be Normal, Optic Disc changes
- Visual Field (+)



◀FIG 10.12 ACUTE ANGLE CLOSURE GLAUCOMA/ ACUTE CONGESTIVE GLAUCOMA

Pupillary dilatation

↓

Pupil Block

↓

Iris Bombe

↓

Complete angle closure

↓

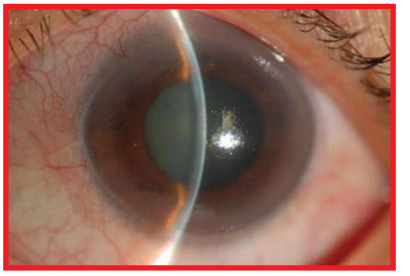
Increased in IOP

❑ **Acute Painful Red Eye**

Ocular emergency

MIST

Active space



◀FIG 10.13 Acute Painful Red Eye

- ✦ Stony Hard ↑↑ IOP
- ✦ Edematous Cornea
- ✦ Shallow AC
- ✦ Mid dilated pupil
- ✦ AC Reaction

❑ **Treatment of Acute Congestive Glaucoma**

1. IV mannitol
2. Oral Acetazolamide
3. Pilocarpine eye drops → Miosis → Breaks Pupillary Block
4. Topical Steroids
5. Other Topical antiglaucoma
6. Laser Iridotomy Both eyes

☞ Treatment Modalities

1. Laser Iridotomy
2. Medical Therapy
3. Trabeculectomy = Chronic ACG

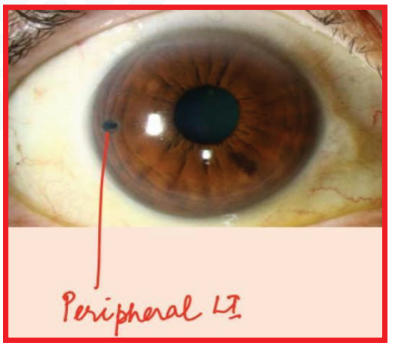
❑ **LASER IRIDOTOMY**

Iris opening

- ✦ Nd Yag Laser 1064nm

↓

Photodisruptive Laser



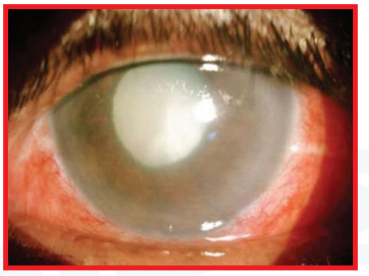
◀FIG 10.14 LASER IRIDOTOMY

Active space

SECONDARY GLAUCOMA

❑ LENS INDUCED GLAUCOMA

✦ Phacomorphic Glaucoma



◀FIG 10.15 PHACOMORPHIC GLAUCOMA

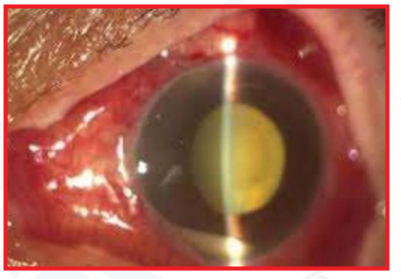
Intumescent Mature Cataract → Swollen

↓

Shallowing of AC

↓

Increased IOP



◀FIG 10.16 PHACOLYTIC / LENS PROTEIN GLAUCOMA

Hypermature Senile Cataract

↓

Morgagnian Cataract

↓

Liquified Cortex

Active space

Topical Steroids = Increased risk of Glaucoma

Systemic Steroids = Increased risk of Cataract Development

NEO VASCULAR GLAUCOMA

- ✦ Rubeosis Iridis
- ✦ New vessel in angle of AC

☞ Causes

Most common cause

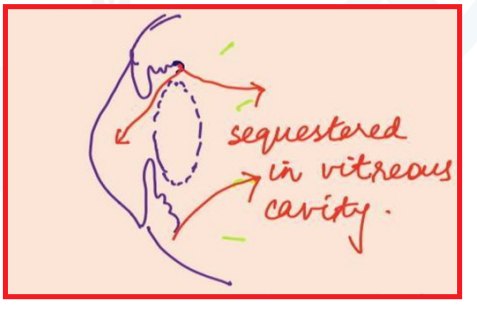
- Diabetic Retinopathy
- CRVO
- Ocular Ischemic Syndrome
- Sickle cell Disease

Treatmentof Choice: Surgery (SETON SURGERY)

- PRP laser
- Anti VEGF
- Medical Therapy

MALIGNANT GLAUCOMA

Aqueous Misdirection Syndrome



◀**FIG 10.17** Malignant Glaucoma

- ✦ Shallowing or Flat AC
- ✦ Increased IOP

Active space

☞ Causes:

- Post Trabeculectomy
- Post Cataract Surgery

☞ Treatment

Atropine eye drops

MEDICAL THERAPY FOR GLAUCOMA

Increased IOP

↓ ↓

↓ Aqueous ↑ Outflow of Aqueous

Production

❑ ANTI GLAUCOMA DRUGS

1. Prostaglandins – PG analogue
2. Beta blockers
3. Alpha agonists
4. Carbonic anhydrase Inhibitors
5. Miotics
6. Hyperosmotics

❑ PROSTAGLANDIN ANALOGUES

✦ First line Drugs for Glaucoma

MOA- Increases UveoScleral outflow

1. Latanoprost
2. Travoprost
3. Bimatoprost

☞ Side effects

- Iris pigmentation
- Hypertrichosis
- Cystoid macular edema
- Anterior Uveitis

Active space

Contra indicated in Uveitis and CME

❑ BETA BLOCKERS

MOA – Decrease Aqueous production by Acting on Beta receptors in ciliary body

1. Timolol maleate 0.5% → Non Selective
3. Betaxolol 0.5% → Selective Blocker

☞ Side effects:

Dry eye, SPK

☞ Systemic Side effects:

Bronchospasm, Bradycardia, Hypotension, Sleep Disorder, fatigue

Contra indicated in : Asthma, COPD, Heart blocks

❑ ALPHA AGONISTS

MOA – Dual action

- ✦ Decrease Aqueous production
- ✦ Increase Uveoscleral outflow

Brimonidine → Safe in Pregnancy

☞ Side effects

- Allergic Conjunctivitis
- Lid retraction
- Cystoid Macular edema
- CNS depression

Contra indication in Children → Sleep Apnea

❑ CARBONIC ANHYDRASE INHIBITORS

MOA – Decrease Aqueous production by inhibition of Carbonic Anhydrase (Aqueous Suppressants)

1. Acetazolamide 250mg oral
2. Dorzolamide } Topical Drugs
3. Brinzolamide } (Eye drops)

Active space

☞ Side effects

(Topical)

Endothelial decompensation – Corneal Edema

❑ **Systemic Side effects – Acetazolamide**

- ✦ Parasthesias – Most common due to Hypokalemia
- ✦ GI upset, Constipation
- ✦ Sulpha Allergy
- ✦ Renal stones
- ✦ Metabolic acidosis
- ✦ Aplastic anemia

❑ **MITOTICS / CHOLINERGICS**

MOA – Increase trabecular Meshwork outflow

Breaking the Pupillary Block by Miosis in ACG

1. Pilocarpine 2%

☞ Side effects

Ciliary Spasm

- Brow achae
- Pseudo Myopia
- Iris cysts
- Retinal detachment
- Increases inflammation

Contra indicated in Uveitis & Macular Edema

❑ **HYPEROSMOTICS**

MOA – Decrease the Vitreous volume by dehydrating vitreous

1. Glycerol 50% oral
2. Mannitol 20% intravenous → Fastest IOP lowering agent

C/I – Pulmonary edema, Renal failure, Right Heart failure

❑ **NEWER DRUGS**

1. Netarsudil
   - Rhokinase Inhibitor

Active space

MIST

- Increases the outflow
- S/E - Cornea VORTICELLATA

☞ Combination of drugs

2 drugs } Maximizing

3 drugs } Medical

4 drugs } Therapy

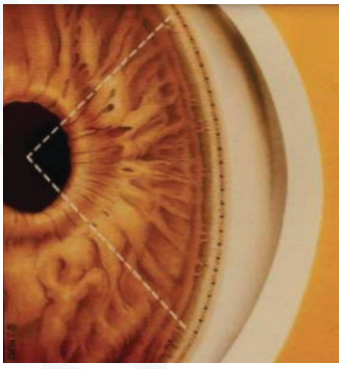
LASER AND SURGICAL TREATMENT OF GLAUCOMA

LASER TRABECULOPLASTY

2 lasers

1. Argon laser
2. Fd Nd yag laser

Helps in increasing outflow through Trabecular Meshwork



◀FIG 10.18 LASER TRABECULOPLASTY

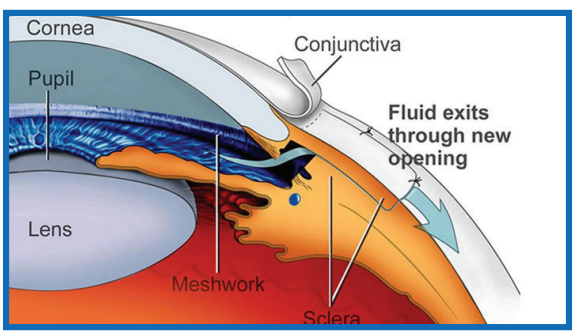
☞ INDICATIONS

- PAOG → Failed Medical therapy

SURGICAL PROCEDURES

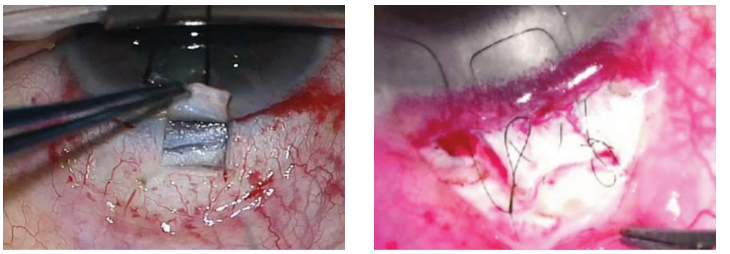
- ✦ Failed Medical therapy
- ✦ Patient intolerant to Medical therapy
- ✦ Patient not compliant to Medical therapy
- ✦ Failed laser Trabeculoplasty
- ✦ Progression of Glaucoma despite Medical treatment

❑ **GLAUCOMA FILTERATION SURGERY**



◀FIG 10.19 GLAUCOMA FILTERATION SURGERY

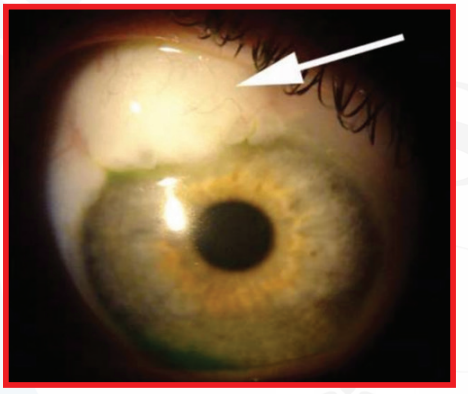
Active space



◀FIG 10.20 TRABECULECTOMY

✦ Fistula between Anterior Chamber and Subtenon space

Conjunctival flap is made → partial thickness Scleral flap is created → Anterior chamber → Full thickness Scleral window → Subtenon space → Aqueous flow in the AC → collection of Aqueous in Subtenon space → Cystic bleb

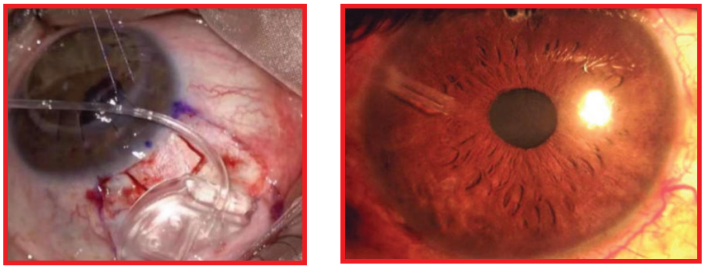


◀FIG 10.21 Cystic Bleb formation in Subtenon space

❑ SETON SURGERY / DRAINAGE IMPLANTS/ GLAUCOMA VALVES

☞ INDICATIONS

1. Failed Trabeculectomy
2. Post RD surgery
3. Post keratoplasty
4. Neo vascular glaucoma
5. Glaucoma with Uveitis



◀FIG 10.22 Ahmed Valve implant

Active space

Workbook Questions

1. Q: All of the following are drugs used in acute congestive glaucoma except
   (a) Atropine
   (b) Pilocarpine
   (c) Mannitol
   (d) Acetazolamide MIST

2. Q: A 65 year old man with hypertension and CAD has been diagnosed with primary open angle glaucoma. Which of the following is the drug of choice for initiating treatment for him?
   (a) Travoprost
   (b) Timolol maleate
   (c) Brimonidine
   (d) Brinzolamide MIST

3. Q: A 50 year old lady presents with acute painful red eye with oval mid-dilated pupil after watching a movie. Most likely diagnosis is:
   (a) Acute retrobulbar neuritis
   (b) Acute angle closure glaucoma
   (c) Acute anterior uveitis
   (d) Acute conjunctivitis MIST

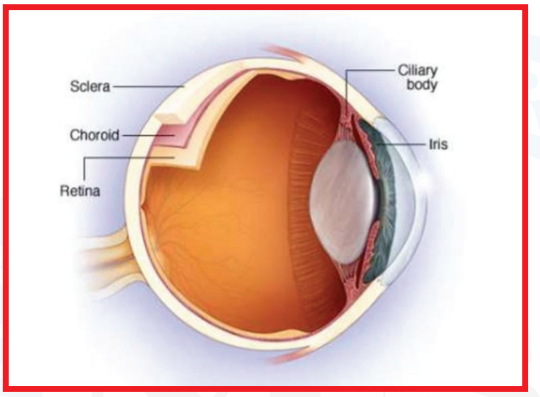
4. Q: Which of the following field defects is not seen in glaucoma?
   (a) Nasal Step
   (b) Arcuate scotoma
   (c) Centrocaecal scotoma
   (d) Siedel scotoma MIST

5. Q: What are the drugs used in acute congestive glaucoma?
   (a) Atropine
   (b) Pilocarpine
   (c) Acetazolamide
   (d) Both b and c MIST

Answers: 1. (a), 2. (a), 3. (b) 4. (c), 5. (d).

Active space

ANATOMY AND CONGENITAL ANOMALIES



◀FIG 11.1 UVEAL TRACT

CILIARY BODY

- Pars plicata 2mm
- Pars plana 4mm

Active space

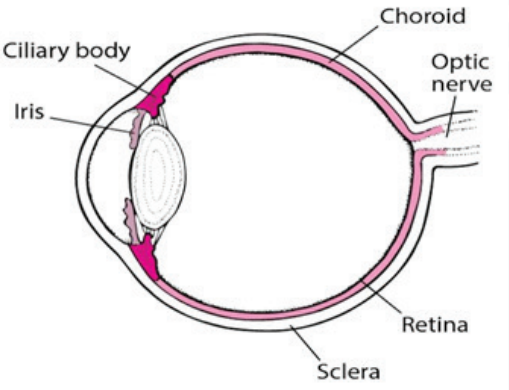
UVEITIS

Inflammation of Uveal tract

CLASSIFICATION OF UVEITIS

- ✦ Anatomical
- ✦ Clinical
- ✦ Pathological
- ✦ Etiological

❑ ANATOMICAL CLASSIFICATION



◀FIG 11.4

1. Iridocyclitis – Anterior Uveitis
2. Pars plana – Pars planitis
   (Intermediate Uveitis)
3. Posterior Uveitis
4. PANUVEITIS- All layers are involved

❑ CLINICAL CLASSIFICATION

Acute: Sudden onset, 6weeks – 3months

Chronic: Insidious onset, > 3months

Recurrent: Episodic, Period of inactivity of 3months

Active space

❑ ETIOLOGICAL CLASSIFICATION

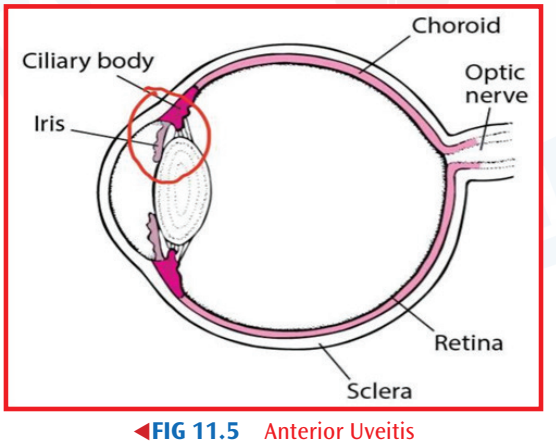
- ✦ Infectious
- ✦ Non – Infectious
- ✦ Idiopathic – Most Common
- ✦ Immune related : Uveitis with arthritis, IBD
- ✦ Uveitis with Non- infectious systemic Diseases

  Sarcoidosis, VKH Syndrome
- ✦ Traumatic
- ✦ MASQUERADE : Neoplastic

❑ PATHOLOGICAL CLASSIFICATION

- ✦ Suppurative – Endophthalmitis
- ✦ Non suppurative
  - Non – Granulomatous
  - Granulomatous

ANTERIOR UVEITIS



◀FIG 11.5 Anterior Uveitis

Iritis

Cyclitis: Pars Plicata

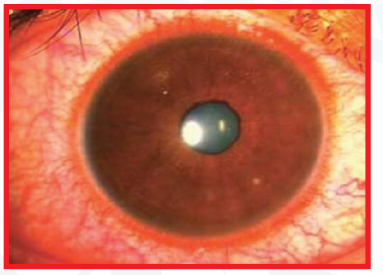
Iridocyclitis

Active space

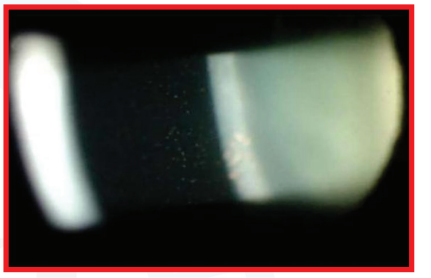
❑ CLINICAL FEATURES

☞ Symptoms

1. Redness
2. Photophobia
3. Blepharospasm
4. Pain
5. Decreased Vision



◀ **FIG 11.6** SIGNS: Circumcorneal congestion



◀ Fig 11.7 Cells Flare

Cells:

Flare:

↓

Ist sign of uveitis

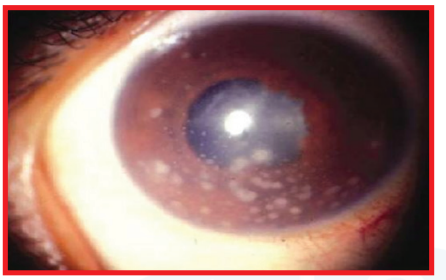
◀**FIG 11.8** Pathognomic of uveitis

◀**FIG 11.9** ARLT'S TRIANGLE

Active space

❑ Mutton fat KPs

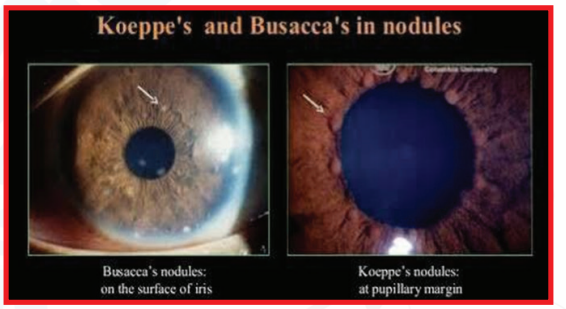
- Granulomatous



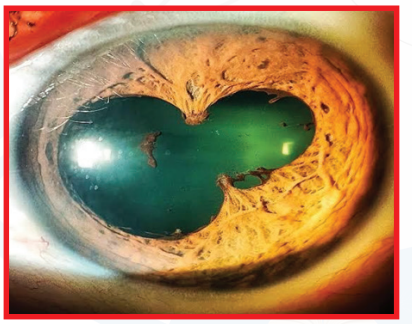
◀FIG 11.10 Mutto Fat KPs

❑ Iris Nodules

- Granulomatous



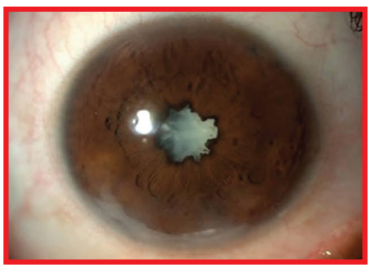
◀FIG 11.11 Iris Nodules



◀FIG 11.12 Posterior Synechiae

❑ Pupil Miosis

- Festooned Pupil



◀FIG 11.13 Festooned Pupils

Active space

CAUSES OF ANTERIOR UVEITIS

- Idiopathic : Most common
- Autoimmune : AS, HLA B-27, JRA, Crohn's disease, Ulcerative colitis

❑ COMPLICATIONS OF UVEITIS

- Posterior Synechiae : Iris Bombe → Angle closure
  Seclusio Pupillae = 360o Synechiae
- Secondary angle closure – Glaucoma
- Posterior Subcapsular cataract – Complicated cataract
  Polychromatic Lustre
- Cystoid Macular Edema – Decreased Vision

❑ TREATMENT OF ANTERIOR UVEITIS

- Topical/ Systemic Steroids – Mainstay
- Cycloplegics – Prevent formation of Posterior Synechiae, Relieve spasm of Ciliary muscle

❑ UVEITIS WITH ARTHRITIS

1. Ankylosing Spondylitis – Most Common
2. Reiter's Syndrome
3. Psoriatic arthritis
4. Juvenile Rheumatoid arthritis

} HLA-B27 ASSOCIATED

❑ ANKYLOSING SPONDYLITIS

- HLA B-27
- Most common uveitis with arthritis
- Acute, Recurrent Anterior Uveitis
- SI joints – Low backache
- Young male

❑ JUVENILE RHEUMATOID ARTHRITIS

- Children below age of 16yrs, seronegative
- Most common with Pauciarticular seronegative RA
- Bilateral chronic Anterior Uveitis

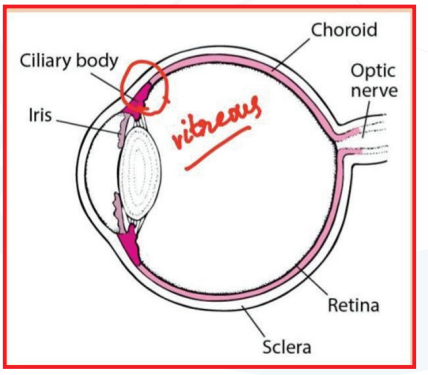
Active space

- Uveitis in white eye – Minimal congestion
- Complicated cataract
- Posterior Synechiae
- Band shaped keratopathy
- IOL Implantation avoided

◀FIG 11.14 Band shaped keratopathy in JRA

INTERMEDIATE UVEITIS/ PARS PLANITIS

- 2nd most common
- Hallmark : Vitreous inflammation



◀FIG 11.15 Location of intermediate uveitis

❑ INTERMEDIATE UVEITIS/ PARS PLANITIS

1. Primary >50%

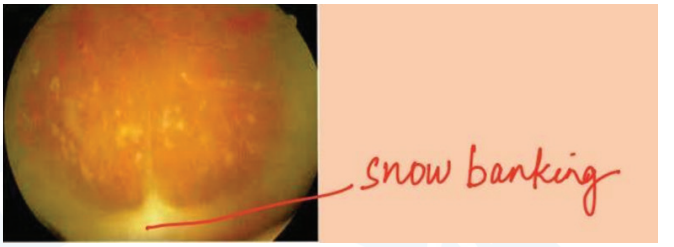
   Idiopathic, Chronic

2. Secondary
   - Sarcoidosis – Most Common

Active space

☞ Clinical features

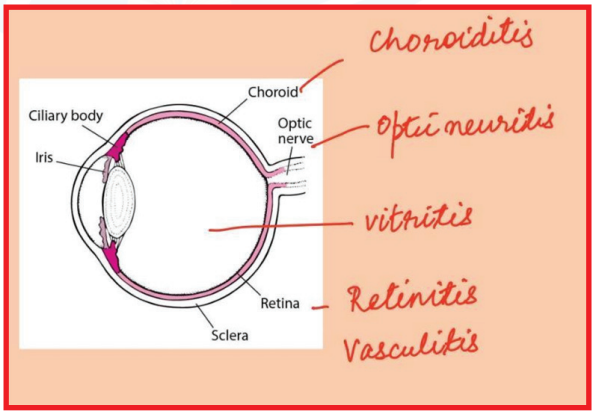
- Floaters – vitreous opacities
- Decreased Vision – Cystoid macular edema (CME)
- Pain, Photophobia Absent



◀**FIG 11.16** Snow Banking

❑ COMPLICATIONS OF INTERMEDIATE UVEITIS

- ✦ Cystoid macular edema – Most Common Decreased Vision
- ✦ Periocular steroid injections
- ✦ Systemic Steroids and cytotoxic drugs



◀**FIG 11.18** POSTERIOR UVEITIS

Active space

☞ Causes

❑ **RETINITIS**

- ✦ Toxoplasmosis – Most Common

❑ **CHOROIDITIS**

- ✦ Sarcoidosis – most common
- ✦ Tuberculosis
- ✦ VKH syndrome

❑ **CLINICAL FEATURES**

- ✦ Decreased Vision
- ✦ Painless

☞ Treatment

Non Infectious: Systemic Steroids
Immuno suppressants

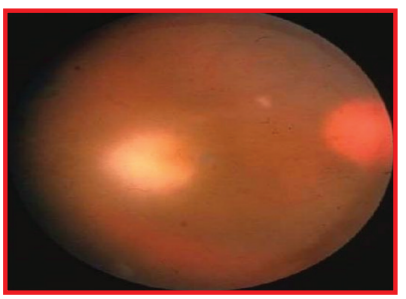
INFECTIOUS UVEITIS

❑ **TOXOPLASMOSIS**

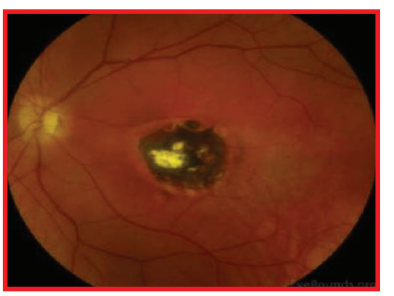
Most common cause of posterior Uveitis

- ✦ Mostly in fetal life, later can recur in chorioretinal scar
- ✦ Acquired
- ✦ Posterior uveitis

  Characteristic – Retino choroiditis
- ✦ Dense vitritis
- ✦ Headlight in fog appearance



◀FIG11.19 Toxoplasmosis



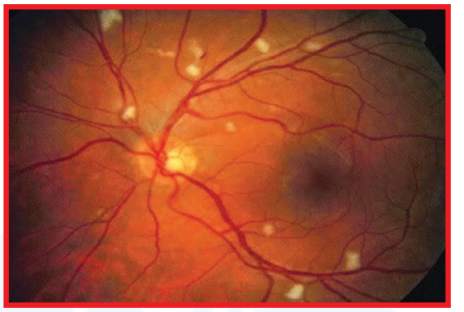
◀FIG 11.20 Toxoplasma scar

Active space

MIST

❑ SYPHILIS

- ✦ Salt and pepper fundus / Ground glass retinitis



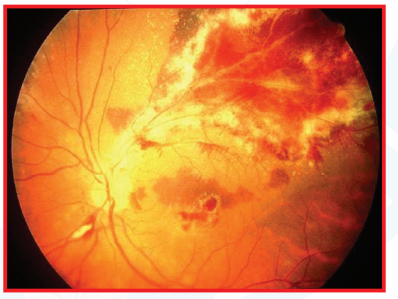
◀FIG 11.21 AIDS

- ✦ HIV microangiopathy – Most common ocular manifestation
- ✦ Cotton wool spots

❑ AIDS

Opportunistic Infections

- ✦ CMV retinitis: most common
  - Pizza cheese retinopathy



◀FIG 11.22

NON – INFECTIOUS UVEITIS

SARCOIDOSIS

- ✦ Multisystem disorder

Active space

- ✦ Non-caseating granulomata in lungs and other organs
  - Uveitis
  - Erythema nodosum
  - B/L hilar lymphadenopathy

} Clinical case scenarios

Candle wax drippings

◀FIG 11.23 Candle Wax Drippings

BEHCET'S DISEASE

HLA-B5 association

☞ C/F

- ✦ Oral ulceration
- ✦ Recurrent genital ulceration
- ✦ Skin lesions – erythema nodosum
- ✦ Eye involvement

} Clinical vignette

VOGT KOYANAGI HARADA SYNDROME (VKH)

- ✦ Idiopathic multisystem disorder
- ✦ Neuro – Oculo – Cutaneous (NOC)
- ✦ HLA-DR4 association

❑ SKIN AND HAIR CHANGES

- ✦ Alopecia

Active space

- Poliosis
- Vitiligo

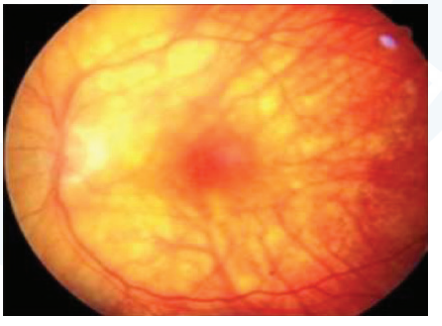
◀FIG 11.24 Harada Syndrome

❑ NEUROLOGICAL FEATURES

- Meningeal irritation
- Tinnitus
- Vertigo
- Deafness

❑ OCULAR FEATURES

- B/L Granulomatous Panuveitis



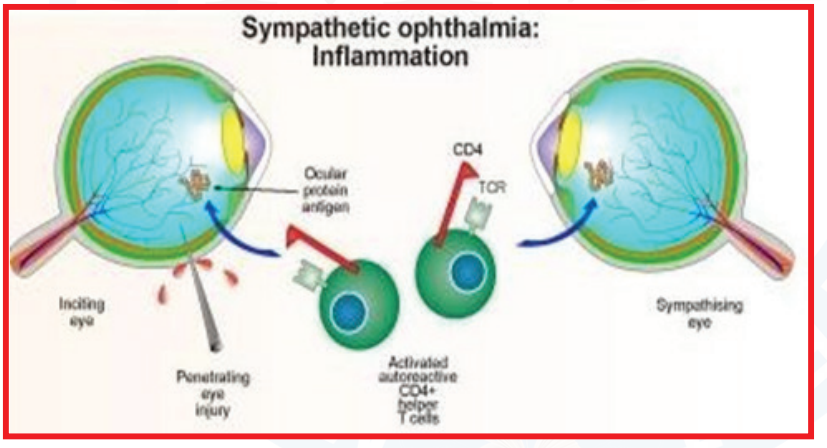
◀FIG 11.25 Sunset Glow Fundus

Active space

SYMPATHETIC OPHTHALMITIS

- B/L granulomatous panuveitis
- Occurs after penetrating trauma

- 65% of cases occur between 2 weeks and 3 months after injury
- 90% of cases occur within first year
- Penetrating wound is the prerequisite
  - Trauma
  - Intraocular surgery
- Wounds involving Ciliary Body , Dangerous Area
- Iris incarceration is also a risk factor
- T cell mediated hypersensitivity reaction
  - INJURED EYE – EXCITING EYE
  - NORMAL EYE – SYMPATHEZING EYE



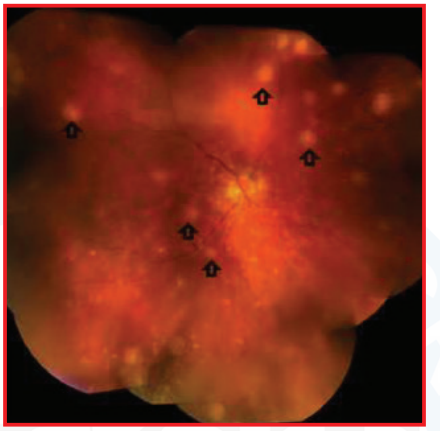
◀**FIG 11.26** Sympathetic Ophthalmia

- Earliest symptom – Weakness of Accommodation, Photophobia
  - Seat of inflammation
- Earliest sign - Retrolental flare – Anterior vitreous

Active space

❑ OTHER SIGNS

- ✦ Anterior andintermediate uveitis
- ✦ Disc edema
- ✦ Macular detachment exudative



◀FIG 11.27 DALEN FUCH'S NODULES

☞ TREATMENT

- ✦ Vigorous steroid therapy – mainstay
- ✦ Immunosuppressive therapy: in severe steroid- resistant cases
  - Primary enucleation severely traumatised
- ✦ Enucleation: within 2 weeks of injury will prevent sympathetic uveitis

SPECIFIC UVEITIC ENTITIES

❑ EALES DISEASE/PERIPHELBITIS RETINAE

- ✦ Young males
- ✦ Bilateral

❑ ETIOLOGY

- ✦ Idiopathic
- ✦ Hypersensitivity to tubercular protein
- ✦ Peripheral retinal vasculitis

Active space

❑ **PRESENTATION:CLINICAL CASE**

- ✦ Young male
- ✦ Sudden blurring of vision with floaters
- ✦ Recurrent vitreous haemorrhage

☞ Treatment: PRP, PPV

- Non resolving VH
- TRD

ENDOPHTHALMITIS

It is an inflammation of inner eye fluids and tissues i.e. uveal tissue, retina, vitreous humor and aqueous humor

☞ TYPES

- Postoperative
- Post – traumatic
- Endogenous

MCC is intraocular surgery

❑ **POSTOPERATIVE**

ACUTE POSTOP (Within 7 days after surgery)

- ✦ Staph Epidermidis, most common

DELAYED ONSET (from a week to months after surgery)

- ✦ Propionibacterium acne-most common
- ✦ Fungi

❑ **POST-TRAUMATIC**

❑ **CLINICAL FEATURES**

- ✦ Severe pain
- ✦ Redness
- ✦ Lacrimation
- ✦ Photophobia
- ✦ Marked loss of vision

Active space

◀FIG 11.28 Post-operative Endophthalmitis

3 days post cataract surgery

❑ TREATMENT OF ENDOPHTHALMITIS

- ✦ Intravitreal antibiotics 1st line
- ✦ Topical and systemic steroids
- ✦ Pars Plana Vitrectomy

Active space

Workbook Questions

1. Q: All of the following are associated with HLA-B27 uveitis except:
   (a) Behcet's disease
   (b) Psoriasis
   (c) Reiter's syndrome
   (d) Ankylosing spondylitis MIST

2. Q: Iridocyclitis is a feature of
   (a) Juvenile RA with systemic features
   (b) Seropositive pauciarticular JRA
   (c) Seronegative pauciarticular JRA
   (d) Seropositive polyarticular JRA MIST

3. Q: Snow banking is typical sign of
   (a) Panophthalmitis
   (b) Anterior uveitis
   (c) Pars planitis
   (d) Posterior uveitis MIST

4. Q: Drug of choice for acute iridocyclitis is
   (a) Steroids
   (b) Acetazolamide
   (c) Antibiotics
   (d) Pilocarpine MIST

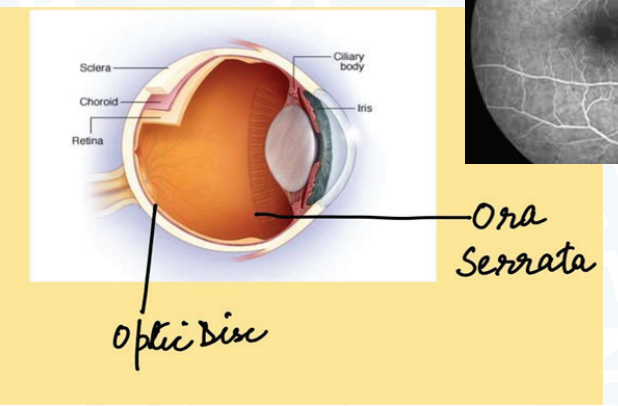
5. Q: Candle wax drippings are seen in retina in which of the following:
   (a) Tuberculosis
   (b) Behcet's disease
   (c) Sarcoidosis
   (d) Sympathetic ophthalmia MIST

Answers: 1. (a), 2. (c), 3. (c) 4. (a), 5. (c).

Active space

RETINA PART -1

ANATOMY



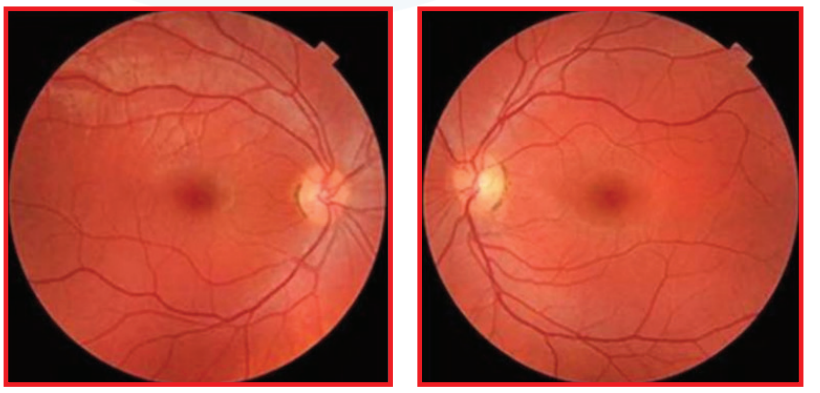
◀**FIG 12.1** Anatomy

CENTRAL RETINA – Direct Ophthalmoscope, Slit lamp +90D

Peripheral Retina = Indirect Ophthalmoscopy

ANATOMY OF FUNDUS

Fundoscopy/ Ophthalmoscopy



◀**FIG 12.2** Fundoscopy/ opthalmoscopy

Active space

❑ FUNDUS LANDMARKS

Optic Disc – Nasally 1.5mm

Retinal vessels

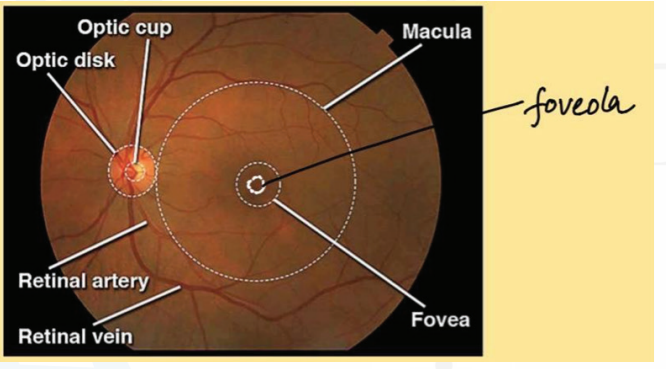
- Arterioles
- Venules

A: V = 2:3

Macula = Temporal to disc 5500u (5.5mm)

Fovea = 1.5mm (1500u)

Foveola = 350u



◀FIG 12.3 MACULA & FOVEA

10 layers

1 RPE (Retinal Pigment Epithelium)

9 Neurosensory Retina

Outer Plexiform layer = Henle's layer

❑ VASCULAR SUPPLY

- ✦ Inner six layers derive it's nutrition from Central Retinal Artery
- ✦ Outer four Layers of Retina get nutrition from choroidal circulation (Posterior ciliary Artery)

Active space

INVESTIGATIONS OF RETINAL DISEASES

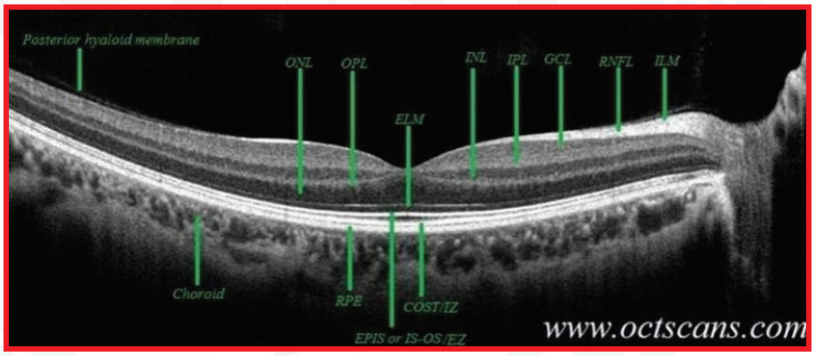
- Fundus Fluroscein angiography FFA
- Optical Coherence Tomography OCT
- Ultrasound B scan (posterior segment)

❑ FUNDUS FLUORESCEIN ANGIOGRAPHY

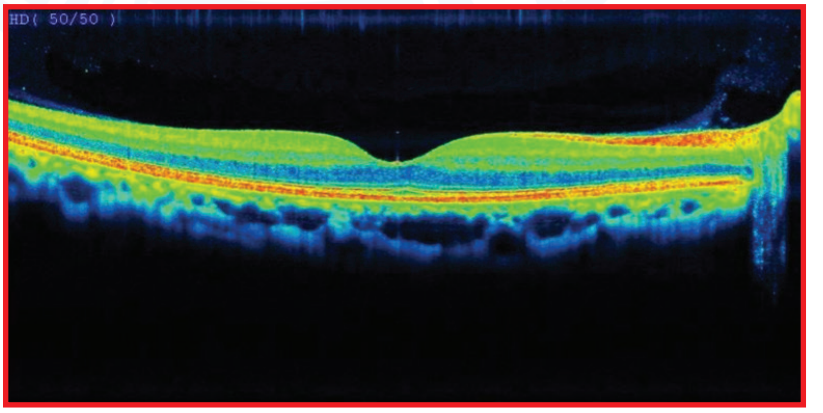
- Sodium fluorescein dye
- Retinal vascular diseases (IOC)
- Optic nerve diseases
- Choroidal diseases

❑ OPTICAL COHERENCE TOMOGRAPHY

- Near Infrared light is used



◀FIG 12.4 IOC for Macular diseases



◀FIG 12.5

Active space

RETINAL DETACHMENT

Separation of Neurosensory Retina from Retinal pigment epithelium (RPE)

Detachment of Retina from it's insertion at Ora serrata

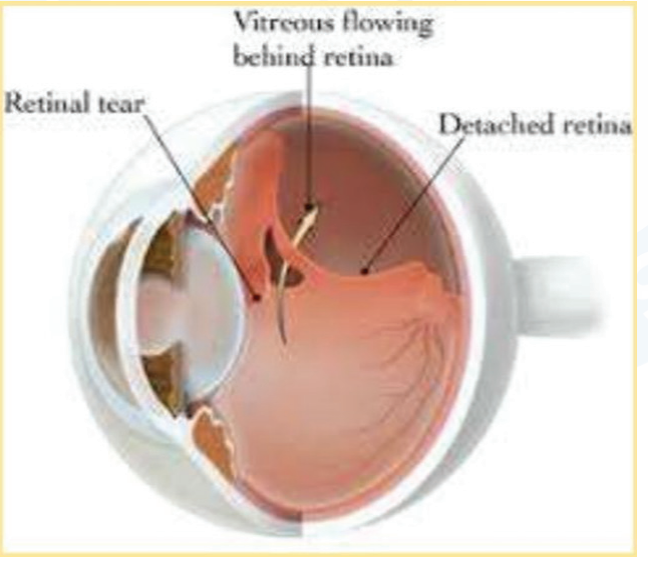
- **Retinoschisis**

  Splitting of two layers of Neurosensory Retina

- **Rhegmatogenous Retinal Detachment**

  ↓

  Retinal Break



◀**FIG 12.6** Rhegmatogenous Retinal Detachment

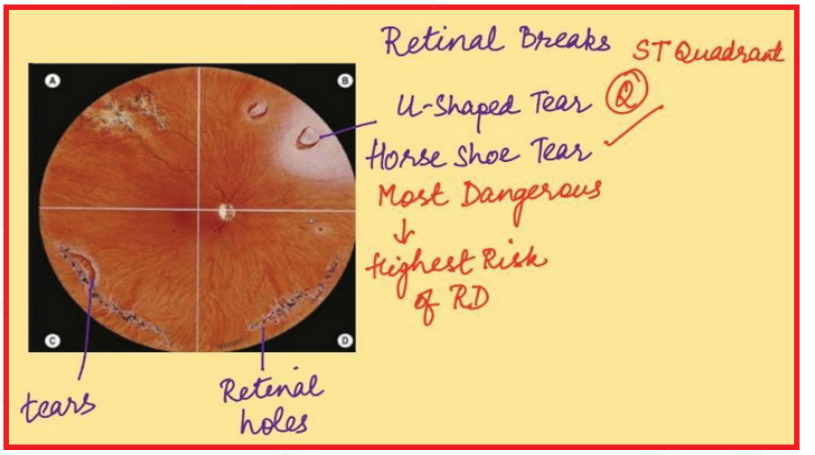
Retinal Detachment – Most Common

- ✦ Development of Retinal break
- ✦ Liquified Vitreous

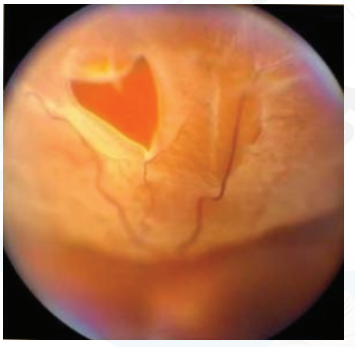
Active space

❑ RISK FACTORS FOR DEVELOPING RETINAL BREAKS

- ✦ High Myopia – Lattice degeneration, holes
- ✦ Trauma
- ✦ Posterior vitreous Detachment (PVD)
- ✦ Pseudophakia, Aphakia
- ✦ Retinoschisis



◀FIG 12.7 TYPES OF RETINAL BREAKS



◀FIG 12.8

◀FIG 12.9

❑ Risk Factors for Rhegmatogenous RD

- ✦ High Myopia (40% of total RRD) Most common
- ✦ Trauma (10%)
- ✦ Age related PVD
- ✦ Loss of Vitreous in Cataract surgery
- ✦ Laser capsulotomy in high Myopic patients

Active space

❑ Symptoms of Rhegmatogenous RD

1. Floaters } PVD
2. Flashes of light (photopsia) } Pre monitory symptoms

Retinal periphery exam by indirect opthalmoscopy

3. Curtain falling in front of eye
4. Decreased Vision (marked)

☞ Signs

- ✦ Decreased Vision
- ✦ RAPD (Relative afferent Pupillary Defect)
- ✦ Hypotony (↓ IOP)
- ✦ Shafer's sign (Tobacco Dusting) → Retinal break
- ✦ Weiss Ring – Solitary floater (Total PVD)
- ✦ Detached Retina
- ✦ Retinal Break

◀FIG 12.10 Rhegmatogenous RD

❑ SURGERY FOR RHEGMATOGENOUS RD

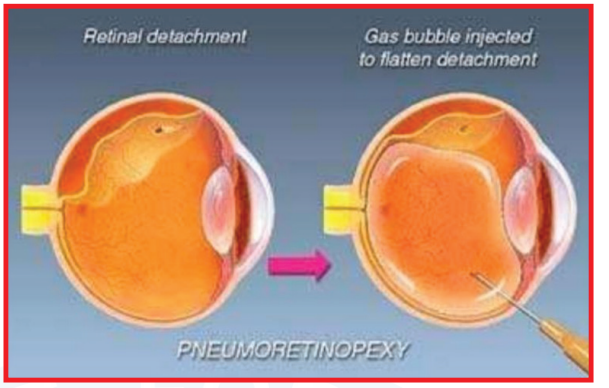
1. Pneumatic Retinopexy
2. Scleral Buckling
3. Pars plana Vitrectomy

❑ PNEUMATIC RETINOPEXY

Gas bubble Injection + Laser/ Cryotherapy

Active space

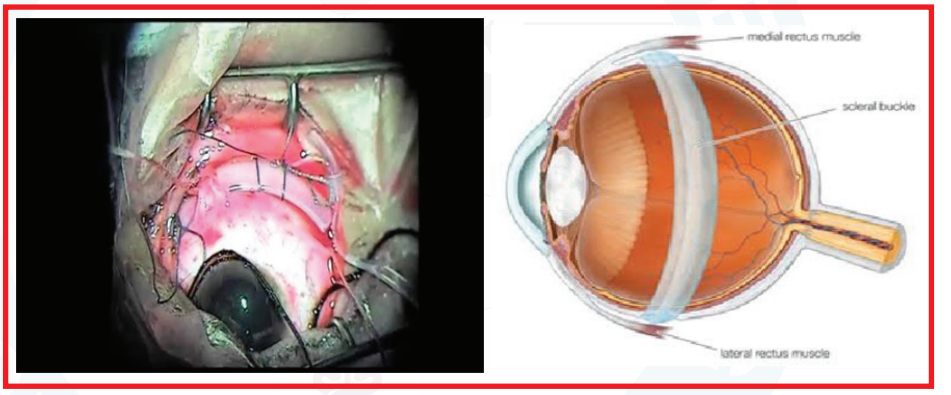
MIST



◀**FIG 12.11** Pneumoretinopexy

Expansile Gas mixture:

- SF6 – Sulphur Hexafluoride
- C3F8 – Perfluoropropane



◀**FIG 12.12** SCLERAL BUCKLING

1. Explant – Silicone
2. Drainage of Subretinal fluid
3. Cryotherapy to break

◀**FIG 12.13** PARS PLANA VITRECTOMY

Active space

Vitreous Substitutes

↓

Tamponade Effect

1. Gases - SF6, C3F8
2. Silicone oil

❑ TRACTIONAL RETINAL DETACHMENT (TRD)

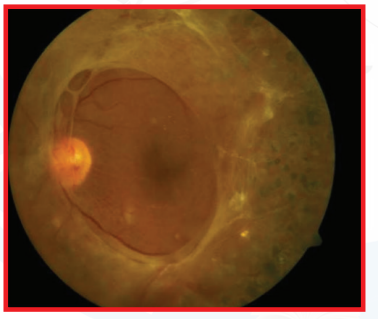
↓

Pulling off Retina

Fibrovascular membrane

☞ CAUSES

1. Most common – Advanced Diabetic Retinopathy
2. Vein occlusions
3. Vasculitis - Eale's disease
4. ROP, Sickle cell Retinopathy



◀FIG 12.14 TRACTIONAL RD

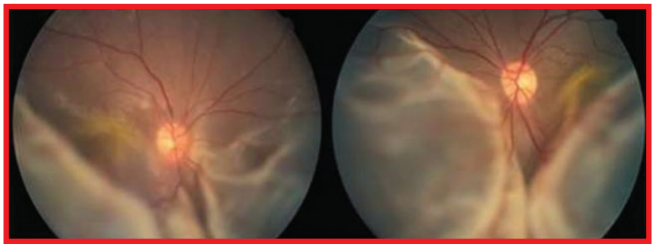
❑ EXUDATIVE RETINAL DETACHMENT / SEROUS RD

☞ Causes

- ✦ Inflammatory – VKH, sympathetic ophthalmitis, posterior scleritis
- ✦ Choroidal tumors
- ✦ Malignant hypertension
- ✦ Eclampsia
- ✦ Coat's disease

Active space

- Idiopathic – Localised – CSR



◀FIG 12.15 EXUDATIVE RD

- Shifting fluid

☞ Treatment

- Treat underlying cause

DIABETIC RETINOPATHY

❑ RISK FACTORS

- Duration of the diabetes – Most important

  20yrs of DM - 90% Type 1

  60% Type 2

  30yrs of DM – 95% of both
- Poor glycemic control
- Type of diabetes

  PDR – Type 1

  DME – Type 2
- Hypertension
- Hyperlipidemia
- Pregnancy, females
- Nephropathy, Anemia

❑ PATHOGENESIS OF DIABETIC RETINOPATHY

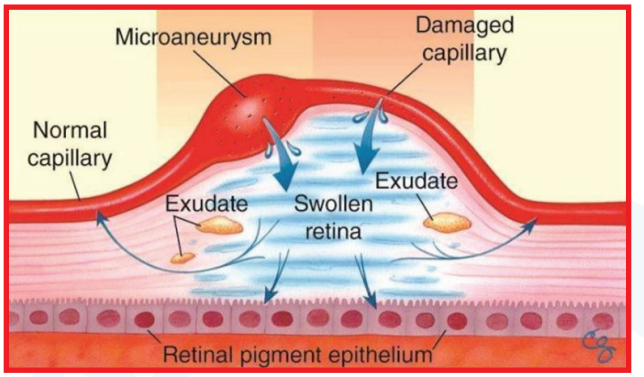
Micro angiopathy

↓

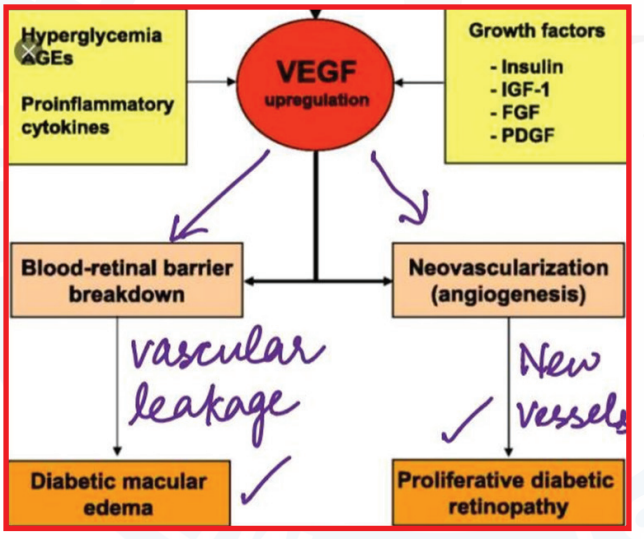
Retinal capillaries

Active space

1. Loss of Pericytes
2. Loss of inner BRB (Endothelial cell junctions)
3. Capillary occlusion



◀FIG 12.16 Micro aneurysm



◀FIG 12.17 VEGF Upregulation

Ischemia

↓

Increased production of Angiogenic Factors

(VEGF – Vascular Endothelial Growth Factor)

❑ CLASSIFICATION OF DIABETIC RETINOPATHY

ETDRS – Early Treatment of Diabetic Retinopathy study

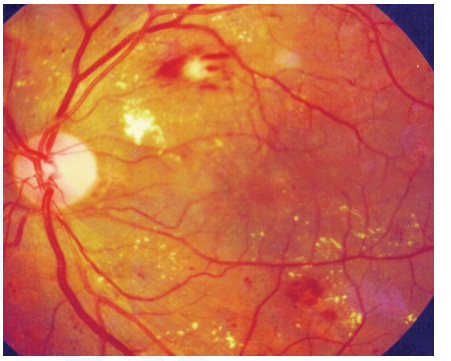
1. Non Proliferative Diabetic Retinopathy
2. Proliferative Diabetic Retinopathy

Active space

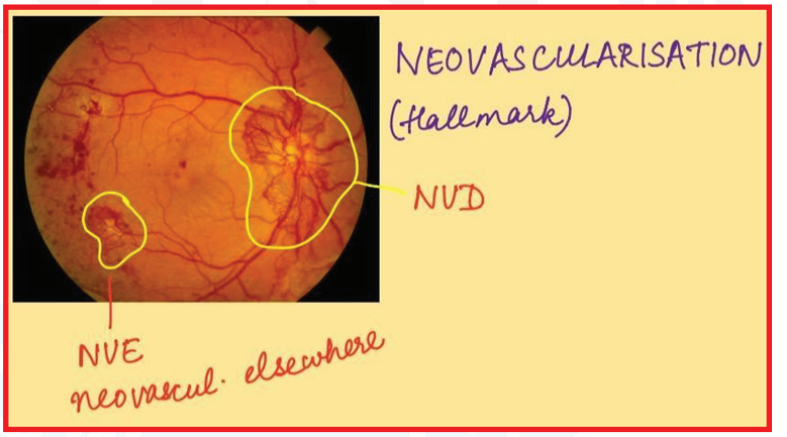
3. Diabetic Maculopathy
4. Advanced diabetic Retinopathy

❑ NON PROLIFERATIVE DIABETIC RETINOPATHY (NPDR) EARLY

1. Micro aneurysm – Hallmark
   (Earliest Sign)
2. Retinal Hemorrhages
   - Flame shaped
   - Deep → Dot & Blot
3. Hard Exudates
   (Lipid deposits)
4. Soft Exudates (Cotton wool spots)



❑ PROLIFERATIVE DIABETIC RETINOPATHY (PDR)

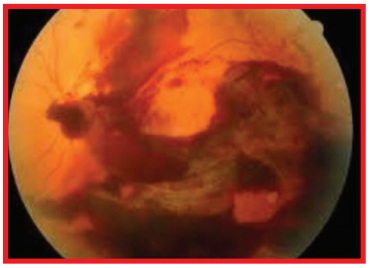


◀FIG 12.18 Neovascularisation

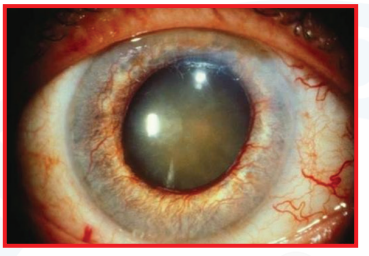
❑ COMPLICATIONS OF PDR

1. Vitreous Hemorrhage
2. Fibrovascular proliferation
   ↓
   Tractional Retinal detachment TRD
3. Rubeosis Iridis
   ↓
   Neovascular Glaucoma

Active space



◀FIG 12.19 COMPLICATIONS OF PDR



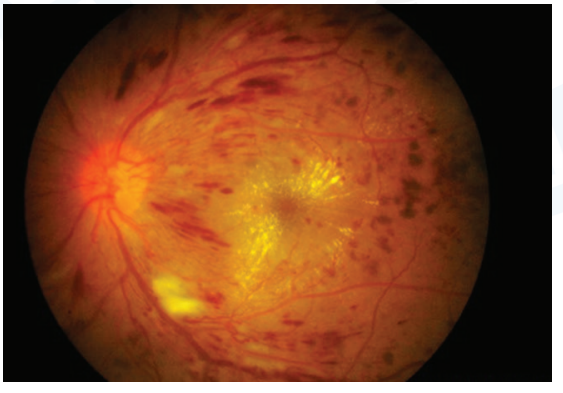
◀FIG 12.20 RUBEOSIS IRIDIS

DIABETIC MACULOPATHY

Macular thickening or edema

Leakage from capillary bed – Focal or diffuse

1. Accumulation of fluid in Retinal areas
2. Hard Exudates



❑ **CSME – CLINICALLY SIGNIFICANT MACULAR EDEMA**

.

Active space

❑ CLASSIFICATION OF NPDR

- Mild NPDR – only microaneurysm present.
- Moderate NPDR – microaneurysms, retinal haemorrhages, hard exudates/cotton wool spots
- Severe NPDR – 4:2:1 rule – haemorrhages in 4 quadrants/ venous beading in 2 quadrants/ IRMA in one quadrant
- Very Severe NPDR – any two of severe NPDR signs present

❑ DIABETIC RETINOPATHY : CAUSES OF VISION LOSS

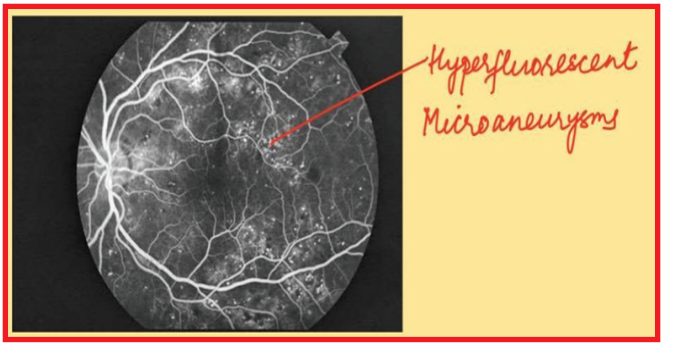
1. Most common cause (over all) – Diabetic macular Edema
2. NPDR – Macular Edema → Moderate Visual loss
3. Vitreous Hemorrhages
4. TRD.
5. Neovascular glaucoma

} Severe Visual Loss PDR

❑ INVESTIGATIONSIN DIABETIC RETINOPATHY

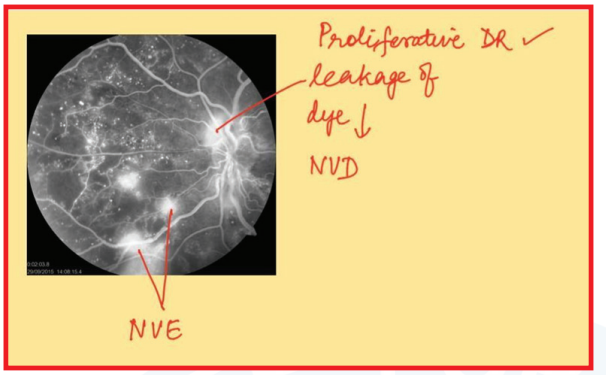
1. Fundus Fluroscein Angiography
   ↓
   Vascular Changes
2. Optical Coherence Tomography
   ↓
   Macular Edema

❑ Non Proliferative Diabetic Retinopathy

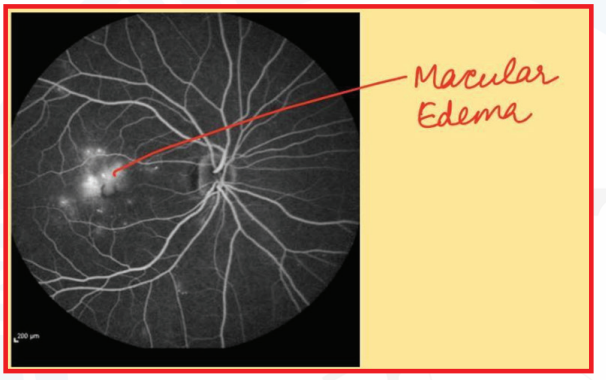


Active space

◀FIG 12.22



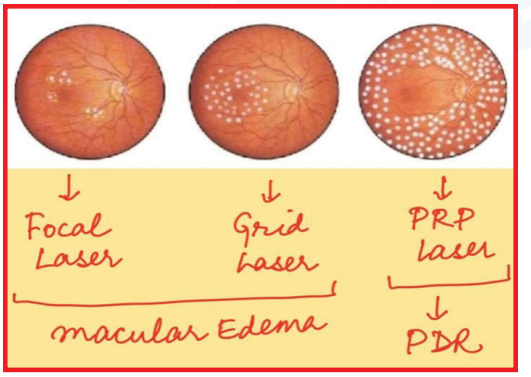
◀FIG 12.23



◀FIG 12.24

❑ TREATMENT OF DIABETIC RETINOPATHY

1. Laser Photo coagulation
2. Intra Vitreal Anti VEGF injections
3. Surgery



◀FIG 12.25 Treatment of Diabetic Retinopathy

Active space

❑ LASER PHOTO COAGULATION

Fd- Nd YAG Laser – 532 nm

❑ PAN RETINAL PHOTOCOAGULATION

↓

Neo vascularisation

↓

Regress

◀FIG 12.26 LASER PHOTO COAGULATION

Converting Ischemic Retina (↑ VEGF)

↓

Anoxic Retina

↓

Decreased production of VEGF

❑ INTRA VITREAL INJECTIONS & IMPLANTS

☞ Indications

1. Diabetic Macular Edema (central)
2. Proliferative DR

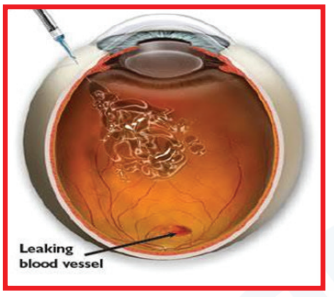
☞ Anti VEGF agents

✦ Ranibizumab

Active space

- Bevacizumab
- Aflibercept

☞ Intravitreal Injection



◀FIG 12.27 Intravitreal Injection

◀FIG 12.28

❑ INTRAVITREAL DEXAMETHASONE IMPLANT

(700u)

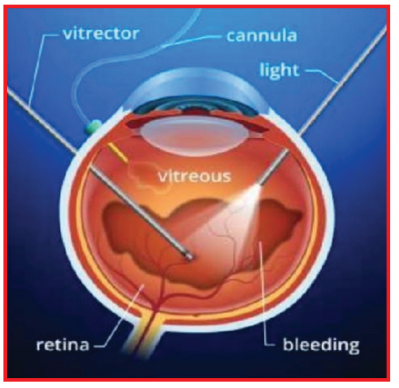
↓

Diabetic macular edema

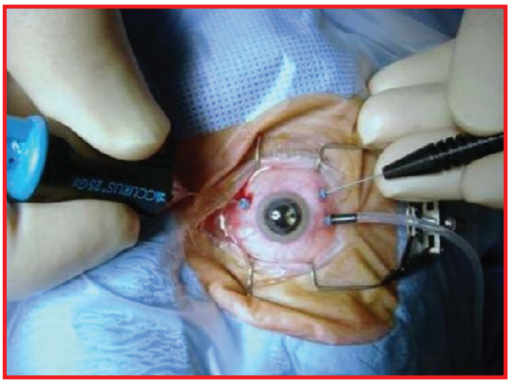
◀FIG 12.29 INTRAVITREAL DEXAMETHASONE IMPLANT

❑ INDICATIONS OF PARS PLANA VITRECTOMY IN DR

1. Non Resolving Vitreous Hemorrhage
2. Tractional RD
3. Recalcitrant Macular edema



◀FIG 12.30 Vitrectomy



◀FIG 12.31

Active space

☞ Examination Schedule (Screening of DR)

- ✦ Type 1 DM – 5 yrs after onset or after puberty
- ✦ Type 2 DM – At the time of Diagnosis
- ✦ Pregnancy with DM – Early in first trimester, each trimester & six weeks postpartum

HYPERTENSIVE RETINOPATHY (HTR)

- Medium sized arterioles

Changes at Arteriovenous crossings (AV nipping)

Thickening of Arterioles walls → Stiff

Hypertensive Retinopathy: Keith & Wagner classification

- ✦ Grade 1: Mild generalized arteriolar attenuation
- ✦ Grade 2: Marked generalized and focal narrowing, A-V nipping, Salu's sign
- ✦ Grade 3: grade 2 + flame shaped hemorrhages, cotton wool spots, hard exudates, Bonnet sign, Gunn sign
- ✦ Grade 4: grade 3 + disc edema

RETINAL VASCULAR OCCLUSIONS

Retinal vein occlusions

Retinal Artery occlusions

❑ RETINAL VEIN OCCLUSION

- ✦ Central Retinal vein occlusion (CRVO)
- ✦ Branch Retinal vein occlusion (BRVO)

☞ Risk Factors

- ✦ Hypertension (Most common Cause)
- ✦ Atherosclerosis
- ✦ Diabetes Mellitus

Active space

☞ Clinical Features

Painless sudden loss of vision

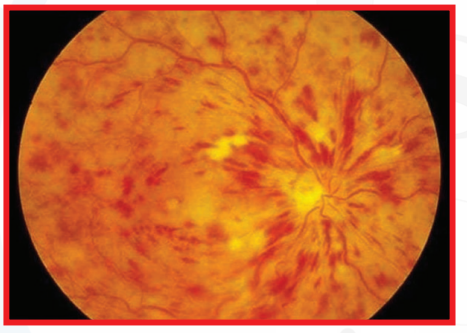
.....

.....

.....

☞ Signs

.....



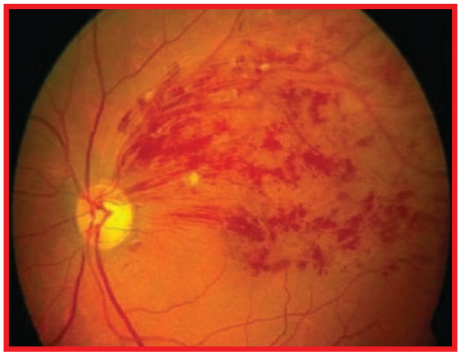
◀FIG 12.32 RETINAL VEIN OCCLUSION

Tomato – Splash appearance = CRVO

(Blood in Thunder)

.....

.....



◀FIG 12.33 Branched Retinal vein occlusion

Active space

MIST

❑ COMPLICATIONS OF RETINAL VEIN OCCLUSION

Macular edema

Ischemic Retina → VEGF angiogenic Factors

- ✦ Retinal Neovascularisation
- ✦ Vitreous Hemorrhage
- ✦ Tractional Retinal Detachment
- ✦ Iris Neovascularisation/Rubeosis Iridis
- ✦ Neovascular Glaucoma

(90 day/ 100 day Glaucoma)

☞ TREATMENT

1. Macular edema – Intra vitreal Anti VEGF Injection (Ranibizumab)
2. Neo vascularisation = PRP laser
3. Tractional RD/ Vitreous Hemorrhage = PPV surgery

❑ RETINAL ARTERY OCCLUSION

Ocular emergency

❑ Risk Factors for CRAO

- ✦ Hypertension, Atherosclerosis
- ✦ Hypercoagulative conditions
- ✦ Atrial fibrillation
- ✦ Emboli

Hollenhorst Plaques

↓

Cholesterol

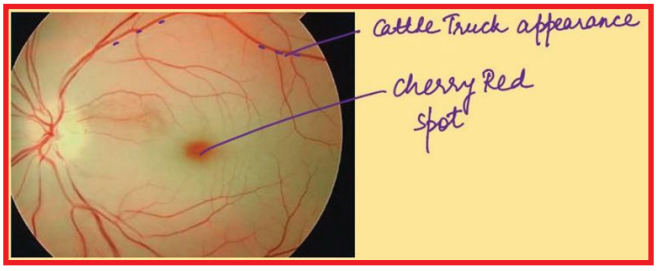
Calcific – Most dangerous

☞ Clinical Features

- ✦ Sudden painless loss of vision

Active space

✦ RAPD → Relative afferent Pupillary Defect



◀FIG 12.34

◀FIG 12.35 CRAO with Sparing of Cilioretinal Artery (15-30%)

❑ TREATMENT OF CRAO

Prognosis is very poor

Ocular emergency

- ✦ Ocular massage (using a Gonioscopic lens)
- ✦ IV mannitol
- ✦ Inhalation of Carbogen (95% o2 + 5% CO2)
- ✦ Paracentesis
- ✦ Antithrombolytic therapy

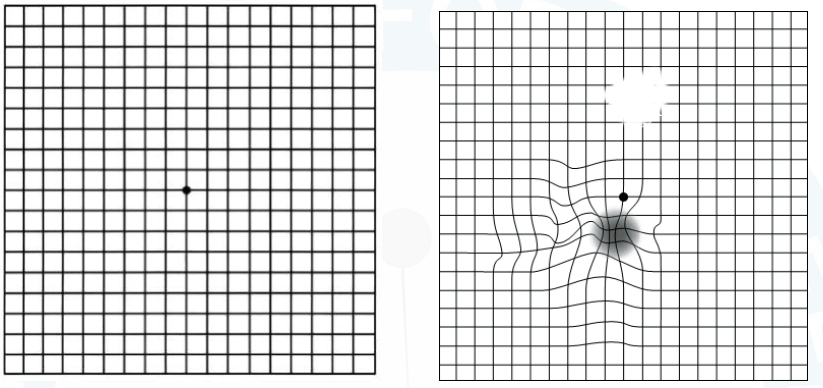
❑ CAUSES OF CHERRY RED SPOT

Christmas - CRAO
Tree – Tay Sach's disease
Never – Neimann Pick disease
Grows - Gaucher's disease
Tall - Trauma
In
Mud - Metachromatic Leukodystrophy
&
Sand - Sandhoff's disease

Active space

RETINA PART -2

❑ AMSLER GRID



CYSTOID MACULAR EDEMA (CME)

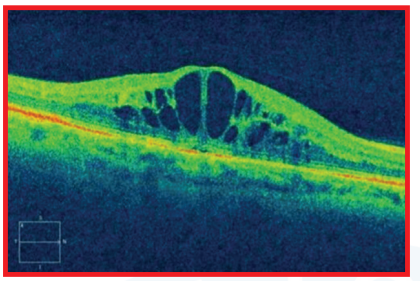
- Accumulation of fluid in OPL/ Henle's layer → INL

☞ Causes

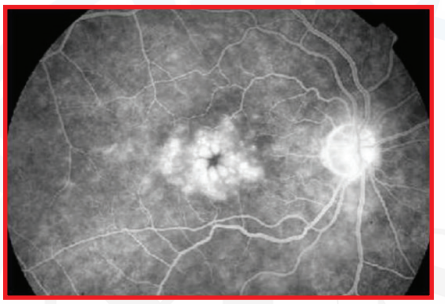
1. Uveitis
2. Diabetic Retinopathy
3. Retinal vein occlusions
4. Irwin – Gass Syndrome
5. RP
6. Drug – Induced
   - Topical PG analogue
   - NIACIN
   - Adrenaline

Active space

❑ FUNDUS EXAM – HONEYCOMB APPEARANCE in MACULA



◀FIG 13.1 OCT Macula



◀FIG 13.2 Flower Petal Appearance FFA

❑ TREATMENT OF CYSTOID MACULAR EDEMA

1. Inflammation – Steroids
   Post Subtenon Injection
   Oral form
2. Post Cataract – Topical NSAIDS
   Peri ocular Steroids
3. Intravitreal Steroids
4. Intravitreal anti VEGF injections
5. ACETAZOLAMIDE – RP

CENTRAL SEROUS RETINOPATHY (CSR)

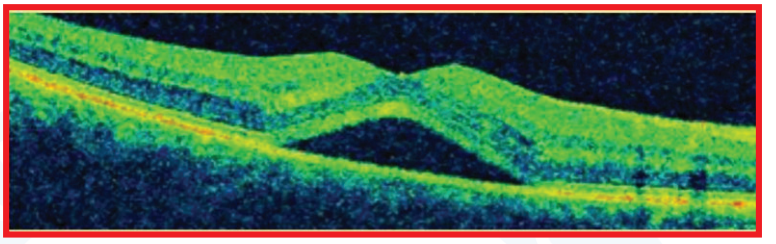
✦ Idiopathic serous detachment in macular region

Metamorphopsia

Active space

☞ Causes

- Young male
- Stress (Type A personalities)
- Smoking
- Steroids



◀FIG 13.3 Central Serous Retinopathy

❑ FFA in CSR

❑ 1. INK BLOT OR ENLARGING DOT

❑ 2. SMOKE – STACK OR MUSHROOM OR UMBRELLA LEAK

☞ Treatment

1. Spontaneous Resolution
2. Laser

AGE RELATED MACULAR DEGENERATION (ARMD)

Senile age > 60yrs age

1. Dry or Non Exudative (more common)
2. Wet or Exudative ARMD

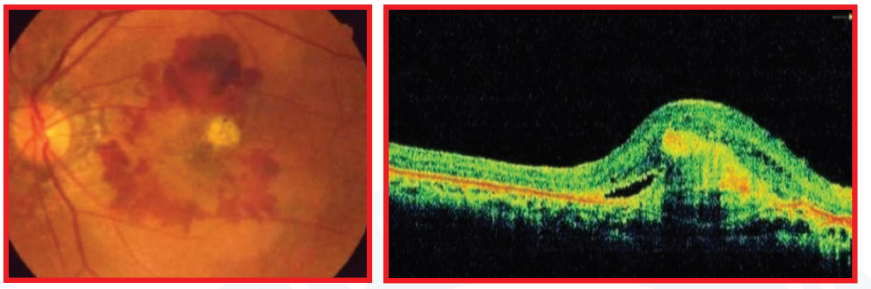
❑ DRY ARMD

◀FIG 13.4 DRUSENS

Active space

- ✦ Vision loss is mild
- ✦ Treatment : Anti oxidants

❑ WET AMD



◀FIG 13.5 WET OR EXUDATIVE ARMD

- ✦ PED pigment Epithelial detachment
- ✦ Choroidal Neovascular membrane (CNVM)

↓ ↓

Classic Occult

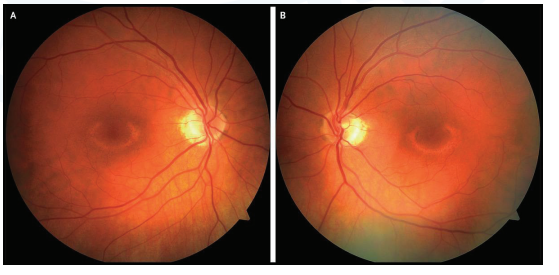
❑ TREATMENT OF WET ARMD

- ✦ Anti VEGF agents : Ranibizumab, Bevacizumab, Aflibercept
- ✦ Photodynamic therapy PDT :

  Photosensitizer dye Verteporfin + Laser
- ✦ Antioxidant therapy : DRY ARMD

❑ CAUSES OF BULL'S EYE MACULOPATHY

1. Chloroquine Toxicity
2. HCQ Toxicity
3. Stargadt's disease
4. Cone Rod Dystrophy
5. Clofazimine

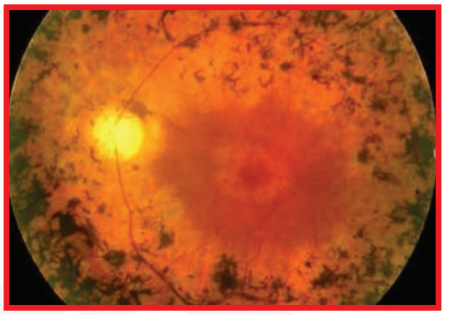


RETINAL DYSTROPHIES

- ✦ Retinitis Pigmentosa
- ✦ Best disease
- ✦ Stargadt's disease

Active space

❑ Retinitis Pigmentosa



◀FIG 13.6 Retinitis Pigmentosa

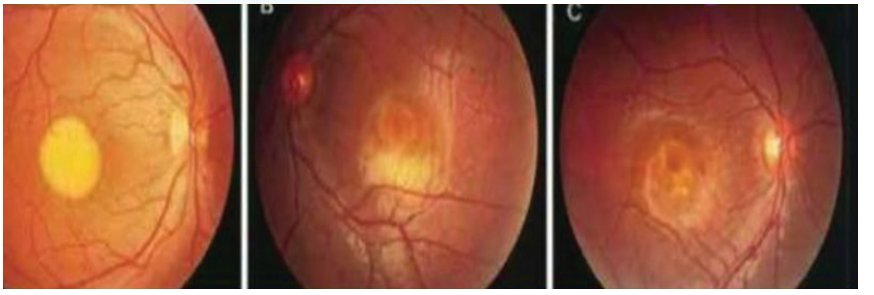
- Nyctalopia : Night Blindness
- Ring Scotoma

☞ Classic Triad

- ✦ Attenuated Arterioles
- ✦ Waxy pallor of Disc
- ✦ Bony spicule pigmentation

❑ BEST'S DISEASE / JUVENILE VITELLIFORM DYSTROPHY

- ✦ Lipofuscin in Macular area
- ✦ AD
- ✦ Mild Decrease in vision



◀FIG 13.7 BEST'S DISEASE / JUVENILE VITELLIFORM DYSTROPHY

EOG abnormal → Best investigation

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ERG is Normal

STARGADT'S DISEASE/ HEREDOMACULAR DEGENERATION

1. Most common macular dystrophy
2. Severe vision loss
3. Excessive deposition of lipofuscin

↓

.....

RETINOPATHY OF PREMATURITY (ROP)

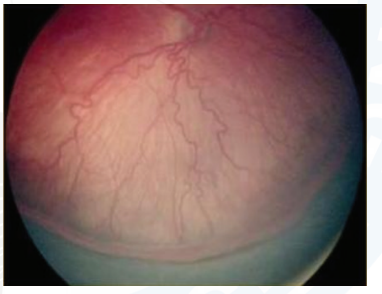
❑ **Proliferative Retinopathy seen in Premature infants**

☞ Risk Factors

1. Low birth weight < 1750gm
2. Prematurity < 34weeks
3. O2 Therapy
4. Respiratory distress
5. Sepsis

VEGF → External 02

↓ (-)



◀FIG 13.8 ROP

Normal vascularisation stops → Excessive VEGF → Abnormal proliferative Retinopathy

❑ **STAGES OF ROP**

Stage 1 – Demarcation line

Stage 2 – Fibrovascular Ridge

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Stage 3 – Extra Retinal proliferation

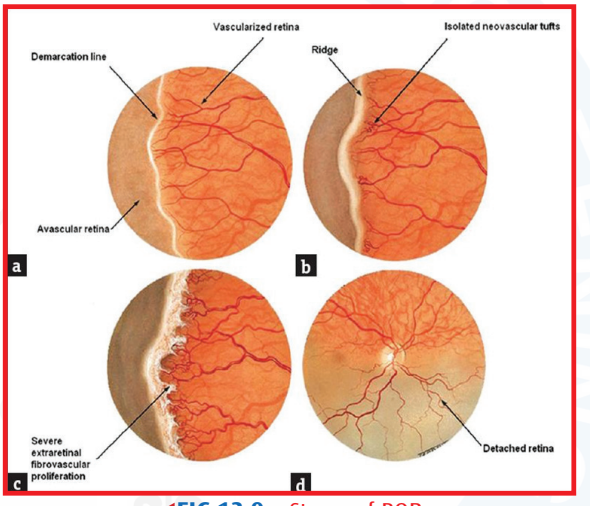
Stage 4 – Partial RD

Stage 4a – Macular attached

Stage 4b – Macular detached

Stage 5 – Total RD (Funnel RD)

Retrolental Fibroplasia



◀**FIG 13.9** Stages of ROP

Peripheral Retinal laser ablation/ Cryotherapy

↓

Ablation of Avascular Retina

Active space

Active space

Workbook Questions

1. Q: Which angiogenic mediator has been implicated mainly in progression of diabetic retinopathy:
   (a) Venous Endothelial Growth Factor
   (b) Vascular Endothelial Growth Factor
   (c) Vitreous Neovascular Growth factor
   (d) Insulin like Growth Factor — MIST

2. Q: Which type of retinal detachment is seen as complication of proliferative diabetic retinopathy:
   (a) Rhegmatogenous retinal detachment
   (b) Tractional retinal detachment
   (c) Exudative retinal detachment
   (d) None of the above — MIST

3. Q: All are features of non-proliferative diabetic retinopathy except:
   (a) Microaneurysms
   (b) Dot & blot haemorrhages
   (c) Cotton wool spots
   (d) Neovascularisation of optic disc — MIST

4. Q: Earliest ophthalmoscopic sign of diabetic retinopathy is
   (a) Hard exudates
   (b) Soft exudates
   (c) Microaneurysms
   (d) Dot hemorrhages — MIST

5. Q: Shaffer's sign is seen in
   (a) Rhegmatogenous RD
   (b) Tractional RD
   (c) Exudative RD
   (d) All of the above — MIST

Answers: 1. (b), 2. (b), 3. (d) 4. (c), 5. (a).

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INTRAOCULAR TUMORS

- Retinoblastoma
- Choroidal Melanoma

RETINO BLASTOMA

- It is a tumor of primitive cellsof retina that occurs in childhood
- Most common intra ocular malignancy of children
- 3rd most common tumor of childhood
- Median age of presentation is 1.5yrs (18months)
  - Usually seen in <3-5yrs of age group

DEMOGRAPHIC FEATURES OF RETINO BLASTOMA

- No sexual predilection
  - Incidence is same between male and female child
- No racial predilection
- Tumor is unilateral in 70-75% of cases
- Tumor is bilateral in 25-30% of cases

GENETICS OF RETINO BLASTOMA

- RB1 gene (anti oncogenic gene)- tumor suppressor gene
  - RB1 gene is present as 14 band on long arm of chromosome 13 (13q 14)

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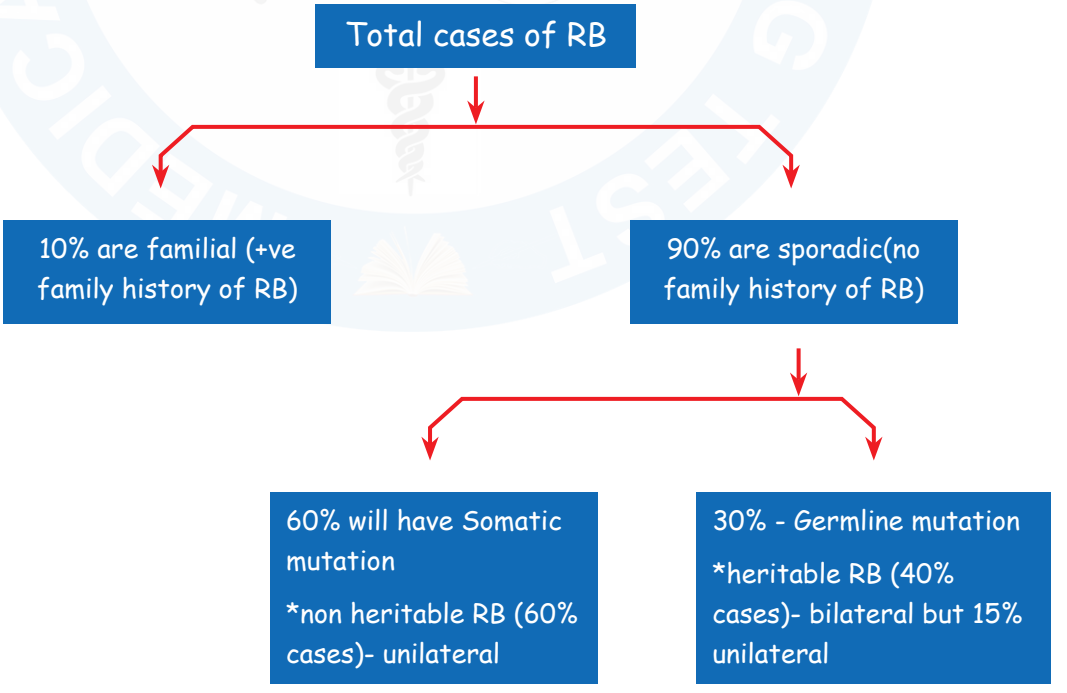
❑ KNUDSON 2 – HIT HYPOTHESIS

- ✦ Both alleles of RB1 gene have to be mutated for the tumor to occur
  - 1st mutation – Somatic cells or Germline cells (cells of gamete)
  - 2nd mutation – Somatic cells of retina
- ✦ Once the 2nd mutation occurs, it leads to the development of Retino Blastoma

❑ DISTRIBUTION OF RETINO BLASTOMA



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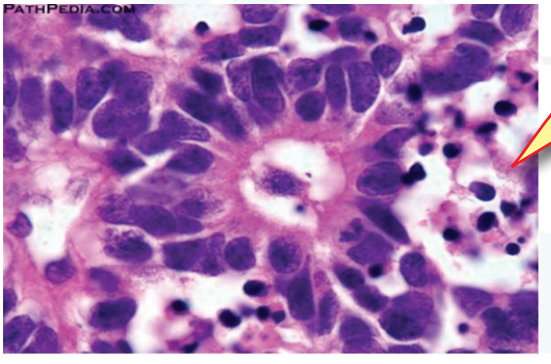
HERITABLE RB	NON-HERITABLE RB
Germline mutation	Somatic mutation
1st mutation in gamete	1st mutation in retinal cells
2nd mutation in retinal cells	2nd mutation in retinal cells
Multifocal, bilateral tumor	Unifocal, unilateral tumor
Passed to offspring 40%	Cannot be passed to offspring
Risk of secondary tumors (osteosarcoma) – most common	No risk of secondary cancer
Development of pineal body tumor (pinealoblastoma)- Trilateral Retinoblastoma	

❑ HISTPATHOLOGY OF RETINOBLASTOMA



Immature retinal cells, when these are arranged in particular Rosettes they have an empty lumen. It is called Flexner-Winter-steiner Rossettes

◀FIG 14.1

- ✦ Poorly differentiated tumor
- ✦ Well differentiated tumor – Most specific of RB- Flexner-Wintersteiner Rosettes
  - Homer – Wright Rossettes (don't have empty lumen)
  - Pseudo Rossettes
  - Fleurettes

MIST

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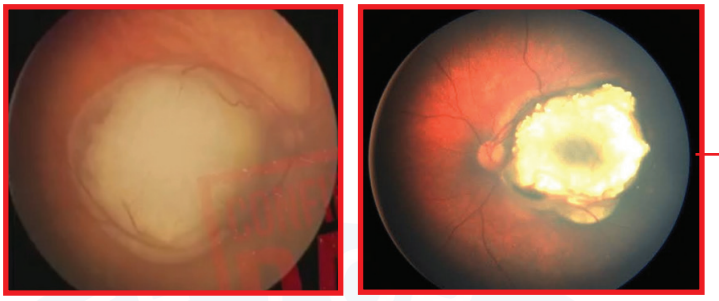
❑ CLINICAL GROWTH PATTERNS OF RB

✦ Endophytic – tumor is growing towards vitreous cavity



◀FIG 14.2 Growth pattern of RB

Exophytic – growing towards choroid → It produces exudative retinal detachment

✦ Mixed type (Endo + Exo)

✦ Diffuse infiltrating tumor

❑ CLINICAL FEATURES OF RETINOBLASTOMA

✦ Most commonpresentation – Leukocoria (white pupillary reflex)

✦ 2nd most common – Strabismus/Squint – esotropia squint

Stage 1 - Leukocoria

◀FIG 14.3

❑ DIFFERENTIALS OF LEUKOCORIA

- Congenital cataract
- Persistent Hyperplastic Primary Vitreous (PHPV)
- Coat's disease
- Stage 5 ROP – Retrolental fibroplasia
- Toxocariasis – severe vitreous inflammation
- Old Vitreous Hemorrhage

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- CR coloboma
- Retinal Detachment

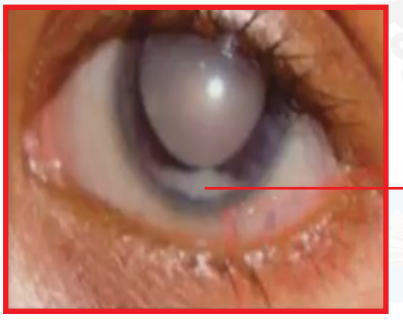
❑ **Stage 2 : painful red eye caused by:**

i) Glaucoma – increased intraocular pressure (IOP)- tumor pushes iris-lens diaphragm leading to the development of secondary glaucoma

- The eye can appear bigger in size leading to development of Buphthalmos

◀FIG 14.4

ii) Uveitis/Pseudohypopyon – it doesn't consist of inflammatory cells. It consists of tumor cells.



◀FIG 14.5

iii) NVI (Neovascularization of Iris), hyphema

iv) Orbital cellulitis like picture

Active space

❑ **Stage 3 : Extra Ocular Extension**

◀FIG 14.6

- Scleral rupture
- Fungating nectrotizing mass

❑ **Stage 4 : Stage of Distant metastatis**

- ✦ Intarcranial – Optic Nerve invasion
- ✦ Lymphnodes – Pre auricular

❑ **Increased risk of Metastatis**

- ✦ Advanced Tumor
- ✦ Optic nerve invasion
- ✦ Choroidal invasion
- ✦ AC involvement
- ✦ Orbital spread

❑ **CLASSIFICATION**

Earlier Classification

↓

Reese Ellsworth classification

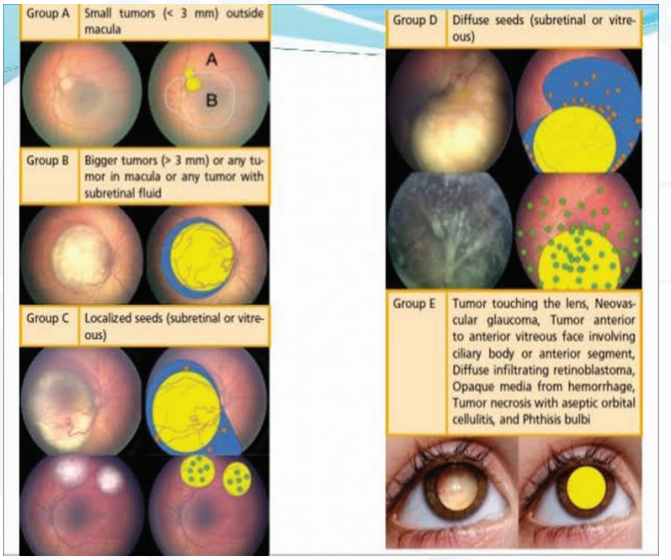
↓

External Beam Radiotherapy (EBRT)

Active space

❑ INTERNATIONAL CLASSIFICATION OF RETINOBLASTOMA (ICRB)

- ✦ Group A : very low risk
- ✦ Group B : low risk
- ✦ Group C : Moderate Risk
- ✦ Group D : High risk
- ✦ Group E : Very high risk



◀FIG 14.7 ICRB

❑ DIAGNOSIS

1. Examination under anesthesia
2. USG B scan
3. X-ray (calcification)
4. CT scan
5. MRI scan
6. Levels of LDH in Aqueous

◀FIG 14.8

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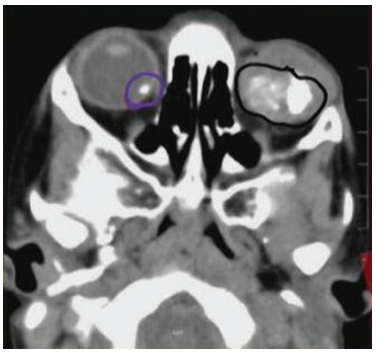
USG: Tumor, calcification

CT scan

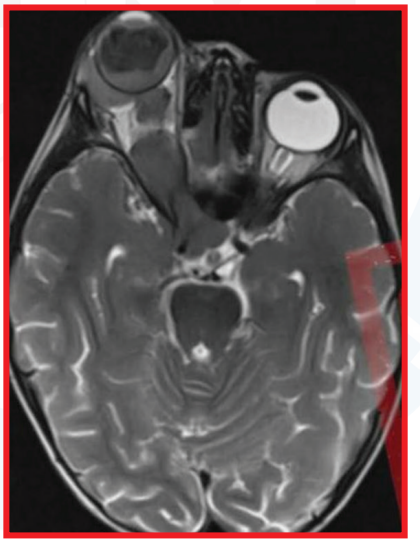
Avoid CT scan – Increased exposure → Secondary Carcinoma



◀FIG 14.9

☞ MRI SCAN

- ✦ IOC
- ✦ Optic nerve invasion
- ✦ Intracranial extension
- ✦ Pineoloblastoma can also diagnosed



◀FIG 14.10

❑ TREATMENT

MODALITIES

1. Focal Therapy
   - ✦ Laser – small posterior tumors
   - ✦ Cryotherapy – small Anterior Tumor

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- ✦ Diode laser TTT- Posterior , away from macula
- ✦ Brachytherapy

Tumors for which laser is not possible

2. Chemotherapy

Globe salvage – Preserve the eye

- ✦ Chemo reduction
- ✦ Optic nerve invasion
- ✦ Palliative treatment
- ✦ VEC – Vincristine, Etoposide, Carboplatin
- ✦ 6-12 cycles chemotherapy

3. ENUCLEATION

Indications

- ✦ Group E Tumor
  - Tumor > 50% globe
  - Anterior segment invasion
  - Orbital after chemoreduction

Optic Nerve 10-15mm stump

↓

Invasion (+) → Chemotherapy & EBRT to orbital apex

4. External Beam Radiation Therapy EBRT
   - only for one eye with extensive Tumor

Group A – D = Focal therapy along with Chemotherapy

❑ REGRESSION PATTERN

Cottage cheese (calcified mass)

Fish – Flesh mass (translucent)

Flat Atrophic scars

Active space

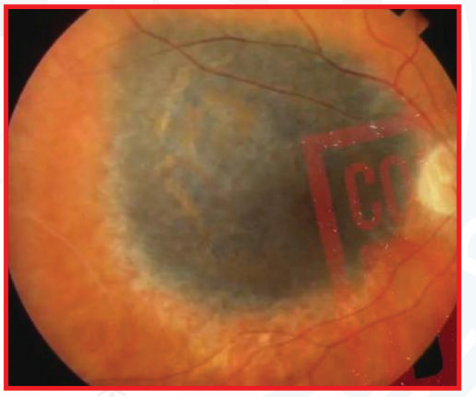
❑ **Poor Prognostic Factors**

- ✦ Optic Nerve invasion
- ✦ Undifferentiated Tumor
- ✦ Massive Choroidal invasion
- ✦ Scleral involvement
- ✦ Associated glaucoma

MALIGNANT MELANOMA

- ✦ Most common Intra ocular malignancy of adults
- ✦ Unilateral

☞ Clinical Features



◀FIG 14.11 MalignantMelanoma

- Collar stud appearance in USG

☞ TREATMENT

Small tumors

- TTT
- Brachy therapy

Large tumors

- Enucleation
- Stereotactic Radio surgery – Gamma knife

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Workbook Questions

1. Q: Most common clinical presentation of retinoblastoma is:
   (a) Leucocoria + strabismus
   (b) Leucocoria + hyphema
   (c) Leucocoria + pseudohypoyon
   (d) Leucocoria + iris heterochromia MIST

2. Q: Most common route of spread of retinoblastoma:
   (a) Lymphatics
   (b) Optic nerve
   (c) Direct spread
   (d) Vascular MIST

3. Q: A one year old child presents with leukocoria and is diagnosed with unilateral RB filling half the globe. What is the management in this case?
   (a) Enucleation
   (b) Photodynamic therapy
   (c) Radiotherapy followed by chemotherapy
   (d) Chemotherapy followed by local therapy MIST

4. Q: Poor prognostic factors for retinoblastoma are all except:
   (a) Size < 2mm
   (b) Size > 4mm
   (c) Scleral involvement
   (d) Undifferentiated tumor cells MIST

5. Q: Most common intraocular tumor in adults
   (a) Retinoblastoma
   (b) Choroidal melanoma
   (c) Choroidal hemangioma
   (d) None of the above MIST

Answers: 1. (a), 2. (b), 3. (a) 4. (a), 5. (b).

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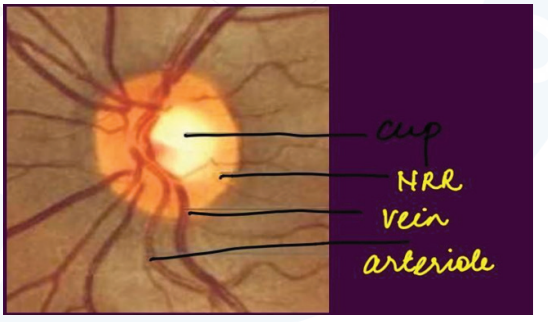
NEUROPHTHALMOLOGY

ANATOMY OF OPTIC NERVE

- Starts at Optic Nerve Head (ONH)
- Ends at Optic Chiasma
- 2nd Cranial Nerve
- Sensory Nerve
- Formed by Axons of ganglion cells of retina
- Contains 1.2 million axons
- Fibres – Visual & Pupillary



◀**FIG 15.1** NORMAL OPTIC DISC

OPTIC NEUROPATHY

Clinical Features of Optic Nerve Disease

- Visual Acuity Decreased
- Contrast sensitivity Decreased
- Brightness Sensitivity Decreased

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- ✦ Colour vision abnormality Red green Deficiency
- ✦ Visual Field : Defects

❑ **RELATIVE AFFERENT PUPILLARY DEFECT RAPD**

MARCUS GUNN PUPIL

OPTIC NEUROPATHY

- ✦ Inflammatory
- ✦ Ischemic
- ✦ Toxic
- ✦ Hereditary
- ✦ Trauma
- ✦ Compressive

❑ **OPTIC NEURITIS**

3 Types

1. Retrobulbar Neuritis
2. Papillitis (Papillae : Optic nerve Head)
3. Neuro Retinitis

❑ **CAUSES OF OPTIC NEURITIS**

1. Demyelinating : Multiple Sclerosis

   Most common in Adults
2. Para – Infectious : Post viral

   Post immunization

   ↓

   Most common cause in children
3. Infections : Sinusitis, Cat- Scratch Disease, Lyme Disease, Syphilitis, HIV
4. Non – Infectious : SLE, PAN, Wegner's, Sarcoid

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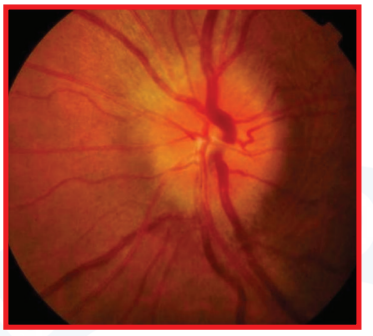
❑ Papillitis

In children

Para Infectious etiology



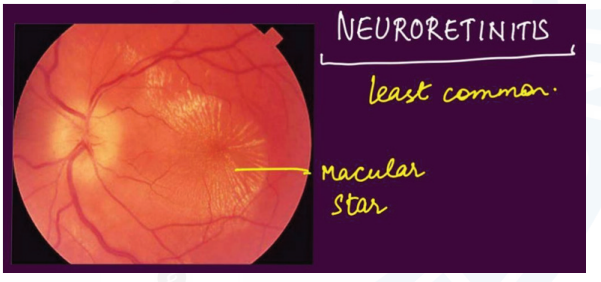
◀FIG 15.2 A. Papillitis B. Retrobulbar neusitis

❑ RETROBULBAR NEURITIS

In adults – Most common Type presentation

Fundus Exam – Normal

❑ NEURORETINITIS



◀FIG 15.3 NEURO RETINITIS

❑ OPTIC NEURITIS IN MULTIPLE SCLEROSIS

❑ TREATMENT OF OPTIC NEURITIS

ONTT → Use of steroids

- ✦ IV Methyl prednisolone Pulse therapy 1gm/day for 3 days.
- ✦ Followed by Oral steroids 1mg/Kg body weight for 11 days

} 14d

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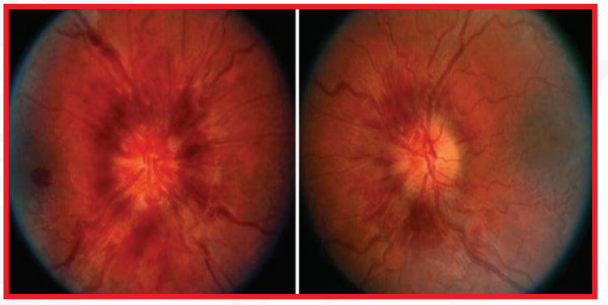
PAPILLEDEMA

Disc edema is Secondary to Increased intracranial pressure

☞ Causes

1. IC SOL (Intracranial Space occupying lesion) – Tumor, Brain abcess, tuberculoma, Aneurysm
2. Infections
3. Intracranial Hemorrhage
4. Obstruction to CSF
5. Idiopathic Intracranial Hypertension (IIH)



◀FIG 15.4 .....

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❑ TOXIC NEUROPATHY

☞ Causes

1. Tobacco – alcohol toxic ambylopia or Nutritional Ambylopia
2. Methyl alcohol Poisoning
3. Drugs
   - Ethambutol
   - Quinine
   - Chloroquine
   - Amniodarone
   - Streptomycin

☞ Clinical Features

- Centro Caecal Scotoma

☞ Treatment

- Vitamin B supplementation
- Withdrawal of drugs

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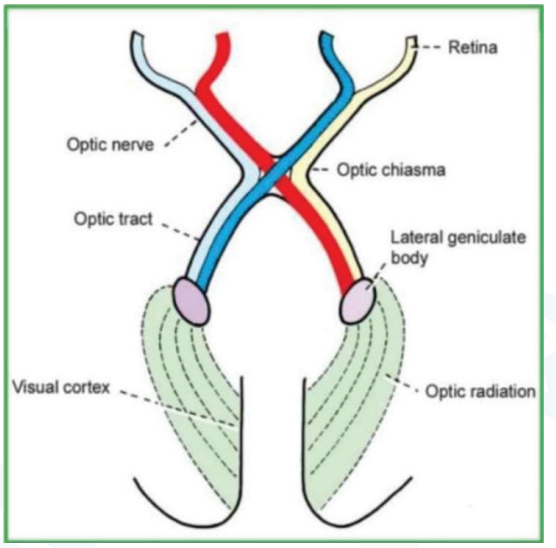
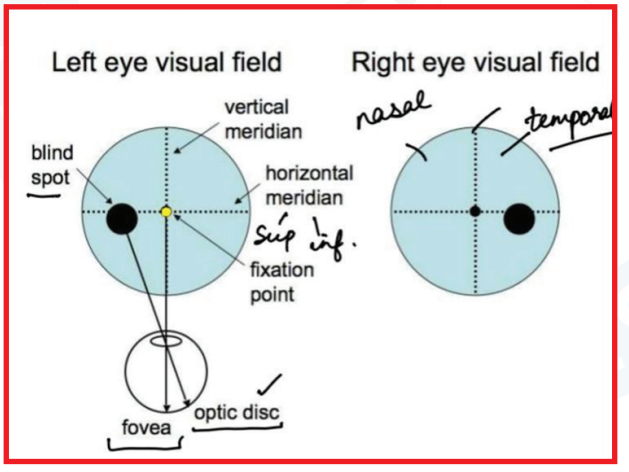
VISUAL PATHWAY



◀FIG 15.5 ANATOMY

❑ NORMAL VISUAL FIELDS

Retina is represented opposite to Visual field

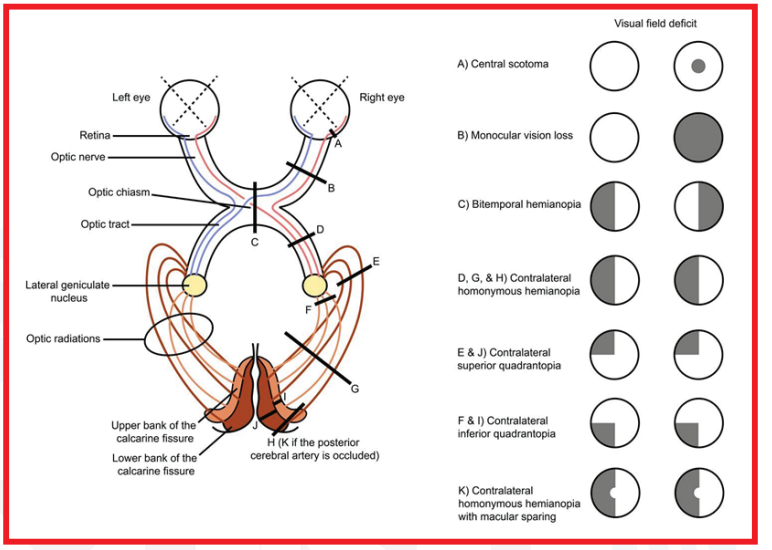


◀FIG 15.6 Normal Visual Fields

Active space

VISUAL PATHWAY LESIONS

VISUAL FIELD DECECTS



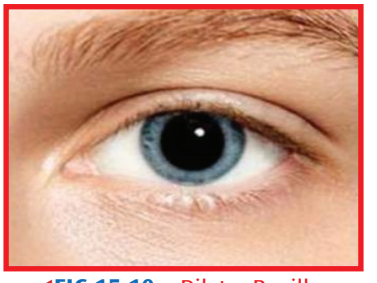
◀**FIG 15.7** Visual Field Defects

PUPIL NORMAL PUPIL AND ITS PATHWAY

Normal pupil → 3-4mm

◀**FIG 15.8** Normal Pupil

◀**FIG 15.9** Sphincter Pupillae



◀**FIG 15.10** Dilator Pupillae

Active space

❑ PUPILLARY REFLEXES

1. Light reflex
   - Direct reflex
   - Consensual reflex
2. Near reflex
   - Convergence
   - Accomodation
   - Miosis
3. Psychosensory
   - Pupillary dilatation

Miosis – constriction of pupil

Mydriasis – Dilatation of pupil

❑ MARCUS GUNN PUPIL

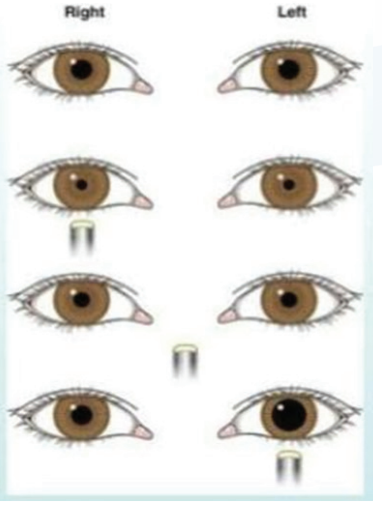
Relative afferent Pupillary defect

Swinging flash light test

- Test to diagnose Marcus gunn pupil

☞ Causes

- Optic nerve lesions
- Disease of Retina



◀FIG 15.11 JUNCTIONAL SCOTOMA

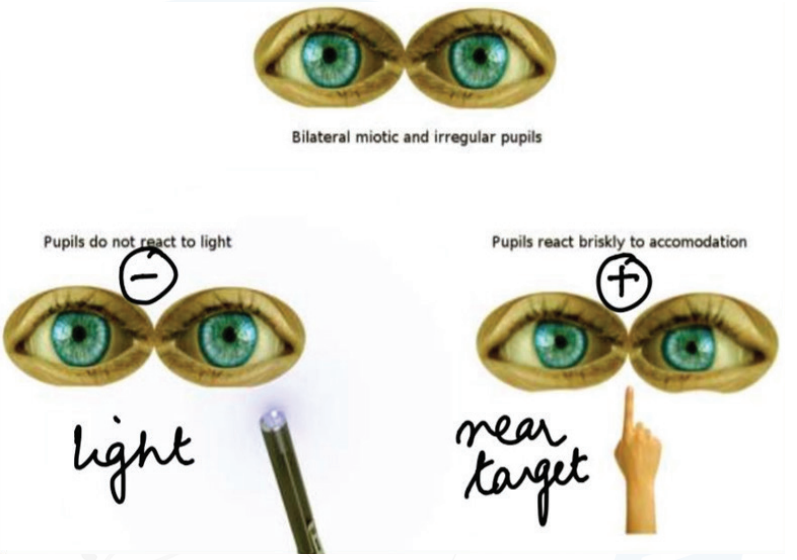
Active space

1. On bringing light to normal side – Both pupils Constrict
2. On bringing light to affected side – Both dilate

❑ LIGHT NEAR DISSOCIATION

Light reflex (-)

Near reflex (+)



◀**FIG 15.12** ARGYLL ROBERTSON PUPIL (AR PUPIL)

Light reflex absent

Accomodation Reflex Present

☞ Causes

- ✦ Neurosyphilis
- ✦ Diabetes

☞ Clinical Examination

- ✦ Bilateral Pupils are smaller, irregular
- ✦ Light reflex (-)
- ✦ Near reflex (+)

Active space

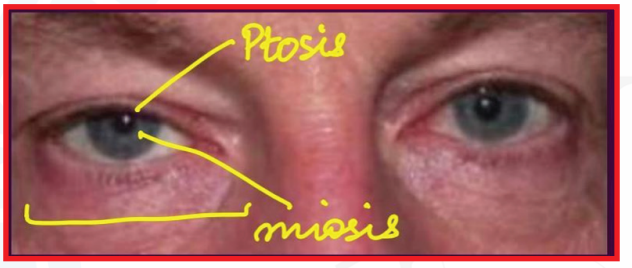
- **ADIE'S TONIC PUPIL**

  (Rigid pupil)

  - Pupil is larger
  - Unilateral
  - Light reflex (-)
  - Near reflex (-)
  - Dilation of Pupil slow

❑ **HOLME – ADIE'S Syndrome**

Horner's Syndrome



◀**FIG 15.13** HORNER SYNDROME

- Muller muscle – Mild Ptosis
- Dilator Pupillae – Miosis
- Anhydrosis –loss of sweating
- Enopthalmos

In Horner Syndrome

- Spincter Pupillae Normal
- Dilatation lag in dark
- Light reflex and near reflex is Normal

☞ Diagnosis

Cocaine in normal eye

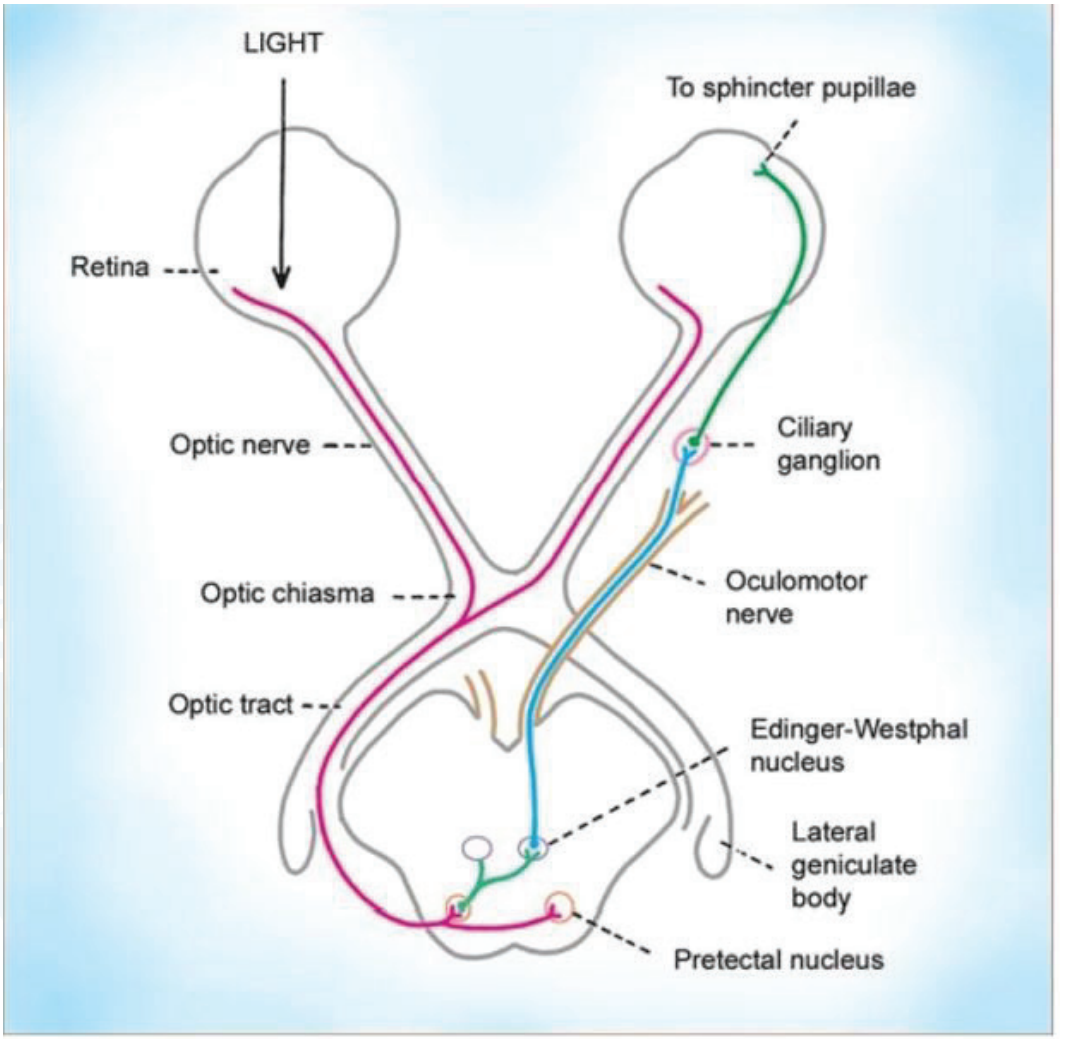
↓

Normal Pupil dilates

Horner pupil → will not dilate with cocaine

Active space

Pupillary Pathway



Active space

Workbook Questions

1. Q: Which of the following is the most common etiology of the optic neuritis?
   (a) Infectious
   (b) Non infectious
   (c) Demyelinating
   (d) Para infectious MIST

2. Q: The typical field defect in tobacco amblyopia is:
   (a) Enlargement of blind spot
   (b) Central
   (c) Centrocaecal
   (d) Altitudinal MIST

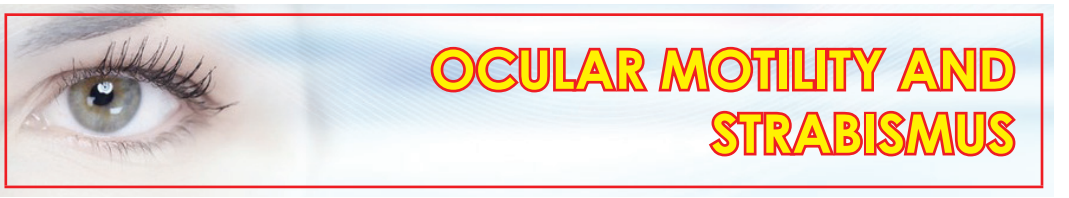
3. Q: Heteronymous hemianopia is seen in:
   (a) Occipital lobe lesion
   (b) Chiasmal lesion
   (c) Optic tract lesion
   (d) Temporal lobe lesion MIST

4. Q: Swinging flash light test is done to check for:
   (a) Adie's pupil
   (b) Marcus Gunn pupil
   (c) Horner's pupil
   (d) Absolute afferent pupillary defect MIST

5. Q: All of the following are features of Horner's syndrome except:
   (a) Miosis
   (b) Enophthalmos
   (c) Ptosis
   (d) Proptosis MIST

Answers: 1. (c), 2. (c), 3. (b) 4. (b), 5. (d).

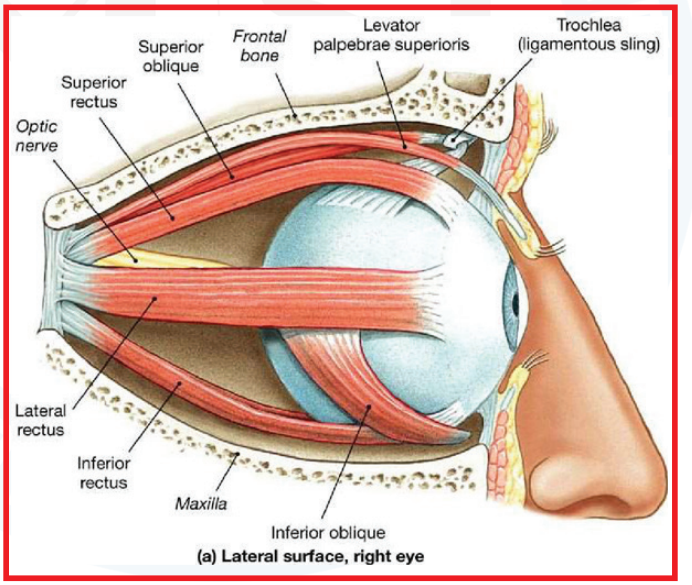
Active space

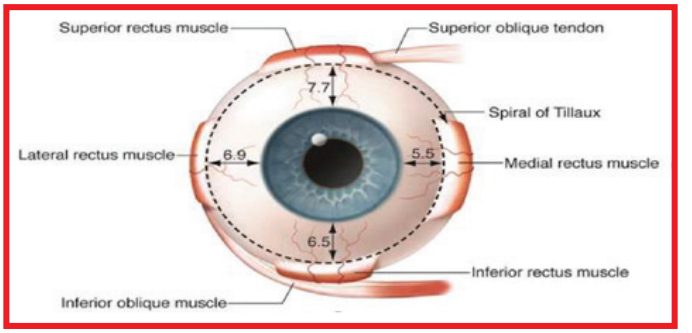


OCULAR MOTILITY AND STRABISMUS

EXTRAOCULAR MUSCLES

- **7 muscles**
  1. Four rectus = superior,inferior,medial,lateral
  2. Two oblique=superior,inferior
  3. Levatorpalpebraesuperioris





◀**FIG 16.1** Muscles of Eye

Active space

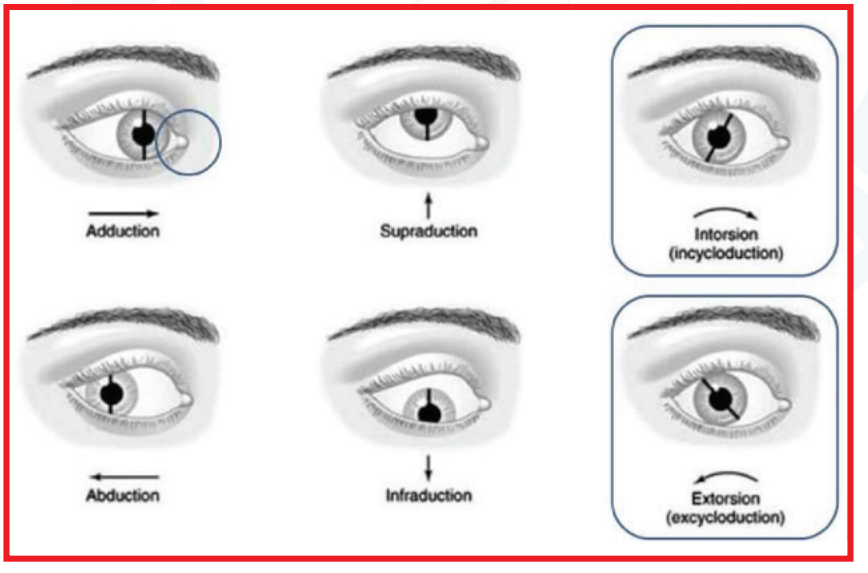
NERVE SUPPLY

- Lateral Rectus = 6th nerve
- Superior oblique = 4th nerve
- Medial Rectus, Inferior Rectus, Superior Rectus, Inferior oblique,LevatorpalpebraeSuperioris = 3rd nerve

  (LR6 SO4 O3)

❑ ACTIONS OF EXTRAOCULAR MUSCLES

- Elevation
- Depression
- Abduction
- Adduction
- Intorsion
- Extortion



◀FIG 16.2 ACTIONS OF EXTRAOCULAR MUSCLES

Active space

MUSCLE	PRIMARY ACTION	SECONDARY ACTION	TERTIARY ACTION
MR	ADDUCTION	—	—
LR	ABDUCTION	—	—
SR	ELEVATION	INTORSION	ADDUCTION
IR	DEPRESSION	EXTORSION	ADDUCTION
SO	INTORSION	DEPRESSION	ABDUCTION
IC	EXTORSION	ELEVATION	ABDUCTION

- All obliques — Abductor
- All recti (except lateral rectus) — Adductor
- All superior muscles — Intortor
- All inferior muscles — Extortor

Active space

❑ TYPE OF EYE MOVEMENTS

- ✦ Ductions = movement of one eye
- ✦ Versions = conjugate movements of two eyes (same direction)
- ✦ Vergence = conjugate movement of two eyes in opposite direction

Vergence

- ✦ Convergence
- ✦ Divergence

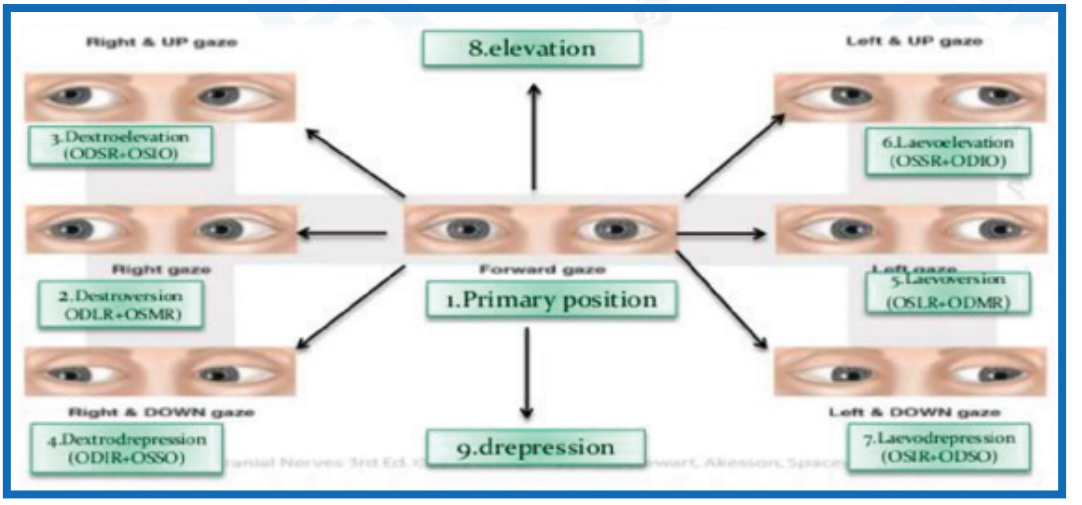
❑ GAZE POSISTIONS

1. Primary position : eyes are looking straight

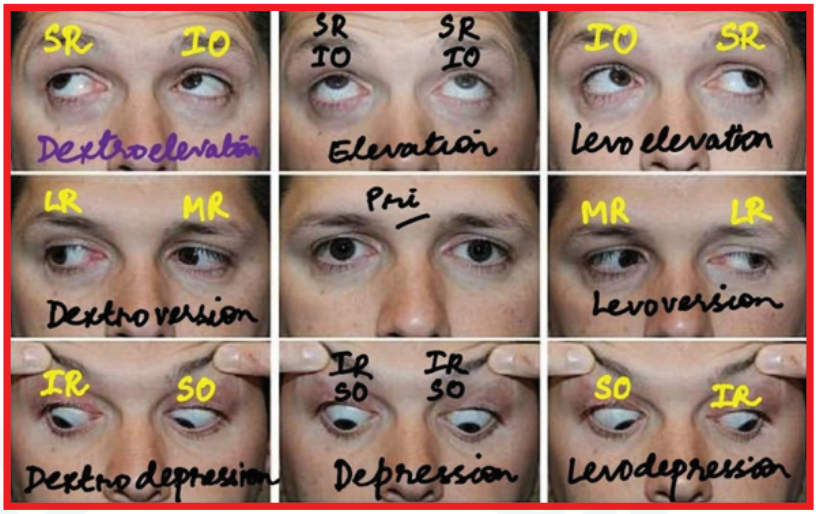
2. Secondary position : upward gaze, downward gaze, right gaze, left gaze

3. Tertiary position : VERTICAL (dextroelevation,dextrodepression) + HORIZONTAL (Levoelevation,Levodepression)

❑ 9 POSITIONS OF GAZE



Active space



◀**FIG16.3** postions of Gaze

- In Abducted position

  Main Elevator - Superior Rectus

  Main depressor- Inferior Rectus

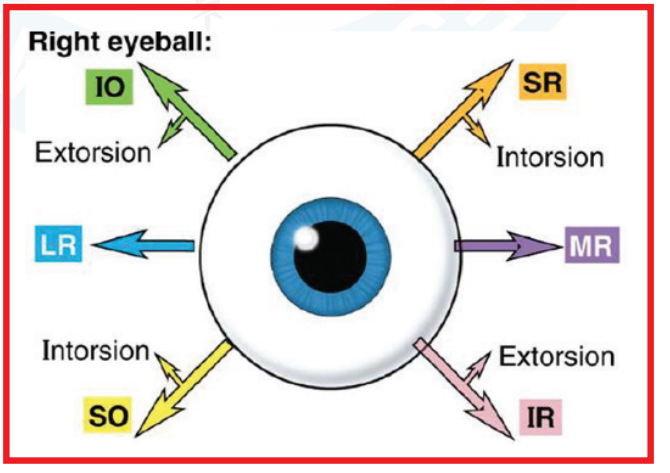
- In Adducted position

  Main elevator – Inferior oblique

  Main depressor – Superior oblique

Synergist Muscle: muscle of same eye acting together eg: 2 elevators of eye – Superior Rectus, Inferior Rectus – synergist for Elevation.

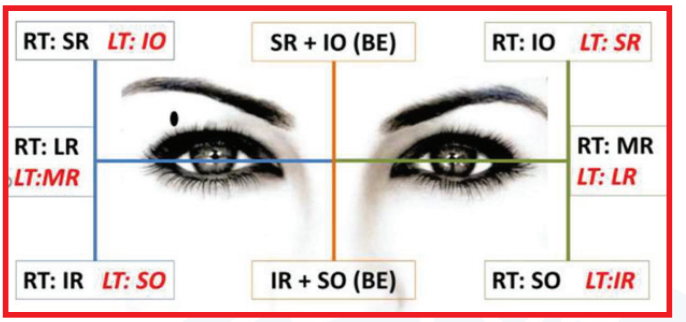
Antagonist muscle: Medial Rectus # Lateral Rectus



◀**FIG 16.4** Eye Muscle

Active space

Yoke muscle = muscle of two eyes - in same gaze movement



◀FIG 16.5 Yoke Muscle

Eg:

Right Superior Rectus &Left Inferior oblique

Left Superior oblique & Right Inferior Rectus

SQUINT/ STRABISMUS

Misalignment of two visual axis

❑ CLASSIFICATION OF SQUINT:

- ✦ Latent squint – Phoria
- ✦ Manifest Squint – Tropia
- ✦ Orthophoria – visual axis completely aligned
- ✦ Heterophoria
- ✦ Manifest Squint = Tropia

- ✦ ESOTROPIA -convergent squint
- ✦ EXOTROPIA – Divergent squint

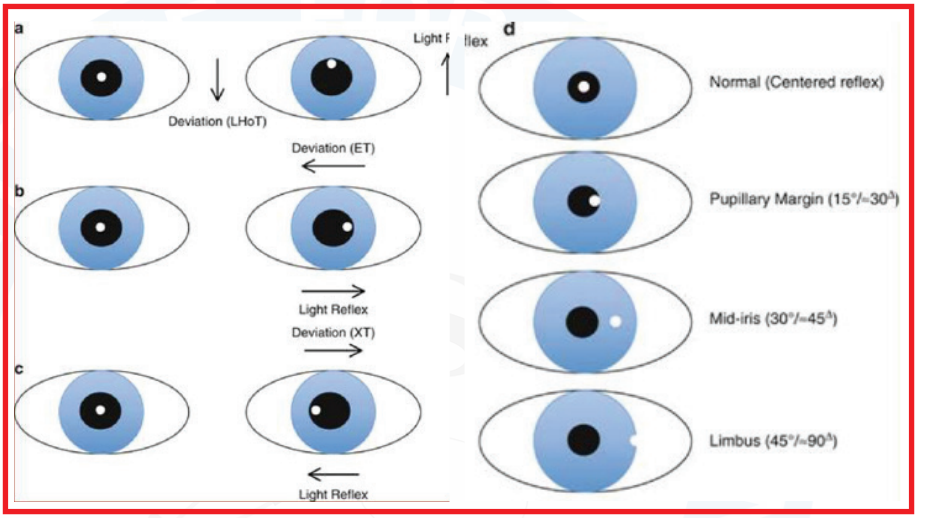
Active space

- HYPERTROPIA –upward
- HYPOTROPIA – downward

TESTS FOR SQUINT

Hirschberg corneal Reflex Test

Normal central corneal reflex = no deviation



◀FIG 16.6 Tests for Squint

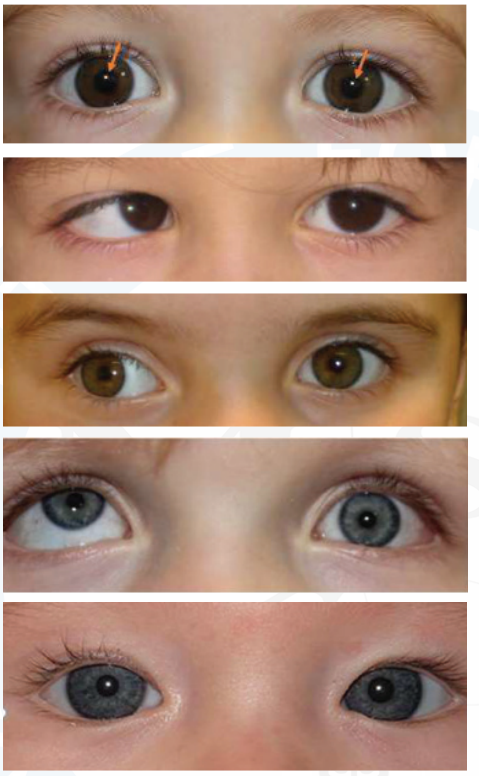
Active space

Normal - central

Pupillary margin (15° = 30 PD)

Mild Iris (30 ° = 60 PD)

Limbus (45° = 90 PD)



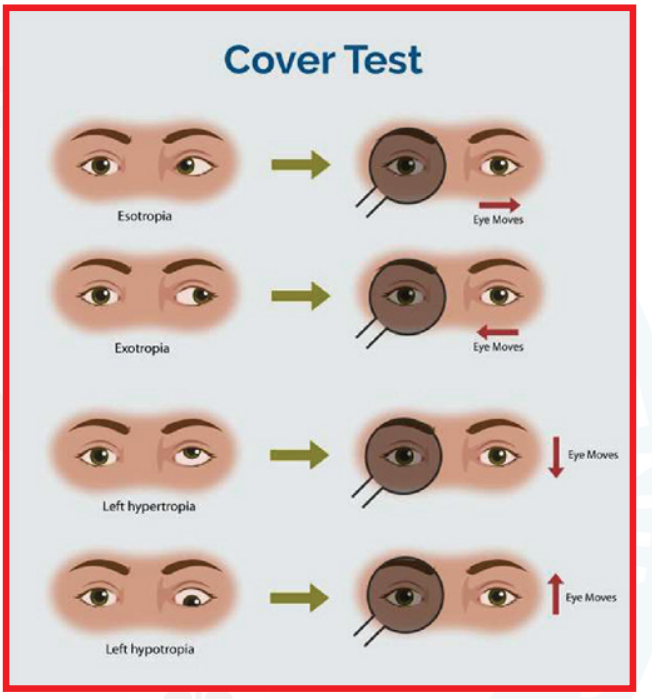
◀**FIG 16.8** Hirschberg Test

Active space

COVER -UNCOVER TEST

❑ COVER TEST

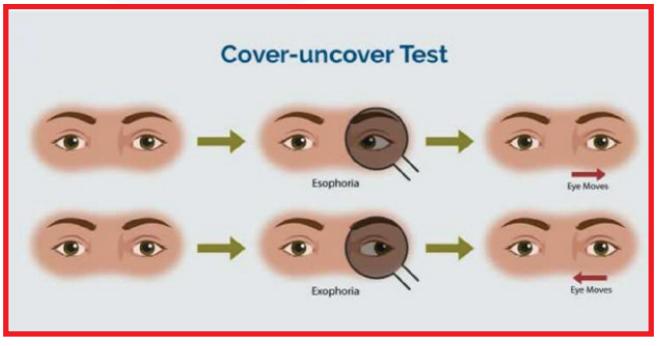
- ✦ Cover the straight eye
- ✦ Look at the uncovered eye
- ✦ Movement of the uncovered eye indicates tropia



◀**FIG 16.10** Cover Test

❑ UNCOVER TEST

- ✦ Uncover the covered eye
- ✦ Movement in just uncovered eye indicates phoria



◀**FIG 16.11** COVER-UNCOVER TEST: FOR PHORIAS

Active space

❑ ALTERNATE COVER TEST

Total squint (Manifest + latent)

- ✦ Alternately cover both eyes and see the movement of the uncovered eye.

SQUINT TYPES

1. Concomitant
2. Incomitant
   - ✦ Paralytic
   - ✦ Restrictive

❑ TREATMENT FOR SQUINT

S = spectacle correction (full)

O = orthoptis exercise

P = Prisms

S = surgery

Resection of muscle = increases strength

Recession of muscle = decreases strength

Resect- recession procedure

Oculo- cardiac Reflex – bradycardia during squint surgery

Stop the surgery immediately

PARALYTIC SQUINT

- ✦ Nerve palsy
- ✦ Myopathy
- ✦ Neuromuscular junction

❑ Features of Paralytic squint

- ✦ Diplopia – double vision

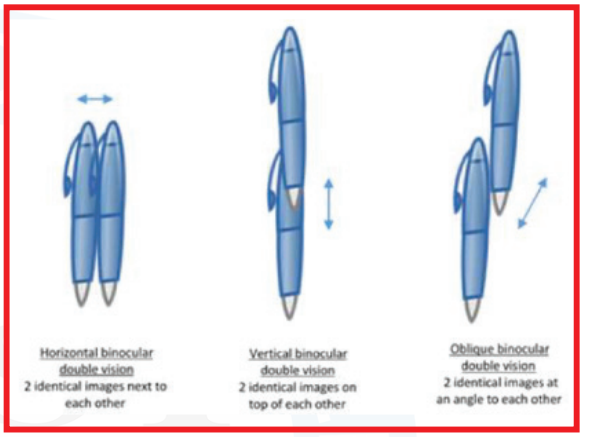
Active space

- Confusion
- Eye Movement restricted
- Compensatory head postures
- Secondary deviation > primary deviation

DIPLOPIA

Double vision

- Horizontal diplopia – horizontal muscle palsy (Medial Rectus, lateral Rectus)
- Vertical diplopia – vertical muscles
- Torsional or oblique diplopia -Vertical muscle palsy.



◀FIG 16.12

❑ CAUSES OF PARALYTIC SQUINT

- Sixth Nerve palsy
- Third Nerve palsy
- Fourth Nerve palsy

❑ Sixth Nerve palsy (Commonest cause)

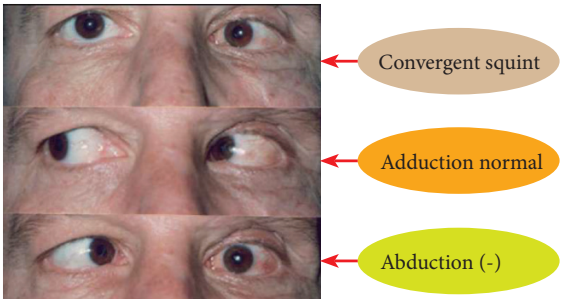
- Ischemia (Hypertension,Diabetes mellitus)
- Trauma, Intracranial disease

6thNerve - Lateral Rectus - Abduction of eye

Clinical features:

- Convergent squint
- Diplopia on Abduction
- Face turn

Active space



◀FIG 16.13 (Left Lateral Rectus palsy / 6th nerve palsy

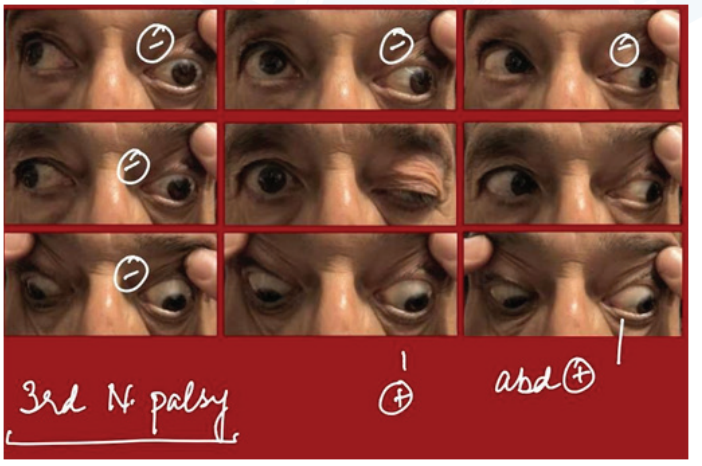
❑ THIRD NERVE PALSY

- ✦ All muscle except Lateral Rectus and Superior oblique palsied
- ✦ Medical cause: Ischemia (Diabetes mellitus, Hypertension) most common cause
- ✦ Surgical cause : Tumor, Trauma, Aneurysm of Posterior communicating Artery

☞ Clinical features:

- ✦ Ptosis
- ✦ Eyeball is downward and outward
- ✦ Pupil reaction

Absent : surgical cause Normal pupils sparing ischemia



◀FIG 16.14 (3rd Nerve palsy)

Active space

❑ FOURTH NERVE PALSY

Superior oblique muscle

- ✦ Congenital * Trauma

☞ Clinical features:

- ✦ Hypertropia, Diplopia, Maximum on looking down
- ✦ Head tilt on opposite side

BINOCULAR SINGLE VISION (BSV)

☞ 3 grades

- ✦ 1. Simultaneous Macular perception
- ✦ 2. Fusion
- ✦ 3. Steropsis
  - Suppression
  - Amblyopia

❑ Tests for Steropsis

1. Titmus Fly Test

❑ Titmus Fly Test

◀FIG 16.15 .....

Active space

AMBLYOPIA

- Vision loss in the absence of any organic cause
- Disruption during critical period of visual development (6-7 years)

TYPES OF AMBLYOPIA

1. Strabismic - most common
2. Anisometropic - hypermetropic> myopic
3. Ametropic - hypermetropic> +SD, Myopia >-8D
4. Meridoneal - Astigmatism > 2.5
5. Sensory deprivation - Congenital cataract

☞ Treatment

Spectacle correction

Surgical treatment of the cause

- Occlusion therapy (main stay) : patch Normal eye
- Penalisation : Atropinise,Plus Glasses - Normal eye
- Active vision Therapy - computerized
- L-Dopa may be useful

Active space

Workbook Questions

1. Q: What is the yoke muscle of right eye lateral rectus in dextroversion?

   (a) Medial rectus left eye

   (b) Superior oblique left eye

   (c) Inferior oblique left eye

   (d) Inferior rectus left eye MIST

2. Q: Esotropia is most commonly associated with

   (a) Hypermetropia

   (b) Myopia

   (c) Presbyopia

   (d) Astigmatism MIST

3. Q: A surgeon was doing strabismus surgery on a child, when there was a severe decrease in heart rate, what immediate steps you will take to manage?

   (a) Lessen the plane of anaesthesia

   (b) Atropine

   (c) Stop the surgery immediately

   (d) Glycopyrrolate MIST

4. Q: Of the following, which is the most common cause of third nerve palsy in adults?

   (a) Trauma

   (b) Tumours

   (c) Ischemic

   (d) Aneurysm of posterior communicating artery MIST

Active space

5. Q: All of the following are seen in oculomotor palsy except:

(a) Ptosis

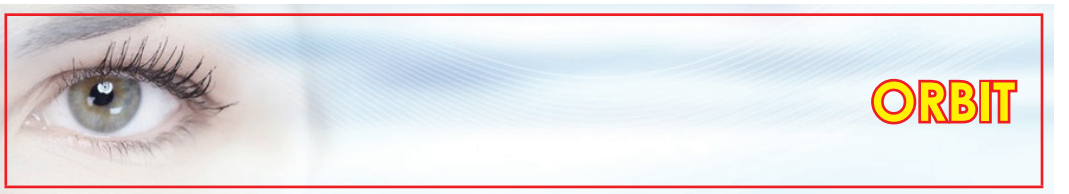
(b) Paralysis of accommodation

(c) Lateral gaze is affected

(d) Upward gaze is affected MIST

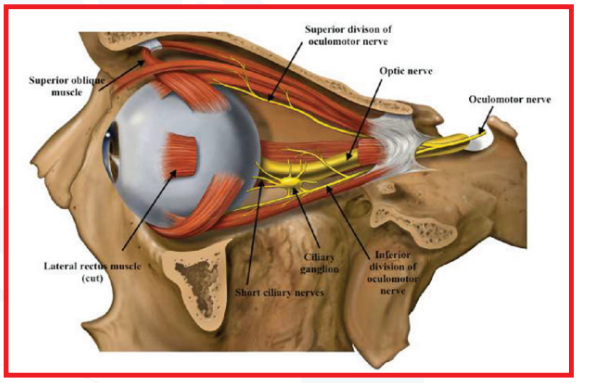
Answers: 1. (a), 2. (a), 3. (c) 4. (c), 5. (c).

Active space



ANATOMY OF ORBIT

◀ FIG 17.1



◀ FIG 17.2

❑ ORBITAL WALLS

- ✦ Weakest wall - medial wall
- ✦ Weakest point – maxillary part of orbital floor
  (Most commonly involved in blow out fracture)
- ✦ Strongest wall – lateral wall (orbitotomy approach)

❑ EXOPHTHALMOMETRY

◀FIG 17.3 Proptosis

Active space

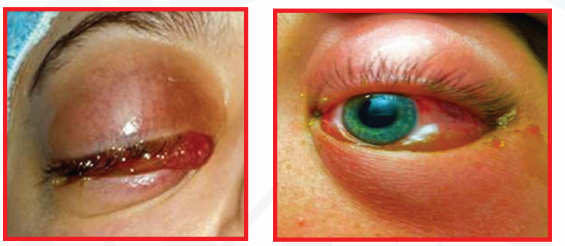
❑ ENOPTHALMOS

(Backward /inward displacement of eye ball)

☞ Orbital cellulitis

Infection of soft tissues behind orbital septum

- Infection from paranasal sinuss
- Post traumatic
- Post operative (DCR,orbitotomy)
- Pneumococcus ,staph, H.influenzae



◀FIG 17.4 Orbital Cellutitis

❑ Complications of orbital cellulitis

✦ Cavernous sinus thrombosis

✦ Intracranial spread

☞ Treatment :

✦ Hospitalized

✦ ENT consultation

✦ I/V antibiotics

ORBITAL MUCORMYCOSIS

◀FIG 17.5 Orbital Mucormycosis

Active space

Most common cause- mucor

Rhizopus

Necrotizing inflammation

☞ Treatment :

- Hospitalisation
- I/V amphoteracin

THYROID EYE DISEASE

- ✦ Goitre opthalmopathy
- ✦ Thyroid associated
  - ✦ Pathophysiology
  - ✦ Hyperthyroid
  - ✦ Euthyroid : normal levels of thyroid hormones
  - ✦ Hypothyroid

☞ RISK FACTORS:

- Smoking
- Female sex
- Middle age
- HLA DR 3 , HLA B 8

☞ Clinical features :

- ✦ Lid retraction
- ✦ Soft tissue involvement
- ✦ Proptosis
- ✦ Restrictive myopathy
- ✦ Optic neuropathy

Active space

❑ **Lid retraction**

- ✦ Earliest
- ✦ Most common presentation

◀FIG 17.6 Dalrymple sign

❑ EYE SIGNS IN THYROID EYE DISEASE

- ✦ Dalrymples sign = lid retraction
- ✦ Kocher sign =staring look
- ✦ Stellwag sign = infrequent blink

Proptosis

◀FIG 17.7

❑ Restrictive thyroid myopathy

Sequence of EOM involvement

IR > MR > SR > LR

☞ Treatment of thyroid eye disease

- Dry eye treatment
- Systemic steroids
- Immunosuppressant
- Orbital radiotherapy

☞ Surgical

- Orbital decompression (sight threatening and severe proptosis)
- Extraocular muscle surgery

Active space

- Eyelid surgery

ORBITAL TUMORS:

❑ ADULT:

- ✦ Most common beningn tumor: CAVERNOUS HEMANGIOMA
- ✦ Most common primary malignant orbital tumor :LYMPHOMA

❑ PEDIATRIC:

- ✦ Most common beningn tumor:DERMOID
- ✦ Most common malignant orbital tumor RHABDOMYOSARCOMA

MEDICAL INSTITUTE FOR SCREENING TEST MIST

Active space

Workbook Questions

1. Q: Which is the weakest wall of the orbit:
   (a) Lateral
   (b) Medial
   (c) Superior
   (d) Inferior MIST

2. Q: Most common orbital wall fractured in blow out fracture orbit is
   (a) Medial wall
   (b) Lateral wall
   (c) Floor
   (d) Roof MIST

3. Q: Most common cause of unilateral proptosis in adults is
   (a) Thyroid ophthalmopathy
   (b) Cavernous sinus thrombosis
   (c) Orbital sinusitis
   (d) Retrobulbar haemorrhage MIST

4. Q: Which of the following is the first sign to appear in thyroid eye disease?
   (a) Lid lag
   (b) Upper lid retraction
   (c) Infrequent blinking
   (d) Staring look MIST

5. Q: Purulent inflammation of the tissues of the orbit is called:
   (a) Orbital cellulitis
   (b) Endophthalmitis
   (c) Panophthalmitis
   (d) Dacryocystitis MIST

Answers: 1. (b), 2. (c), 3. (a) 4. (b), 5. (a).

Active space

EYELIDS AND LACRIMAL SYSTEM

EYELID ANATOMY AND MUSCLES

◀**FIG 18.1** Eylid Anatomy

❑ LEVATOR PALPEBRAE SUPERIORIS

❑ GLANDS OF EYELID

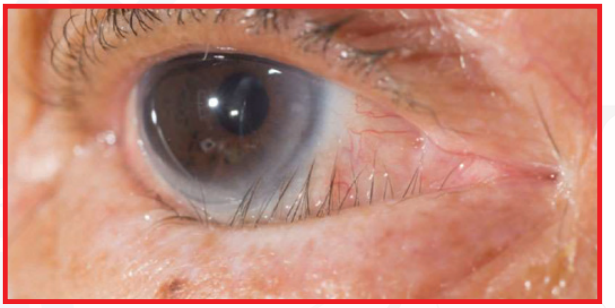
1. Meibomian glands

Active space

2. Glands of zeis
3. Glands of moll: modified sweat glands
4. Accessory lacrimal glands

ABNORMALITIES OF EYELASHES

- Trichiasis -misdirected eyelashes
- Distichiasis -extra/second row of eyelashes
- Poliosis- premature greying
- Madarosis -loss of eye lashes

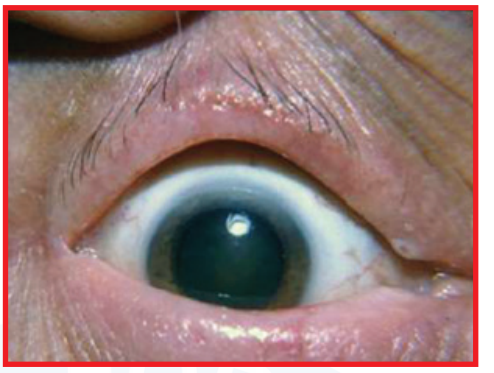


◀FIG 18.3 Trichiasis

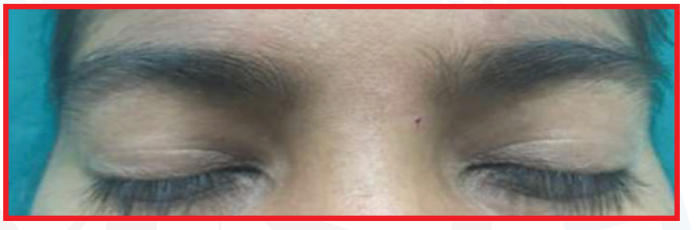
◀FIG 18.4 Distichiasis

◀FIG 18.5 Poliosis

Active space



◀FIG 18.6 Madarosis



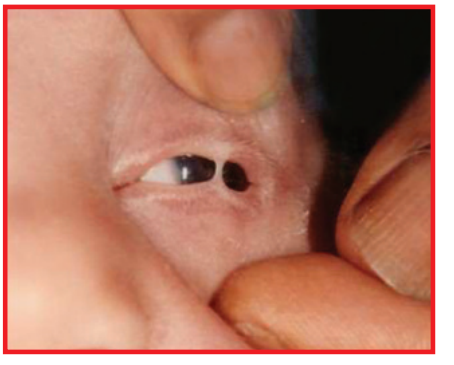
◀FIG 18.7 TRICHOMEGALY

- **Drug induced trichomegaly**
  - Topical prostaglandins
  - Phenytoin
  - Cyclosporine
  - Erlotinib

ABNORMALITIES OF EYELIDS

Ankyloblepharon – adhesion of two lids

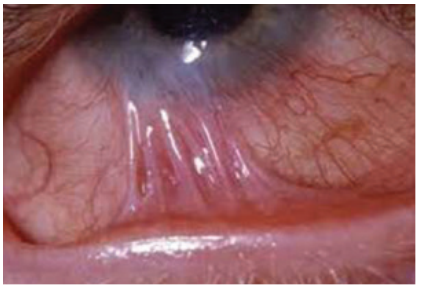
Symblepharon – adhesion between palpebral and bulbar conjunctiva



◀FIG 18.8 Ankyloblepharon

Active space

- Chemical burns
- Thermal injuries
- s-j syndrome
- ocp



◀FIG 18.9 Symblepharon

INFLAMMATORY DISORDERS

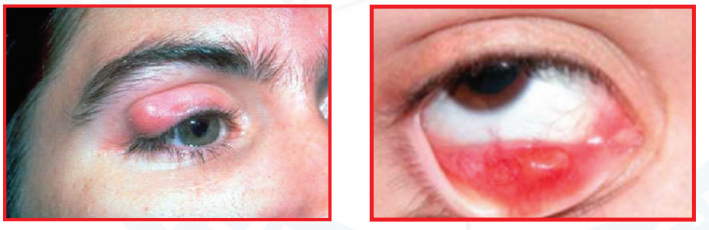
❑ BLEPHARITIS

- .....
- .....
- .....
- .....

.....

❑ Chalazion

Chronic lipogranulomatous inflammation of meibomian gland



◀FIG 18.10 Chalazion

☞ Treatment

1. Smallchalazion: hot fomentation antibiotic steroid ointment
2. Intralesionalsteroid: triamcinalone acetate
3. Incision and curettage

◀FIG 18.11 INCISION AND CURETTAGE OF CHALAZION

Active space

❑ External hordeolum/stye

Suppurative inflammation of glands of zeis /moll

◀FIG 18.12 External hordeolum/stye

(Painfultender, treatment – systemic antibiotic)

Epilate the involved eyelash

Internal hordeolum/infected chalazion

Suppurative Inflammation of meibomian gland

◀FIG 18.13 .....

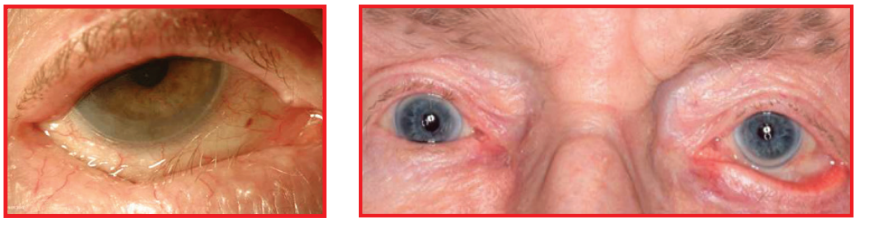
(Painful tender swelling)

(Treatment: topical +systemic antibiotics)

ABNORMALITIES OF EYELID MARGIN

Entropion – inward turning

Ectropion – outward turning



◀FIG 18.14 Entropion

Active space

❑ Entropion

1. Senile entropion (most common)
2. Congenital entropion
3. Cicatricidal entropion
4. Spastic entropion

☞ Surgeries

- ✦ Jones procedure
- ✦ Weis procedure
- ✦ Wheeler's procedure
- ✦ Quickert

☞ Ectropion

4 causes:

- Senile/ Involutional (most commom)
- Congenital
- Cicatricial
- Paralytic (7th nerve palsy)

☞ Treatment

- ✦ Medial conjunctivoplasty
- ✦ Modified kuhnt procedure – syzmanowsky
- ✦ Lazy T procedure
- ✦ Z – plasty
- ✦ v-y operation

Active space

PTOSIS

Drooping of eyelid

- ✦ Congenital ptosis
- ✦ Acquired ptosis

❑ Congenital ptosis

2 types

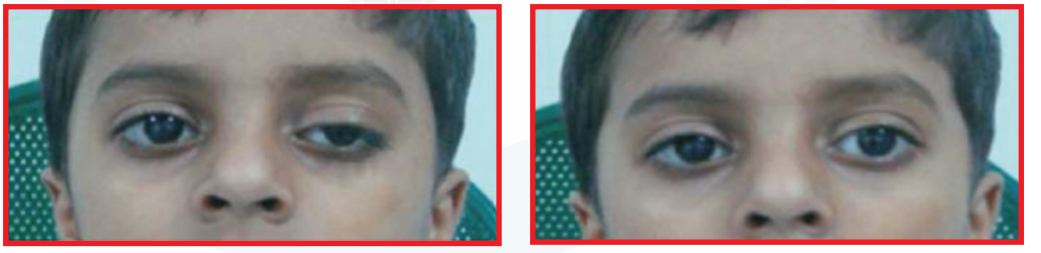
Simple – only congenital ptosis

Complicated – associated with each other abnormalities

1. Weakness of superior rectus muscle
2. Marcus gun synkinetic ptosis
3. Blepharophimosis syndrome

❑ Marcus gunn synkinetic ptosis

- ✦ Aberrant connection between 3rd (levator palpebral muscle) and 5th nerve (pterygoid muscle)



◀**FIG 18.15** Marcus gunn synkinetic ptosis

❑ Blepharophimosis epicanthus inversus syndrome (BPES)

1. Phimosis –shortened horizontal aperture
2. Epicanthus inversus
3. Ptosis
4. Telecanthus
5. Ectropion of lower lid

Active space

ACQUIRED PTOSIS

- **Types**
  - Aponeurotic – LPS aponeurosis (senile and trauma)
  - Mechanical – cyst , tumor
  - Neurogenic – 3rd nerve palsy ,horner syndrome
  - Myogenic – myasthenia gravis
    Myotonic dystrophy
    Myopathies
- **SENILE PTOSIS /INVOLUNTIONAL PTOSIS**
  - Usually bilateral ,symmetrical

◀**FIG 18.16** SENILE PTOSIS /INVOLUNTIONAL PTOSIS

- **Mild ptosis and good LPS action**
  - Fasanella servat operation (conj-tarso-mullerectomy)
- **Moderate ptosis and moderate LPS action**
  - LPS resection
- **Severe ptosis and poor LPS action**
  - Frontalis suspension
    Frontalis suspension or sling surgery
- **Myasthenia Gravis**
  - Deficiency of acetylcholine at neuromuscular junction
  - Ptosis – 1st ocular sign (bilateral asymmetrical variable)
  - Orbicularis oculi weakness (eye peek sign)

Active space

- EOM weakness (medial rectus most common)

◀FIG 18.17 Myasthenia Gravis

Tests for myasthenia gravis

- Ice pack test
- Edrophonium /neostigmine test
- Muscle biopsy
- Thoracic imaging

◀FIG 18.18 Edrophonium or tensilon test

MALIGNANT EYELID TUMORS

- Basal cell Carcinoma
- Squamous cell carcinoma
- Sebaceous cell carcinoma

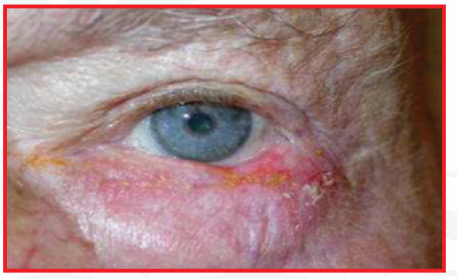
Active space

1. Basal cell carcinoma

◀**FIG 18.19** MALIGNANT EYELID TUMORS

Rodent ulcer

2. Squamous cell carcinoma



◀**FIG 18.20** Squamous Cell

Most common: lower lid

3. Sebaceous cell carcinoma

◀**FIG 18.21** Sebaceous cell

Meibomian gland can mimick –chalazion (recurrent, elederly)

◀**FIG 18.22** Meibomian gland

Active space

LACRIMAL APPARATUS

LACRIMAL APPARATUS

- ✦ Lacrimal glands
- ✦ Lacrimal passages

❑ **Main lacrimal gland**

- ✦ Reflex tear secretion

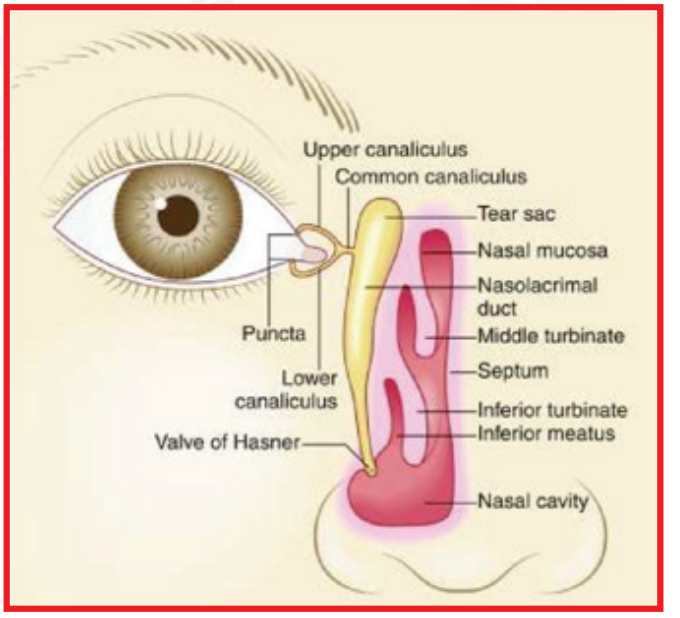
◀**FIG 18.23** Main Lacrimal Gland

◀**FIG 18.24**

❑ **Accessory lacrimal glands**

Glands of Krause and wolfring

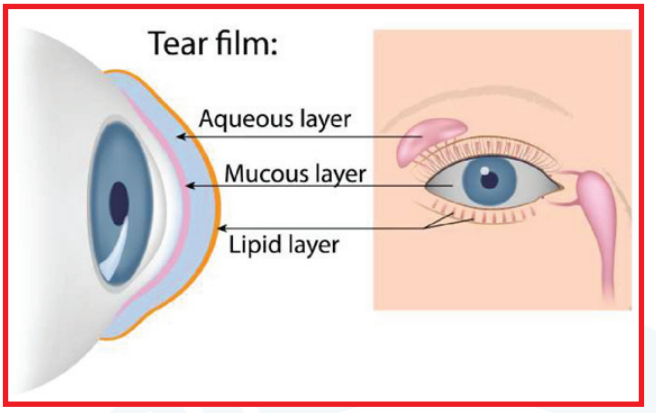
- ✦ Basal tear secretion



◀**FIG 18.25** Lacrimal passages

Active space

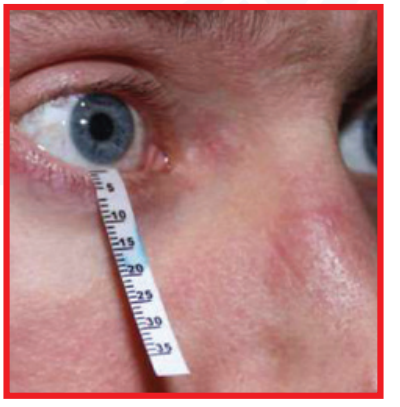
Total nasolacrimal duct 18mm (12mm –bony, 6mm –membranous)



◀FIG 18.26 PECORNEAL TEAR FLIM

DRY EYE

- **Evaporative dry eye-**
  - Lipid layer abnormality (meibomitis, exposure, keratitis)
- **Aqueous dry eye**
  - Lacrimal glands (sjogrens disease, rheumatic arthritis, SLE, KCS)
- **Mucin deficiency dry eye**
  - Goblet cells of conjunctiva (chemical trauma, ocp, SJ syndrome, trachoma)
- **DRY EYE WORKUP**
  - Schimmer test-normal - 10-25mm
  - Vital staining (rose Bengal stain)



◀FIG 18.27 Dry Eye Workup

☞ Treatment :

- Tear substitutes
- Treatment of cause

Active space

WATERING OF EYES

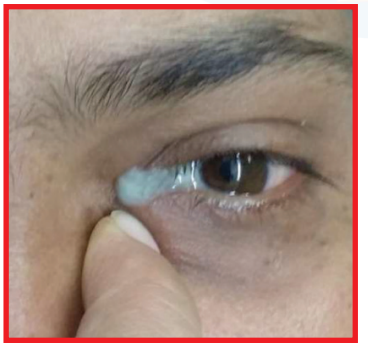
- Lacrimination :hypersecretion of tears
- Epiphora :obstruction to outflow (anatomical , physiological-lacrimal pump failure)

❑ Causes of blockage

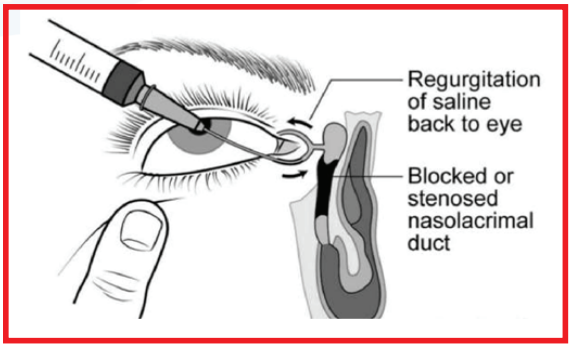
1. Anatomical:
   - Strictures
   - Obstruction
   - Foreign body
   - Tumor
2. Physiological
   - Lacrimal pump failure

❑ Tests for lacrimal drainage system

- Regurgitation test
- Syringing
- Jones dye test
- Dacryocystography
- Dacryoscintigraphy



◀FIG 18.28 Watering of eyes



◀FIG 18.29 Regurgitation test

Active space

A, Complete canalicular obstruction: The cannula is advanced with difficulty, and irrigation fluid refluxes from the same canaliculus.

B, Complete common canalicular obstruction: A "soft stop" is encountered at the level of the common canaliculus, and irrigated fluid refluxes through the opposite punctum.

C, Complete nasolacrimal duct obstruction (NLDO): The cannula is easily advanced to the medial wall of the lacrimal sac, then a "hard stop" is felt, and irrigation fluid refluxes through the opposite punctum. Often, the refluxed fluid contains mucus and/or pus.

D, Partial NLDO: The cannula is easily placed, and irrigation fluid passes into the nose as well as refluxing through the opposite punctum.

E, Patent lacrimal drainage system: The cannula is placed with ease, and most of the irrigation fluid passes into the nose

❑ **Congenital dacrocystitis :**

Failure of canalization of nasolacrimal duct end of gestation

Site of obstruction – valve of hasner

Epiphora

Positive regurgitation

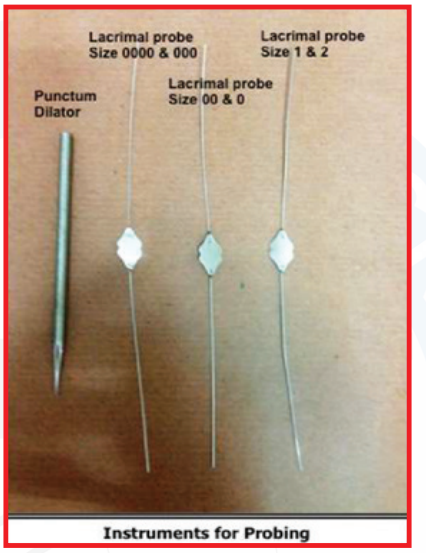
Swelling over sac area

◀**FIG 18.30** Congenital dacrocystitis

Active space

☞ Treatment

- Upto 9 months (sac massage +topical antibiotics)
- 9 months to 4 years (probing and syringing)
- After 4 years :DCR (dacrocystorhinostomy)



◀FIG 18.31

❑ Acquired dacrocystitis :

Acute : acute suppurative inflammation (most common st.pnemococcus)

☞ Treatment :

- Cellulitis stage – hot compress +antibiotics
- Lacrimal abscess stage : drain abscess under DCR
- External lacrimal fistula :fistulectomy + DCR
- Chronic dacrocystitis

Active space

More common

Causes: chronic inflammation of nasal mucosa or by polyp pressure

Treated by DCR

◀FIG 18.32 DACROCYSTORHINOSTOMY

Dacryocystorhinostomy DCR:

1. Conventional External DCR for NLD obstruction
2. Endonasal DCR for NLD obstruction
3. Endocanalicular laser DCR for NLD obstruction: uses Holmium YAG laser
4. Conjunctivo-DCR with Jones tube for canalicular obstruction

Active space

Workbook Questions

1. Q: Distichiasis is
   (a) Increased length of eyelashes
   (b) Loss of eyelashes
   (c) Second row of eyelashes
   (d) Increased number of eyelashes MIST

2. Q: Glands of Zeis are:
   (a) Modified sweat glands
   (b) Modified sebaceous glands
   (c) Modified meibomian glands
   (d) None of the above MIST

3. Q: Lid lag is feature of which of the following
   (a) Congenital ptosis
   (b) Senile ptosis
   (c) Synkinetic ptosis
   (d) Myasthenia gravis MIST

4. Q: What is the most common ocular finding in myasthenia gravis?
   (a) Ptosis
   (b) Lagophthalmos
   (c) Proptosis
   (d) Enophthalmos MIST

Active space

5. Q: The test shown in the picture is done for which of the following conditions?

(a) Dry eye

(b) Epiphora

(c) Congenital NLD blockade

(d) Lacrimal pump failure MIST

Answers: 1. (c), 2. (a), 3. (a) 4. (a), 5. (a).

Active space

OCULAR TRAUMA

NOMENCLATURES

Closed Globe injuries: .....

.....

Open Globe Injuries: Full thickness injury

To cornea/ Sclera

Blunt trauma Sharp object

Globe Rupture Laceration

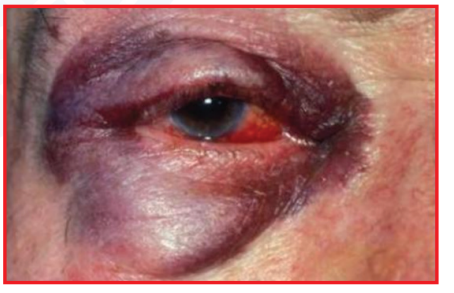
Penetrating injury: Only entry wound

Perforating injury: Entry wound and Exit wound

BLUNT TRAUMA

❑ ORBITAL BLOW OUT FRACTURE

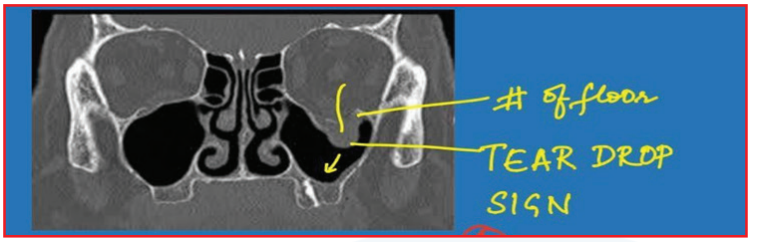
- ✦ Most common wall : Floor of orbit
- ✦ 2nd most common : Medial wall
- ✦ Lid edema
- ✦ Orbital ecchymosis
- ✦ Subconjunctival Hemorrhages
- ✦ Diplopia (Muscle Inferior Rectus entrapment)
- ✦ Double Diplopia
  - Upgaze
  - Down gaze



◀FIG 19.1 ORBITAL BLOW OUT FRACTURE

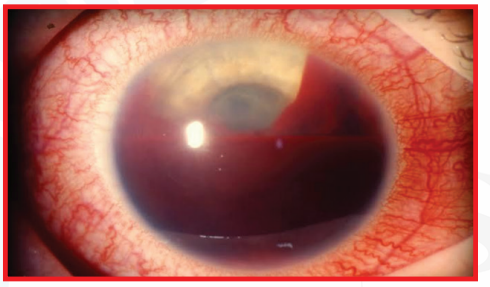
Active space

- Anesthesia of cheek
- Enaphthalmos



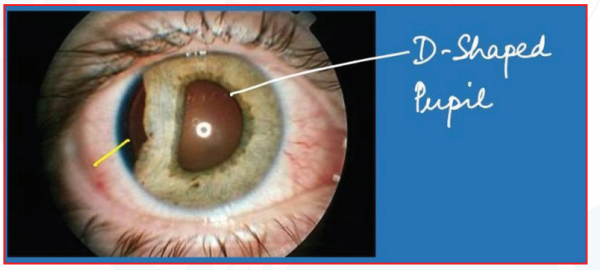
◀FIG 19.2 Blow out fracture of floor of the orbit

❑ Hyphema



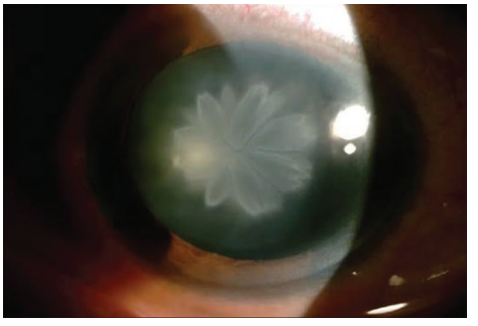
◀FIG 19.3 Blood in Anterior Chamber

❑ Irido dialysis

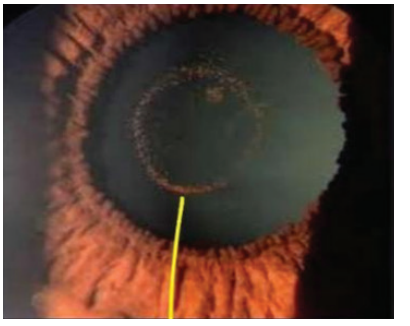


◀FIG 19.4 Irido dialysis

❑ Cataract due to Blunt Trauma

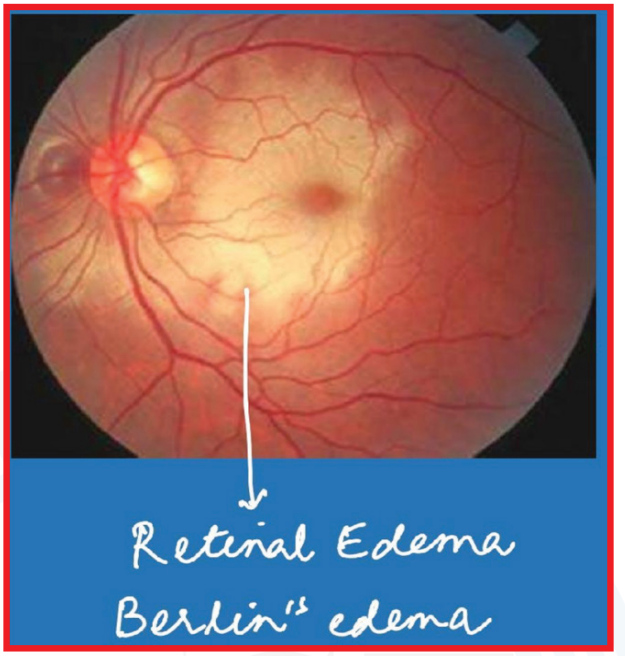


◀FIG 19.5 Rosette Cataract



◀FIG 19.6 Vossius Ring

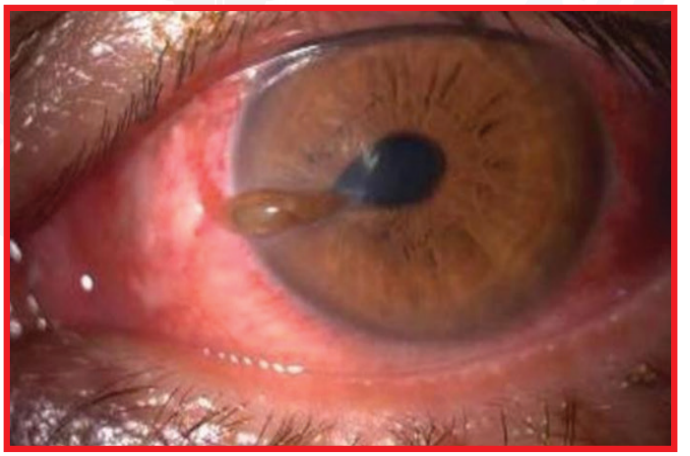
Active space



◀FIG 19.7 COMMOTIO RETINAE

PENETRATING EYE INJURY

Caused by sharp objects

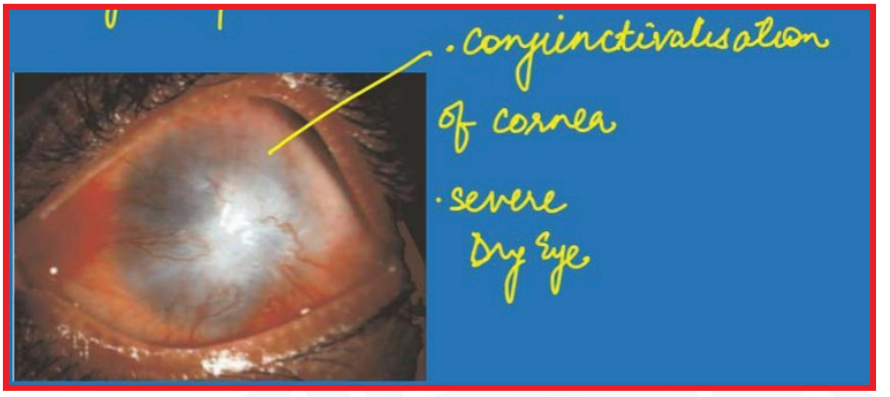


◀FIG 19.8 PENETRATING EYE INJURY

❑ CHEMICAL INJURY

Active space

✦ Symblepharon



◀**FIG 19.9** OPHTHALMIC EMERGENCY

Active space

Workbook Questions

1. Q: Vossius ring is seen:
   (a) Penetrating trauma
   (b) Blunt trauma
   (c) Iridocyclitis
   (d) Acute angle closure glaucoma MIST

2. Q: A patient presents with history of chuna particles falling in the eye. Which of the following should not be done immediately?
   (a) Repeated irrigation of the conjunctival sac with normal saline
   (b) Frequent instillation of sodium citrate drops
   (c) Thorough slit lamp examination
   (d) Double eversion of the lids to remove the chuna particles MIST

3. Q: Pre-requisite for sympathetic ophthalmia is
   (a) Penetrating eye trauma
   (b) Blunt ocular trauma
   (c) Chemical injury
   (d) Urinary tract infection MIST

4. Q: Berlin's edema is seen in
   (a) Penetrating eye trauma
   (b) Blunt ocular trauma
   (c) Chemical injury
   (d) All of the above MIST

5. Q: Which area is known as dangerous area of eye
   (a) Optic nerve
   (b) Lens
   (c) Ciliary body
   (d) Sclera MIST

Answers: 1. (b), 2. (c), 3. (a) 4. (b), 5. (c).

Active space

COMMUNITY OPHTHALMOLOGY

❑ DEFINITION OF BLINDNESS by WHO

- ✦ Visual Acuity < 3/60 on snellen's chart
- ✦ Economic Blindness < 6/60 to 3/60 in better eye
- ✦ Social Blindness < 3/60 to 1/60
- ✦ Legal Blindness < 1/60 to Light perception
- ✦ Total Blindness → No light Perception

❑ AVOIDABLE BLINDNESS

- ✦ Preventable Blindness
  - Vitamin A deficiency
  - Trachoma
- ✦ Curable Blindness
  - Cataract

❑ VISION 2020 – RIGHT TO SIGHT (WHO)

- ✦ Cataract
- ✦ Childhood Blindness
- ✦ Trachoma
- ✦ Refractive Errors & Low vision
- ✦ Onchocerciasis (River Blindness) → Endemic in Africa

❑ VISION 2020 – INDIA

- ✦ Cataract

Active space

- Childhood Blindness
- Trachoma
- Refractive Errors & Low vision
- Corneal Blindness
- Diabetic Retinopathy
- Glaucoma

❑ CHILDHOOD BLINDNESS

- Vitamin A Deficiency
- Measles
- Conjunctivitis
- Ophthalmia neonatorum
- Injuries
- Congenital cataract
- ROP
- Childhood Glaucoma

Active space